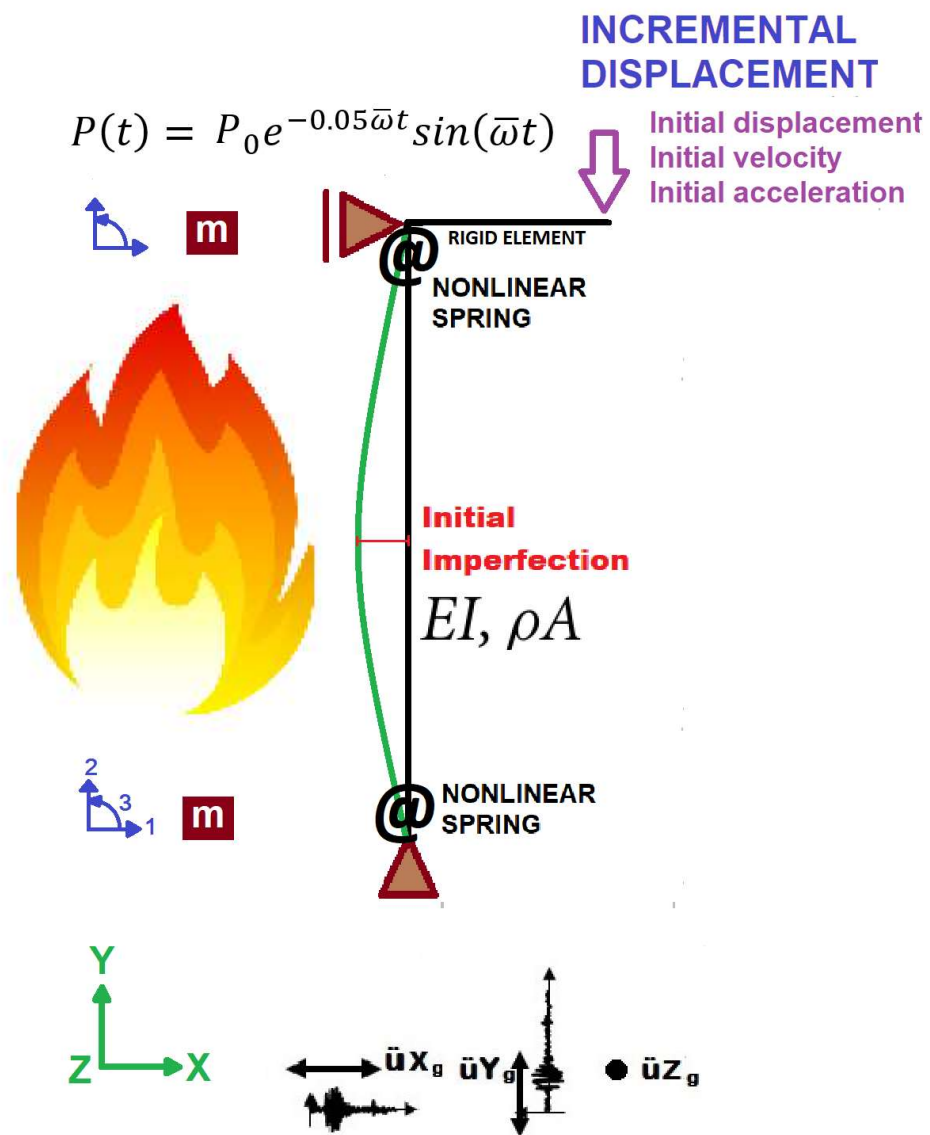


>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF MULTI-MODE POST- BUCKLING PHENOMENA IN A POST-FIRE STEEL COLUMN WITH SEMI-RIGID CONNECTIONS: AN OPENSEES FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY, AND DYNAMIC TIME-HISTORY ANALYSES

THIS PROGRAM IS WRITTEN BY SALAR DELAVAR GHASHGHAEE (QASHQAI)



$$W(x,t) = P(t) = W_0 e^{-0.05\bar{\omega}t} \sin(\bar{\omega}t)$$

$$m\ddot{u} + c\dot{u} + ku = P(t)$$

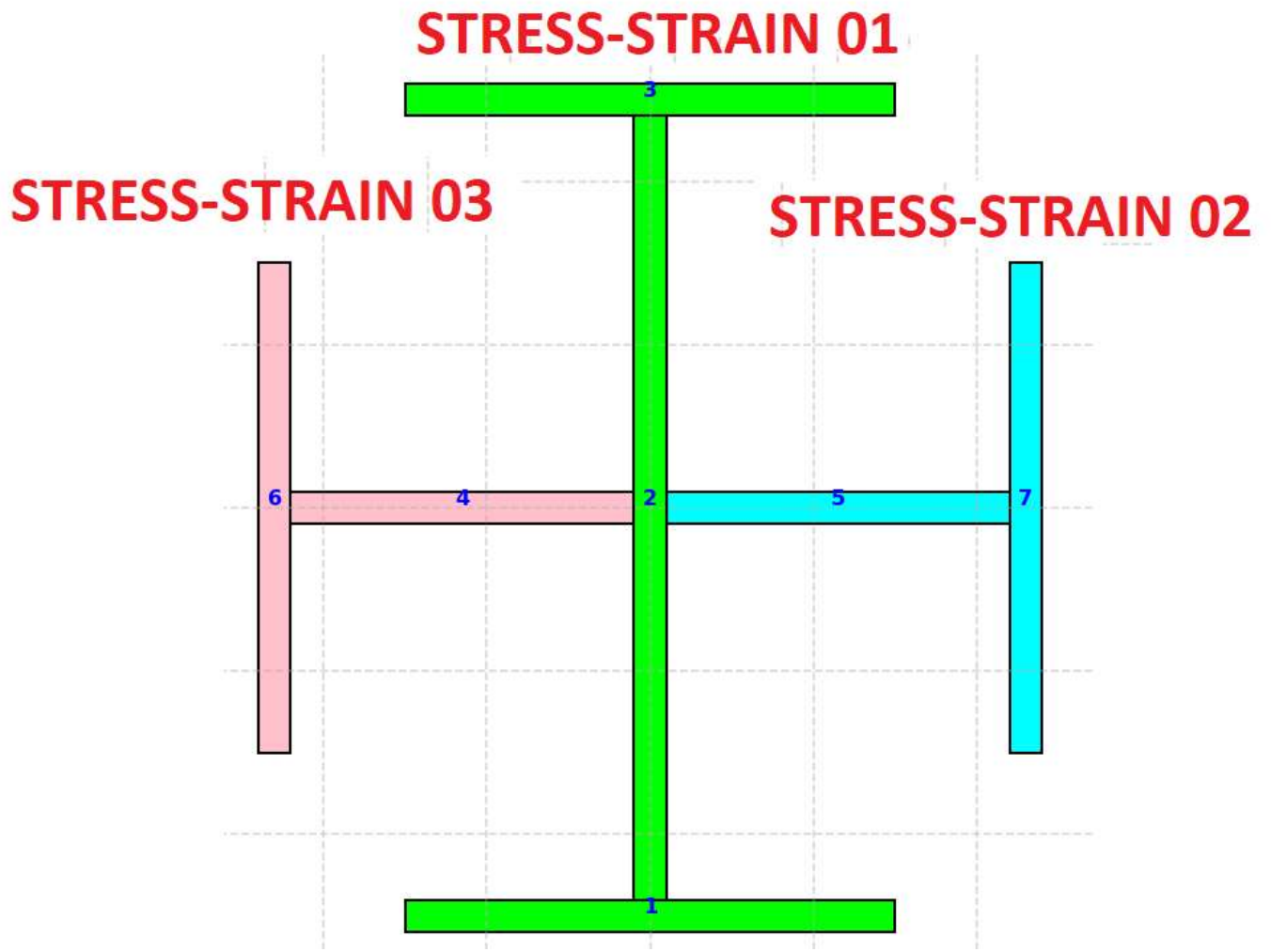
$$Structure\ Ductility\ Damage\ Index = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

$$\rho A \frac{\partial^2 v}{\partial t^2} + EI \frac{\partial^4 v}{\partial x^4} = w(x,t).$$

boundary conditions

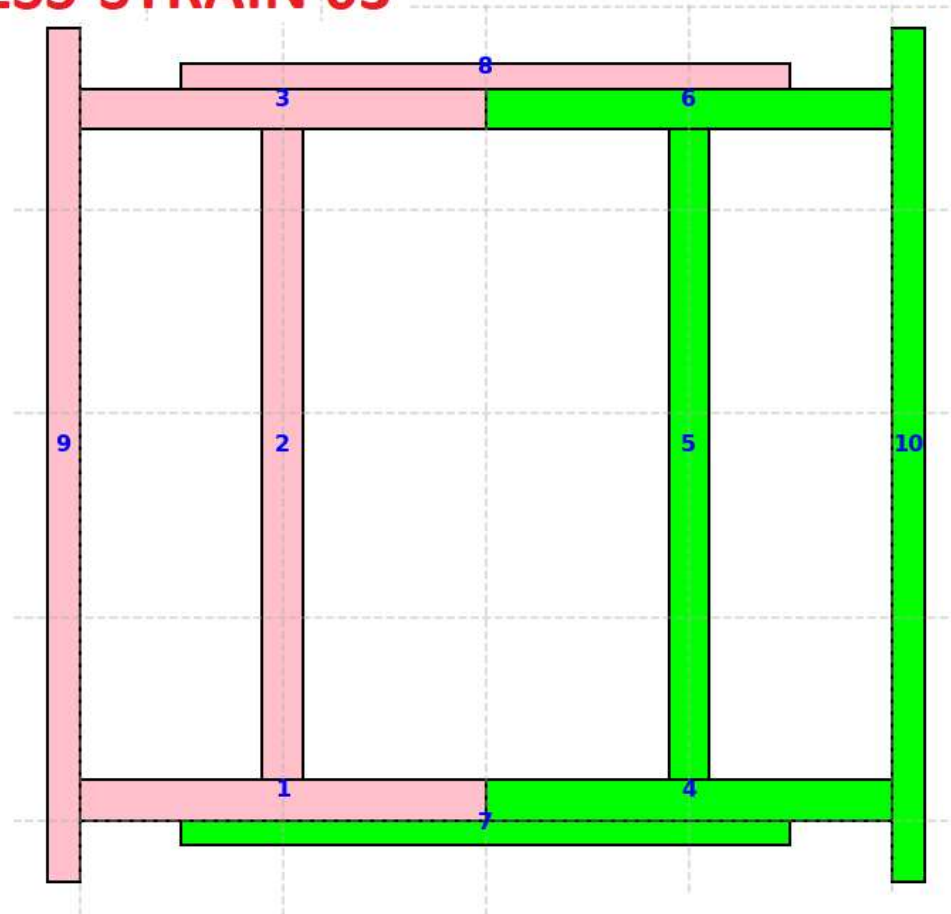
$$v(0,t) = v(L,t) = 0, \quad \left. \frac{\partial^2 v(x,t)}{\partial x^2} \right|_{x=0} = \left. \frac{\partial^2 v(x,t)}{\partial x^2} \right|_{x=L} = 0.$$

CROSS SHAPED STEEL SECTION WITH DIFFERENET STRESS-STRAIN



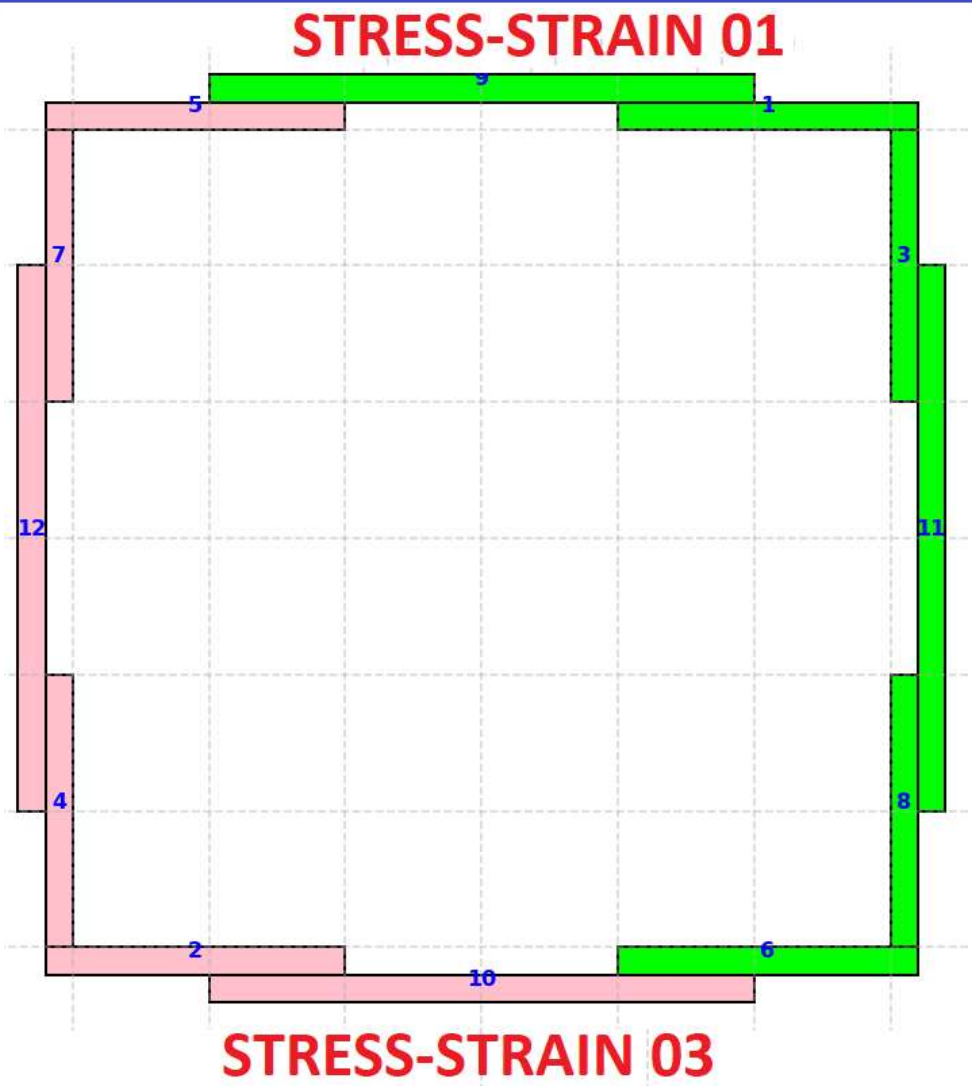
CROSS SHAPED STEEL SECTION WITH DIFFERENT STRESS-STRAIN

STRESS-STRAIN 03

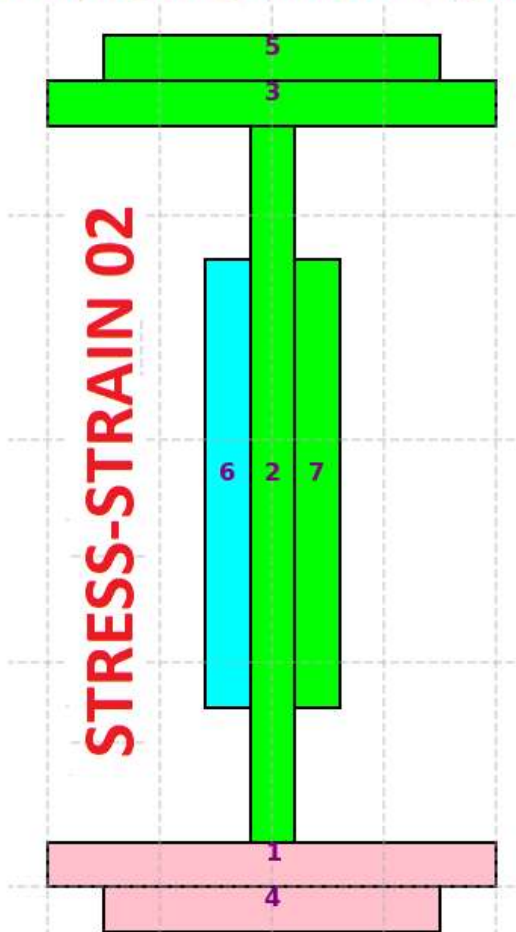


STRESS-STRAIN 01

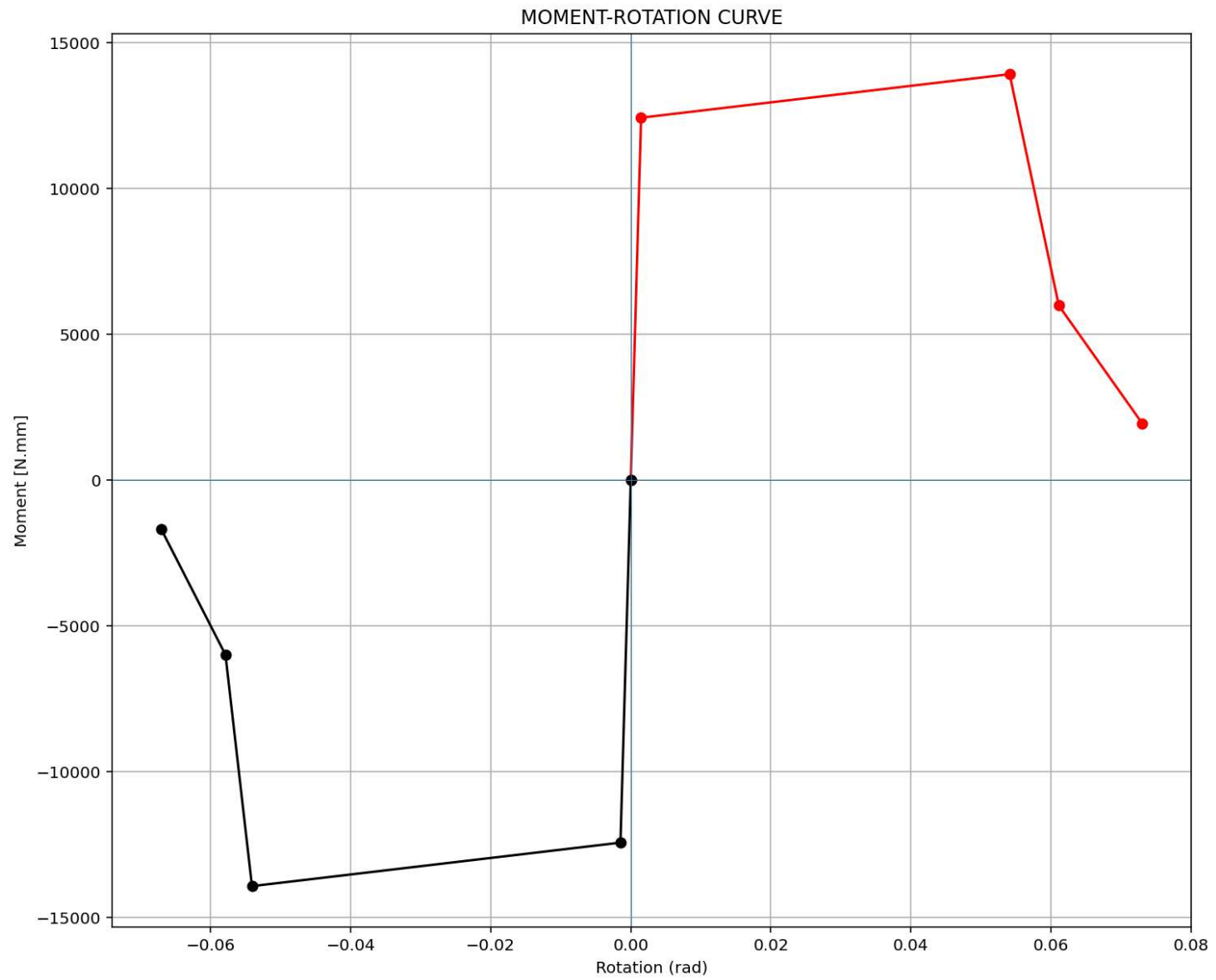
CROSS SHAPED STEEL SECTION WITH DIFFERENT STRESS-STRAIN



STRESS-STRAIN 01



STRESS-STRAIN 03



Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...S_AXIAL_AND_MOMENT\AFTER_FIRE_01_STEEL_SECTION.py

P(t)_AM_STEEL_CONC...-BUCKLING_8_ANA.py AFTER_FIRE_01_STEEL_SECTION.py

```
1 def AFTER_FIRE_01_STEEL_SECTION DIFF WIDTH HEIGHT FUN EXTRA_TWO(secTag, STEEL
2 """
3 Create a steel with different widths and heights, fiber section (OpenSee
4 if PLOT=True, draw a simple 2-D sketch of the section.
5
6 Parameters
7 -----
8 secTag      : int - section identifier
9 STEEL_TYPE  : str - 'ELASTIC' or 'INELASTIC'
10 fc          : float - unconfined concrete compressive strength (MPa,
11 Kfc         : float - ratio of confined to unconfined concrete stren
12 CONCRETE_DENSITY: float - pc (kg/m³) - will be converted to N·s²/mm³
13 PLOT        : bool - draw the section if True
14
15 Thermal Effects of Material Stres-Strain relation
16 THIS PYTHON SCRIPT WRITTEN BY SALAR DELAVAR GHASHGHAIEI (QASHQAI)
17 """
18 import numpy as np
19 import openseespy.opensees as ops
20 import matplotlib.pyplot as plt
21 import matplotlib.pyplot as plt, matplotlib.patches as patches
22
23 # Define materials for nonlinear elements
24 # Define parameters (units: mm, N)
25 # -----
26 # CONCRETE          tag  f'c      ec0  f'cu      ecu
27 # Cover concrete (unconfined)
28 fcU = -fc          # [N/mm²] Concrete Compressive Strength
29 Ec = 4700 * np.sqrt(-fcU) # [N/mm²] Concrete Elastic Modulus
30 ec0U = 2*fcU/Ec     # [mm/mm] Concrete Compressive Strain
31 fcUU = 0.2*fcU      # [N/mm²] Concrete Compressive Ultimate Streng
32 ecuU = 5*ec0U       # [mm/mm] Concrete Compressive Ultimate Strain
33 LambdaU = 0.1;      # ratio between unloading slope
34 # Core concrete (confined)
```

After Fire Effects Steel Section - Section Tag: 1

Height (mm)

Width (mm)

IPython Console Files Help Variable Explorer Debugger Plots History

Inline Conda: anaconda3 (Python 3.12.7) ✓ LSP: Python Line 1, Col 1 UTF-8 CRLF RW Mem 50%

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\Dell\Desktop\OPENSEES_FILES\+HEIGHT_ANALY...)_AM_STEEL_CONCRETE_COLUMN_POST-BUCKLING_8_ANA.py

P(t)_AM_STEEL_CONC...-BUCKLING_8_ANA.py X

```
#####
1  # >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
2  # COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF MULTI-MODE POST-BUCKLING PHENOMENA IN A POST-
3  # COLUMN WITH SEMI-RIGID CONNECTIONS: AN OPENSEES FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGRA-
4  # STATIC TIME-HISTORY, AND DYNAMIC TIME-HISTORY ANALYSES
5  #
6  #-----
7  # NONLINEAR ANALYSIS OF POST-FIRE STEEL COLUMN STRUCTURES SUBJECTED TO PUSHOVER, IMPACT, AND
8  #
9  # THERMAL EFFECTS ON STEEL MATERIAL STRESS-STRAIN REALTION
10 #-----
11 # EQUIVALENT VISCOUS DAMPING RATIO:  $\xi_{eq} = 100 * E_d / (4 * \pi * E_s)$ 
12 #
13 # ENERGY DISSIPATION CAPACITY INDEX =  $100 * E_d(\text{earthquake}) / E_d(\text{cyclic displacemen}$ 
14 #-----
15 #  $P(t) = P_0 \sin(\omega t)$ 
16 #  $P(t) = P_0 \exp(-0.05\omega t) \sin(\omega t)$ 
17 #-----
18 # ASSESSMENT OF DUCTILITY DAMAGE INDICES FOR STRUCTURAL ELEMENTS AND SYSTEMS AND EVAL
19 # OF ENERGY DISSIPATION CAPACITY INDEX
20 #-----
21 # THIS PYTHON SCRIPT IS WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
22 # EMAIL: salar.d.ghashghaei@gmail.com
23 #####
24 """
25 Assume we have a structure where, after a fire, certain regions of
26 the structural members have lost their initial material properties
27 (yield strength and modulus of elasticity), and the entire column
28 member has undergone lateral buckling. Now, we assess the strength
29 of this structure by performing eight different analyses.
30
31 Nonlinear Seismic Performance Assessment of a column:
32 An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Static, Cyclic, and Ear
33
```

After Fire Effects Steel Section - Section Tag: 1

Height (mm)

Width (mm)

IPython Console Plots Variable Explorer Help Debugger History Files

Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 1919, Col 2 UTF-8 CRLF RW Mem 51%

STATIC ANALYSIS

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\Dell\Desktop\OPENSEES_FILES\HEIGHT_ANALY..._AM_STEEL_CONCRETE_COLUMN_POST-BUCKLING_8_ANA.py

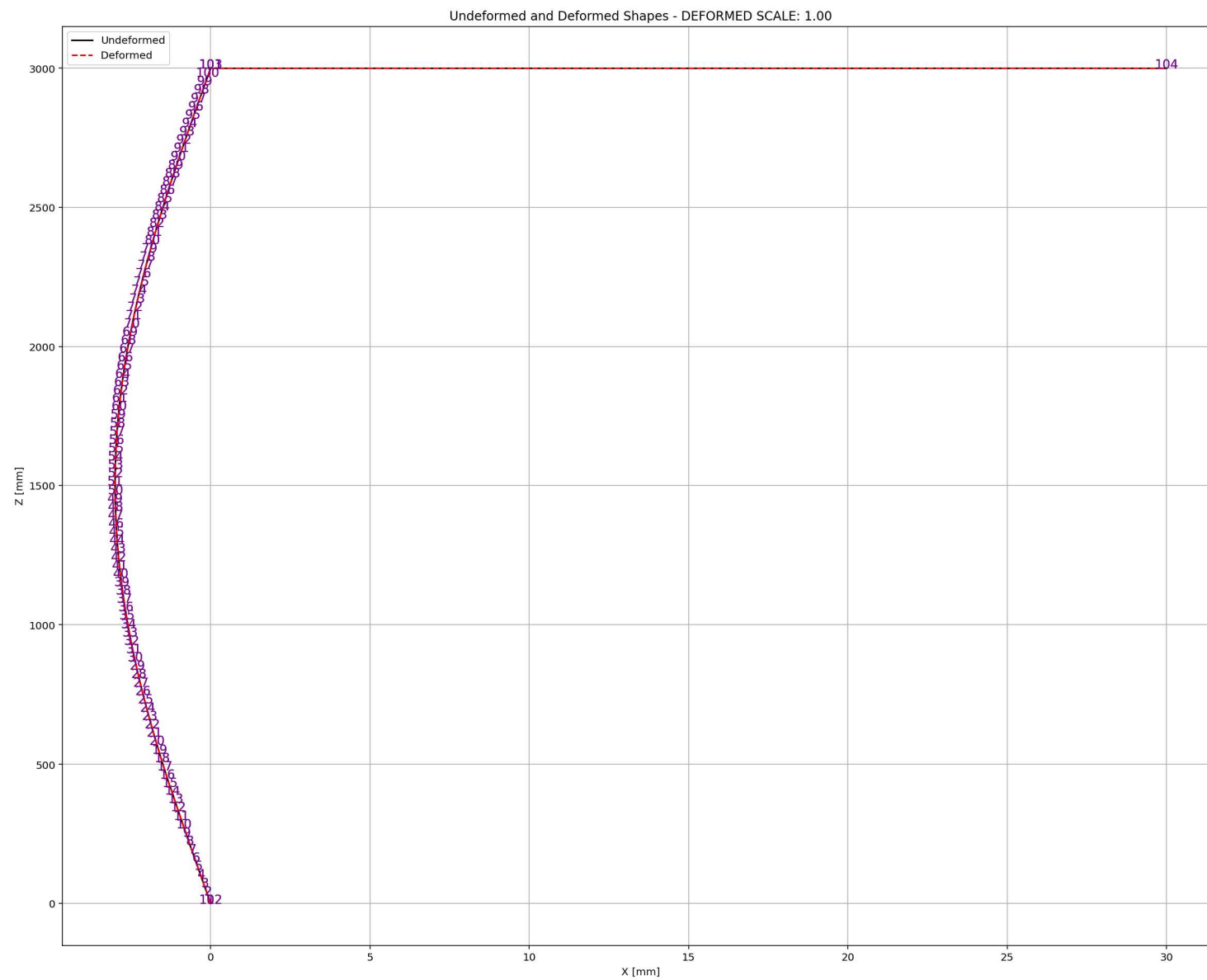
P(t)_AM_STEEL_CONC...-BUCKLING_8_ANA.py X AFTER_FIRE_01_STEEL_SECTION.py X

```
1187 # STATIC ANALYSIS
1188 #ELE_TYPE = 'elasticBeamColumn' # MATERIAL LINEARITY
1189 ELE_TYPE = 'nonlinearBeamColumn' # MATERIAL AND GEOMETRIC NONLINEARITIE
1190 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1191 ANAL_TYPE = 'STATIC'
1192
1193 DATA = POST_BUCKLING_COLUMN(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MA
1194 (reactionX, reactionY, reactionZ, disp_mid,
1195 ele_axialforce, ele_shearforce, ele_momentforce,
1196 node_displacementsX, node_displacementsY,
1197 dispX, dispY) = DATA
1198
1199 S01.PLOT_2D_FRAME(deformed_scale=1.0)
1200
1201 #%%-----
1202 # PUSHOVER ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1203 #ELE_TYPE = 'elasticBeamColumn' # MATERIAL LINEARITY
1204 ELE_TYPE = 'nonlinearBeamColumn' # MATERIAL AND GEOMETRIC NONLINEARITIE
1205 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1206 ANAL_TYPE = 'PUSHOVER'
1207
1208 DATA = POST_BUCKLING_COLUMN(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MA
1209 (reactionX_PUSH, reactionY_PUSH, reactionZ_PUSH, disp_mid_PUSH,
1210 ele_axialforce_PUSH, ele_shearforce_PUSH, ele_momentforce_PUSH,
1211 node_displacementsX_PUSH, node_displacementsY_PUSH,
1212 dispX_PUSH, dispY_PUSH,
1213 PERIOD_MIN_PUSH, PERIOD_MAX_PUSH,
1214 STRESSb_PUSH, STRAINb_PUSH, STRESSt_PUSH, STRAINT_PUSH, CURVATURE_PUSH, NA
1215
1216 XDATA = disp_mid_PUSH
1217 YDATA = reactionY_PUSH
1218 XLABEL = 'Displacement [mm]'
1219 YLABEL = 'Base Reaction [N]'
1220 TITLE = 'Base Reaction and Displacement of Structure During Pushover Analys
```

Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00

IPython Console Files Help Variable Explorer Debugger Plots History

Inline Conda: anaconda3 (Python 3.12.7) ✓ LSP: Python Line 73, Col 19 UTF-8 CRLF RW Mem 49%



PUSHOVER ANALYSIS

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\ DELL\Desktop\OPENSEES_FILES\+EIGHT_ANALY..._AM_STEEL_CONCRETE_COLUMN_POST-BUCKLING_8_ANA.py

P(t)_AM_STEEL_CONC...-BUCKLING_8_ANA.py X

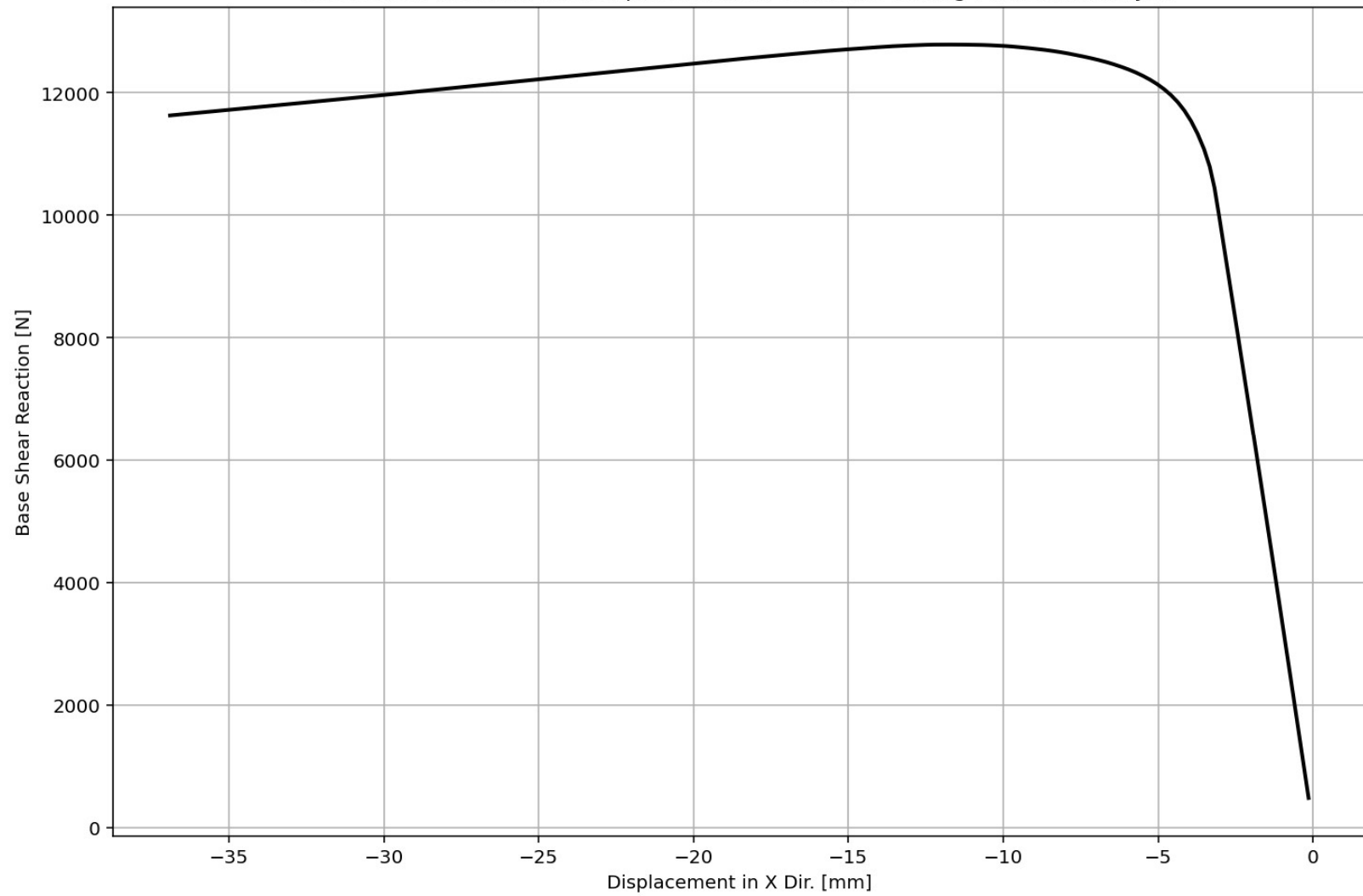
```
1224 # PUSHOVER ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1225 #ELE_TYPE = 'elasticBeamColumn' # MATERIAL LINEARITY
1226 ELE_TYPE = 'nonlinearBeamColumn' # MATERIAL AND GEOMETRIC NONLINEARITIES
1227 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1228 ANAL_TYPE = 'PUSHOVER'
1229
1230 DATA = POST_BUCKLING_COLUMN(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS, MODE_NUM, SEMI
1231 (reactionX_PUSH, reactionY_PUSH, reactionZ_PUSH, disp_mid_PUSH,
1232 ele_axialforce_PUSH, ele_shearforce_PUSH, ele_momentforce_PUSH,
1233 node_displacementsX_PUSH, node_displacementsY_PUSH,
1234 dispX_PUSH, dispY_PUSH,
1235 PERIOD_MIN_PUSH, PERIOD_MAX_PUSH,
1236 STRESSb_PUSH, STRAINb_PUSH, STRESSt_PUSH, STRAINt_PUSH, CURVATURE_PUSH, NA_PUSH, CD_PUSH) =
1237
1238
1239 XDATA = disp_mid_PUSH
1240 YDATA = reactionY_PUSH
1241 XLABEL = 'Displacement [mm]'
1242 YLABEL = 'Base Reaction [N]'
1243 TITLE = 'Base Reaction and Displacement of Structure During Pushover Analysis'
1244 COLOR = 'black'
1245 SEMILOGY = False
1246 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1247
1248 DATA = S07.BILINEAR_CURVE(np.abs(disp_mid_PUSH), np.abs(reactionX_PUSH), SLOPE_NODE=10)
1249 (X_PUSH, Y_PUSH, Elastic_ST, Plastic_ST, Tangent_ST, Ductility_Rito, Over_Strength_Factor) =
1250
1251 # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1252 plt.figure(0, figsize=(12, 8))
1253 plt.plot(disp_mid_PUSH, PERIOD_MIN_PUSH, linewidth=3)
1254 plt.plot(disp_mid_PUSH, PERIOD_MAX_PUSH, linewidth=3)
1255 plt.title('Period of Structure During Pushover Analysis')
1256 plt.ylabel('Structural Period [s]')
1257 plt.xlabel('Displacement [mm]')
```

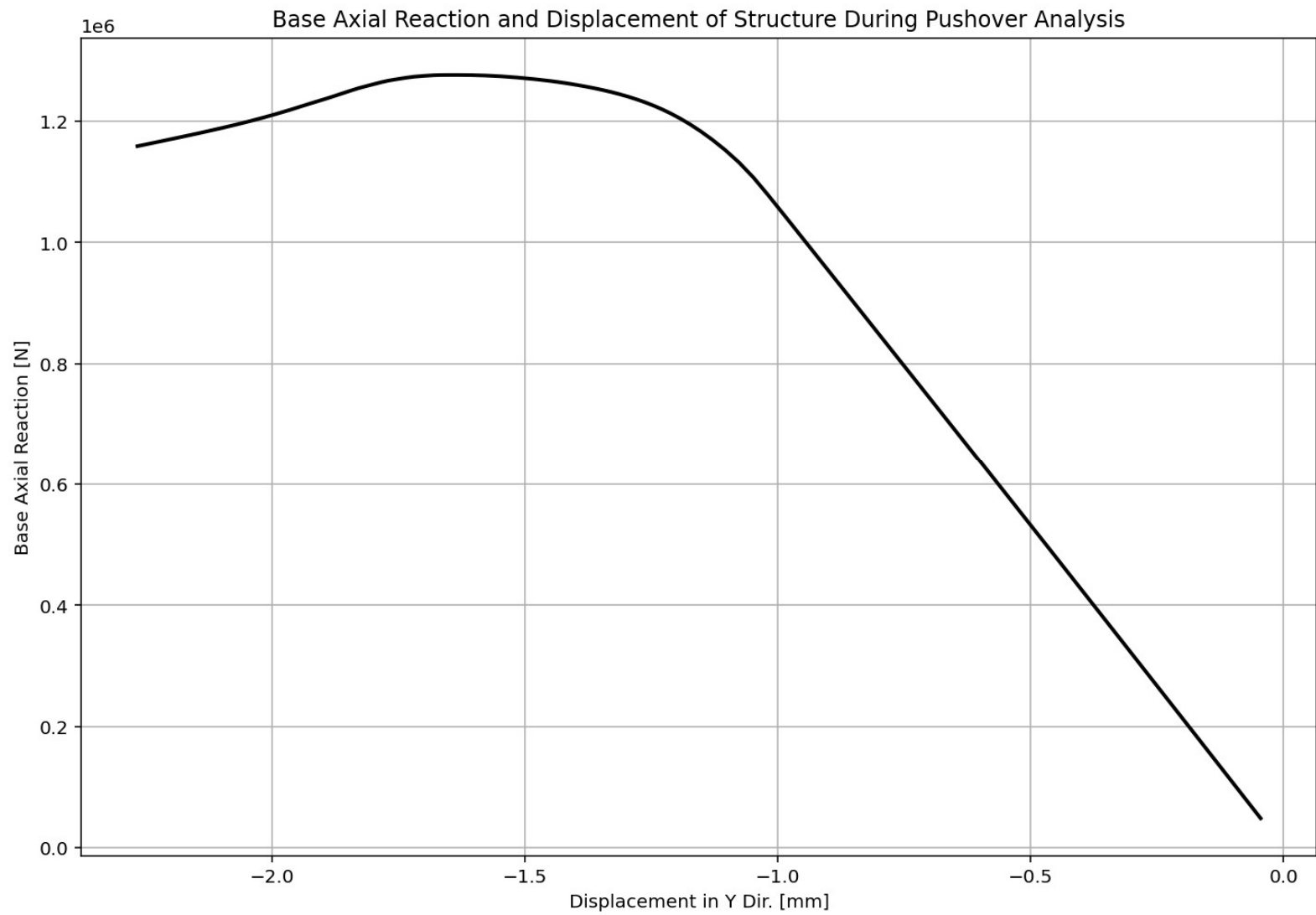
Base Reaction and Displacement of Structure During Pushover Analysis

IPython Console Plots Variable Explorer Help Debugger History Files

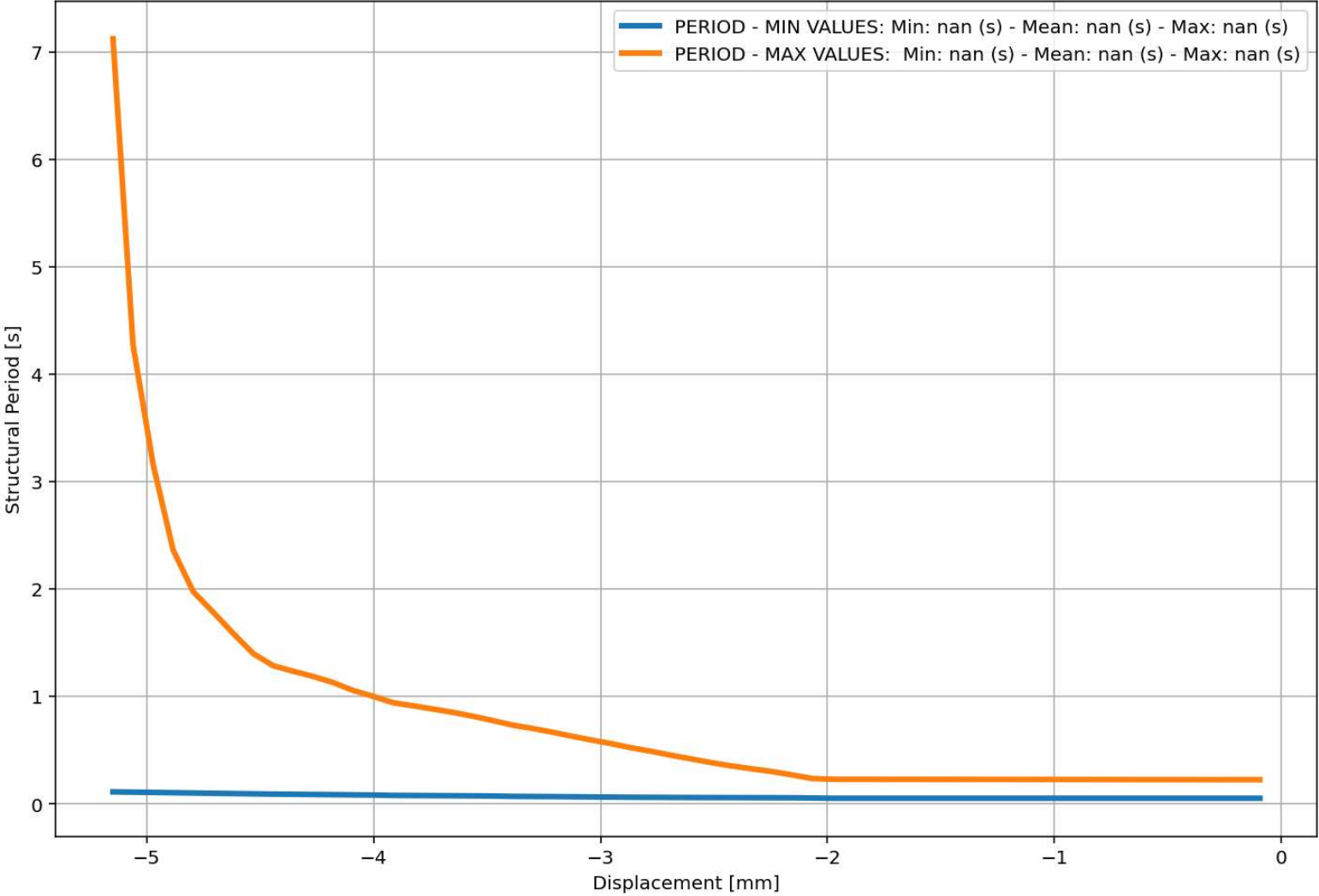
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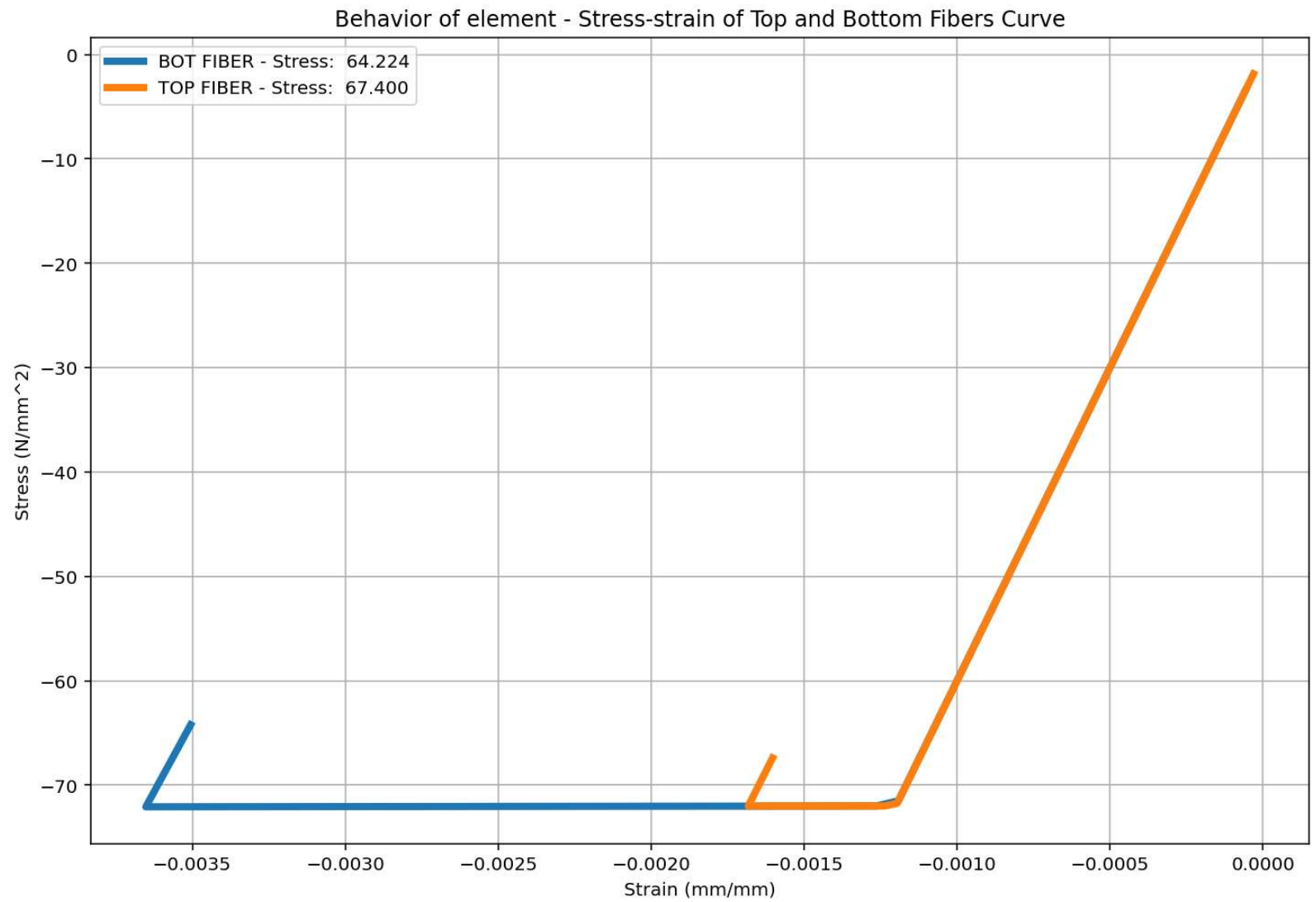
Base Shear Reaction and Displacement of Structure During Pushover Analysis

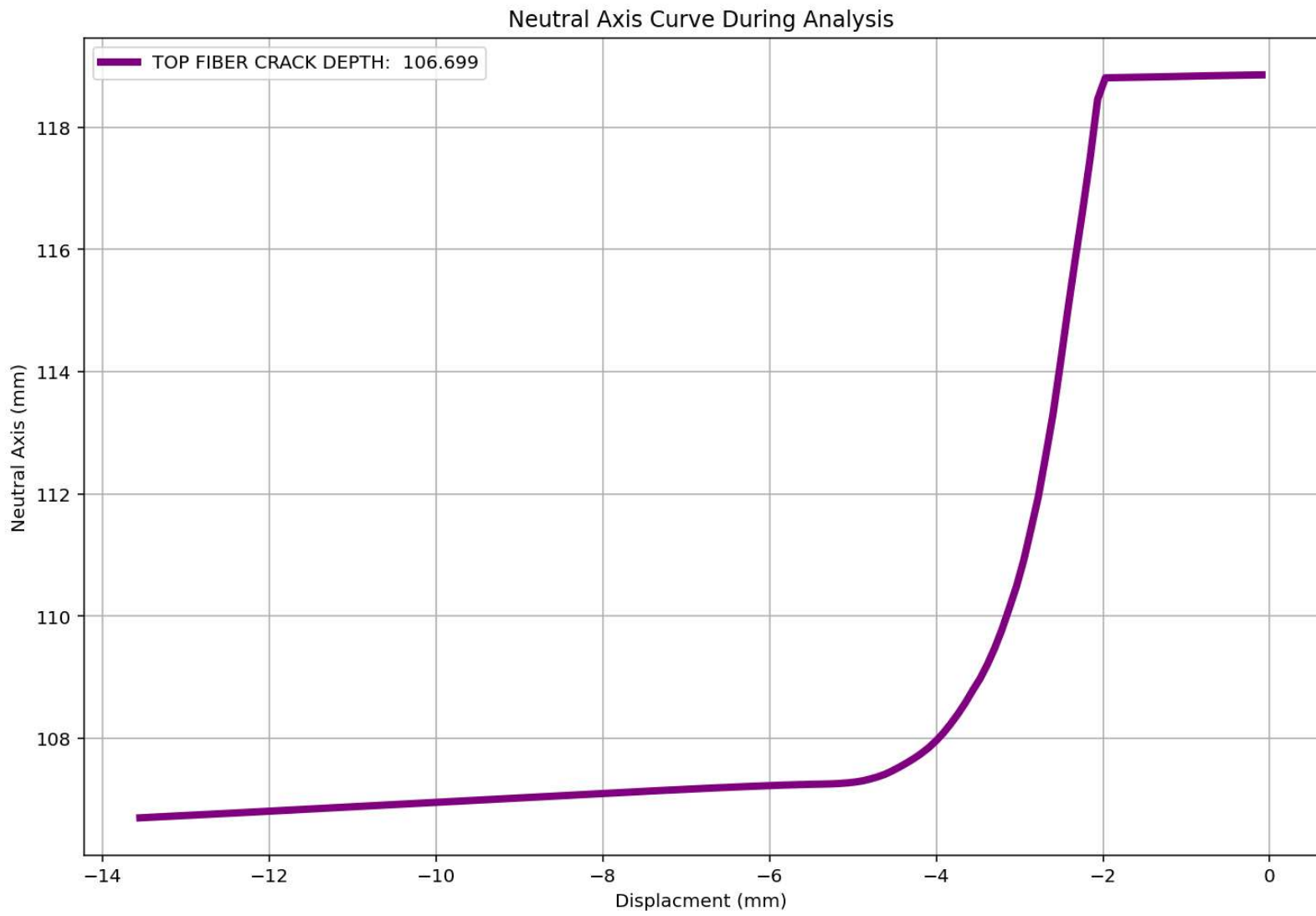


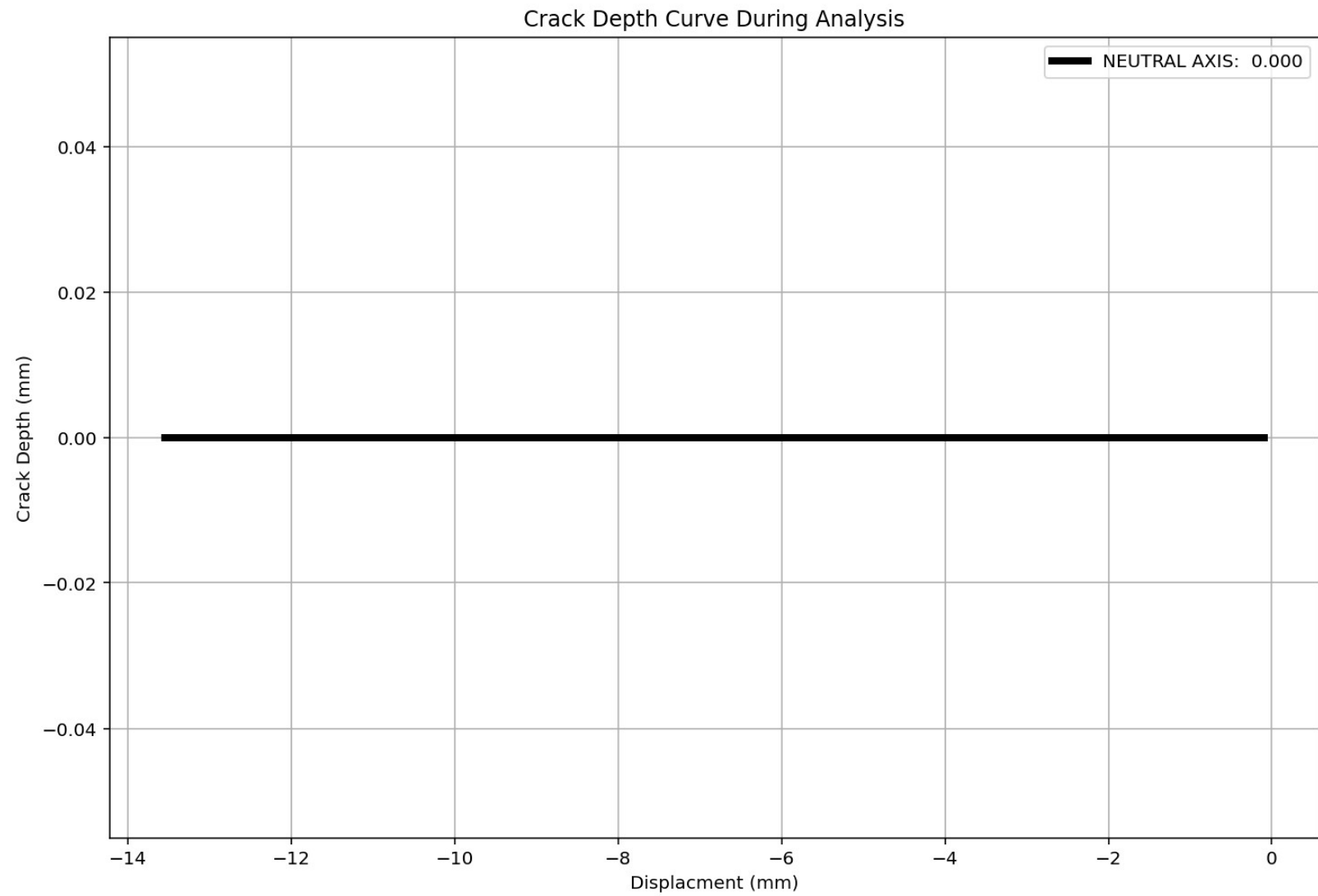


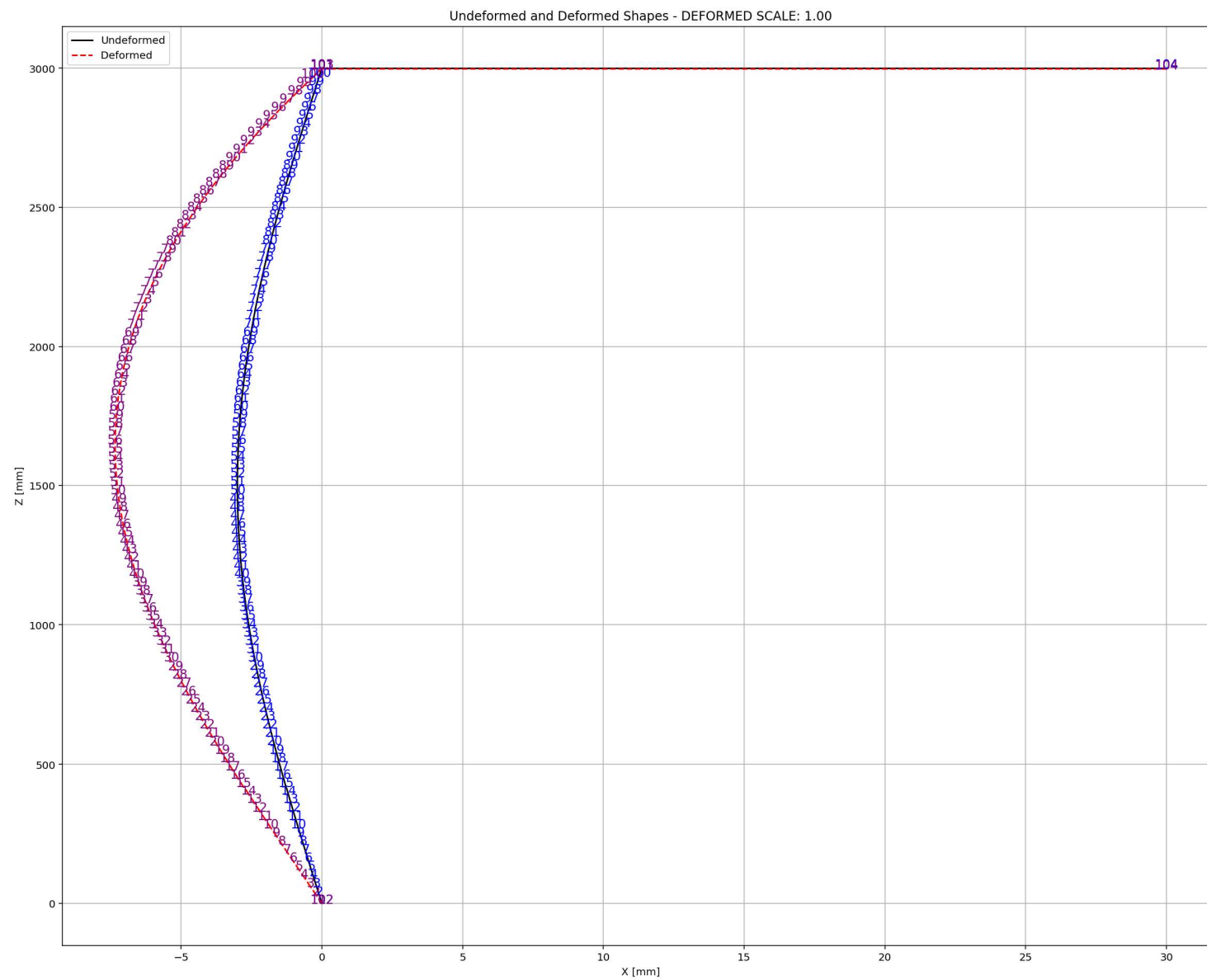
Period of Structure During Pushover Analysis











CYCLIC DISPLACEMENT ANALYSIS

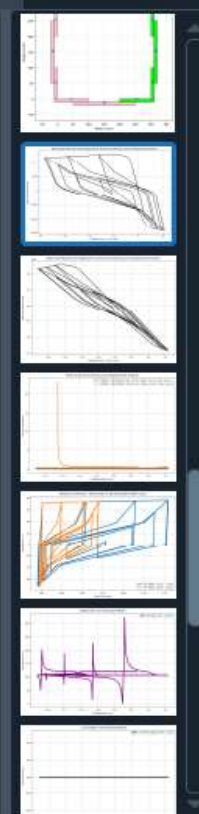
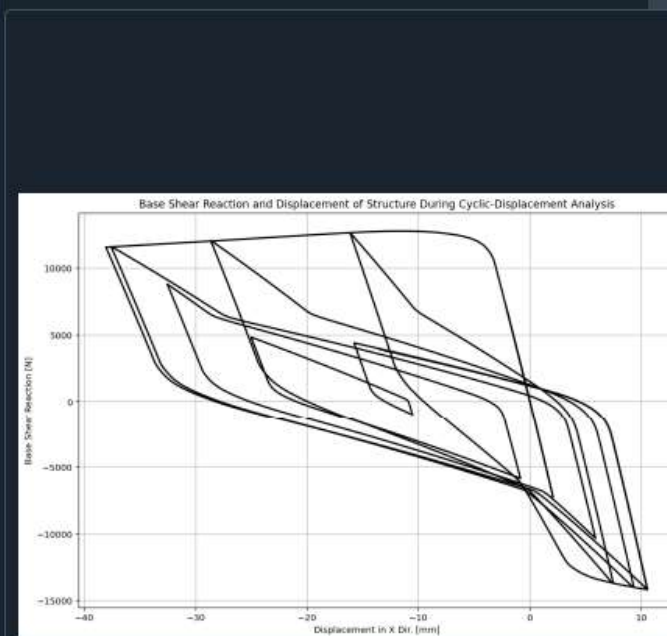
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...)_AM_STEEL_CONCRETE_COLUMN_POST-BUCKLING_8_ANA.py

P(t)_AM_STEEL_CONC...-BUCKLING_8_ANA.py X

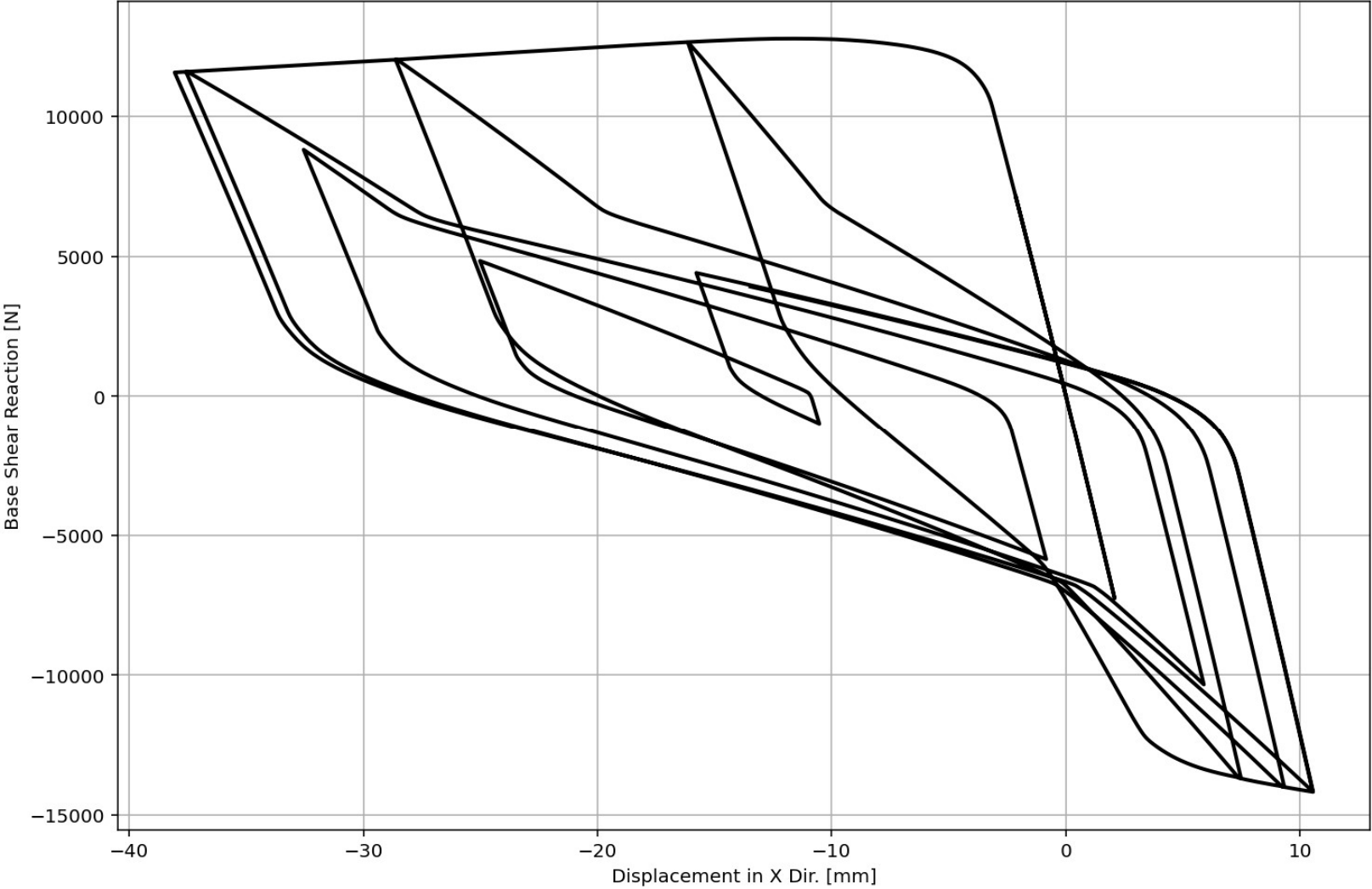
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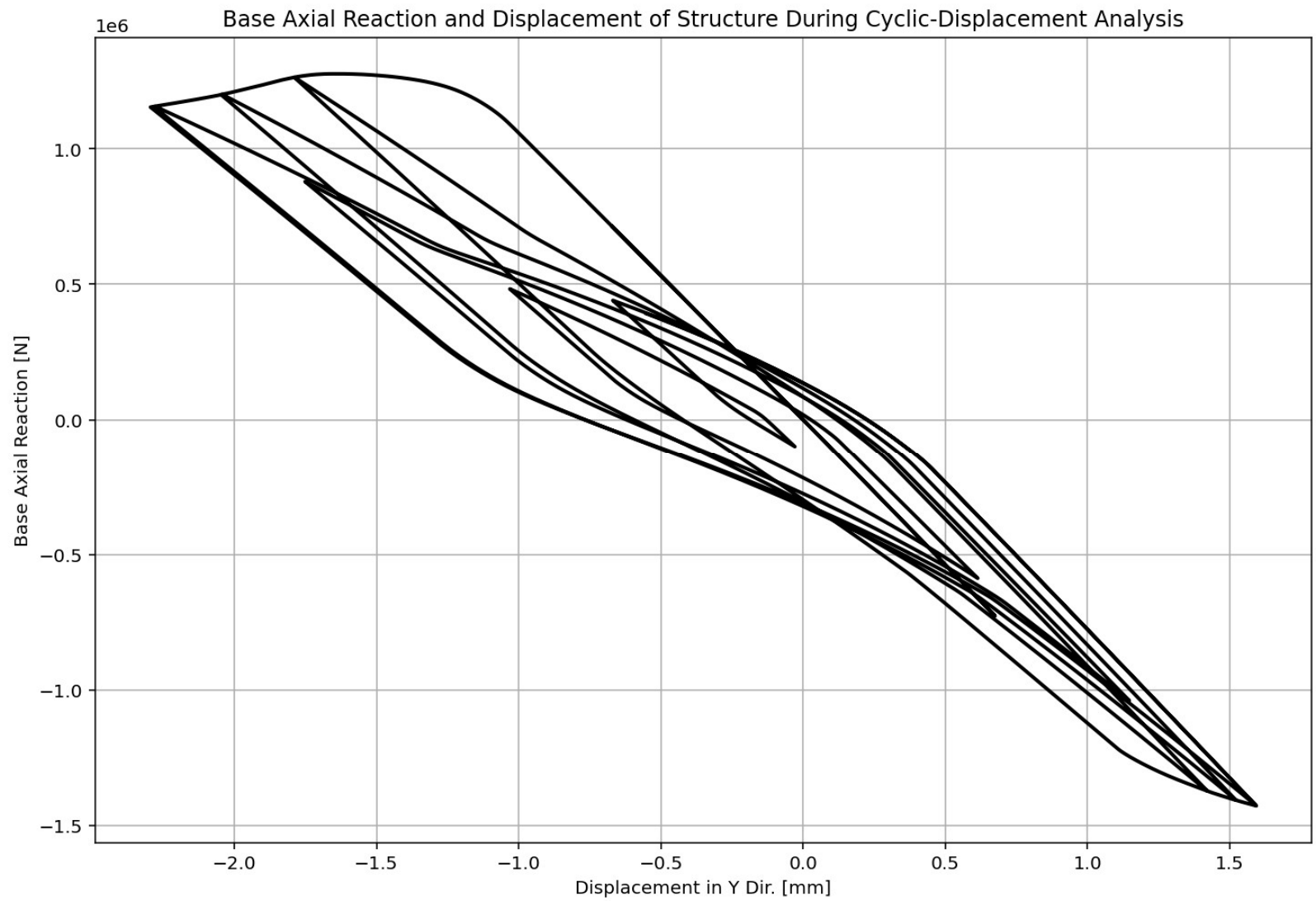
1 #####
2 # >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
3 # COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF MULTI-MODE POST-BUCKLING PHENOMENA IN A POST-
4 # COLUMN WITH SEMI-RIGID CONNECTIONS: AN OPENSEES FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGRA
5 # STATIC TIME-HISTORY, AND DYNAMIC TIME-HISTORY ANALYSES
6 #
7 # NONLINEAR ANALYSIS OF POST-FIRE STEEL COLUMN STRUCTURES SUBJECTED TO PUSHOVER, IMPACT, AND
8 #
9 # THERMAL EFFECTS ON STEEL MATERIAL STRESS-STRAIN REALTION
10 #
11 # EQUIVALENT VISCOUS DAMPING RATIO:  $\xi_{eq} = 100 * E_d / (4 * \pi * E_s)$ 
12 #
13 # ENERGY DISSIPATION CAPACITY INDEX =  $100 * E_d(\text{earthquake}) / E_d(\text{cyclic displacement})$ 
14 #
15 #  $P(t) = P_0 \sin(\omega t)$ 
16 #  $P(t) = P_0 \exp(-0.05\omega t) \sin(\omega t)$ 
17 #
18 # ASSESSMENT OF DUCTILITY DAMAGE INDICES FOR STRUCTURAL ELEMENTS AND SYSTEMS AND EVAL
19 # OF ENERGY DISSIPATION CAPACITY INDEX
20 #
21 # THIS PYTHON SCRIPT IS WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
22 # EMAIL: salar.d.ghashghaei@gmail.com
23 #####
24 """
25 Assume we have a structure where, after a fire, certain regions of
26 the structural members have lost their initial material properties
27 (yield strength and modulus of elasticity), and the entire column
28 member has undergone lateral buckling. Now, we assess the strength
29 of this structure by performing eight different analyses.
30
31 Nonlinear Seismic Performance Assessment of a column:
32 An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Static, Cyclic, and Ear
33

```

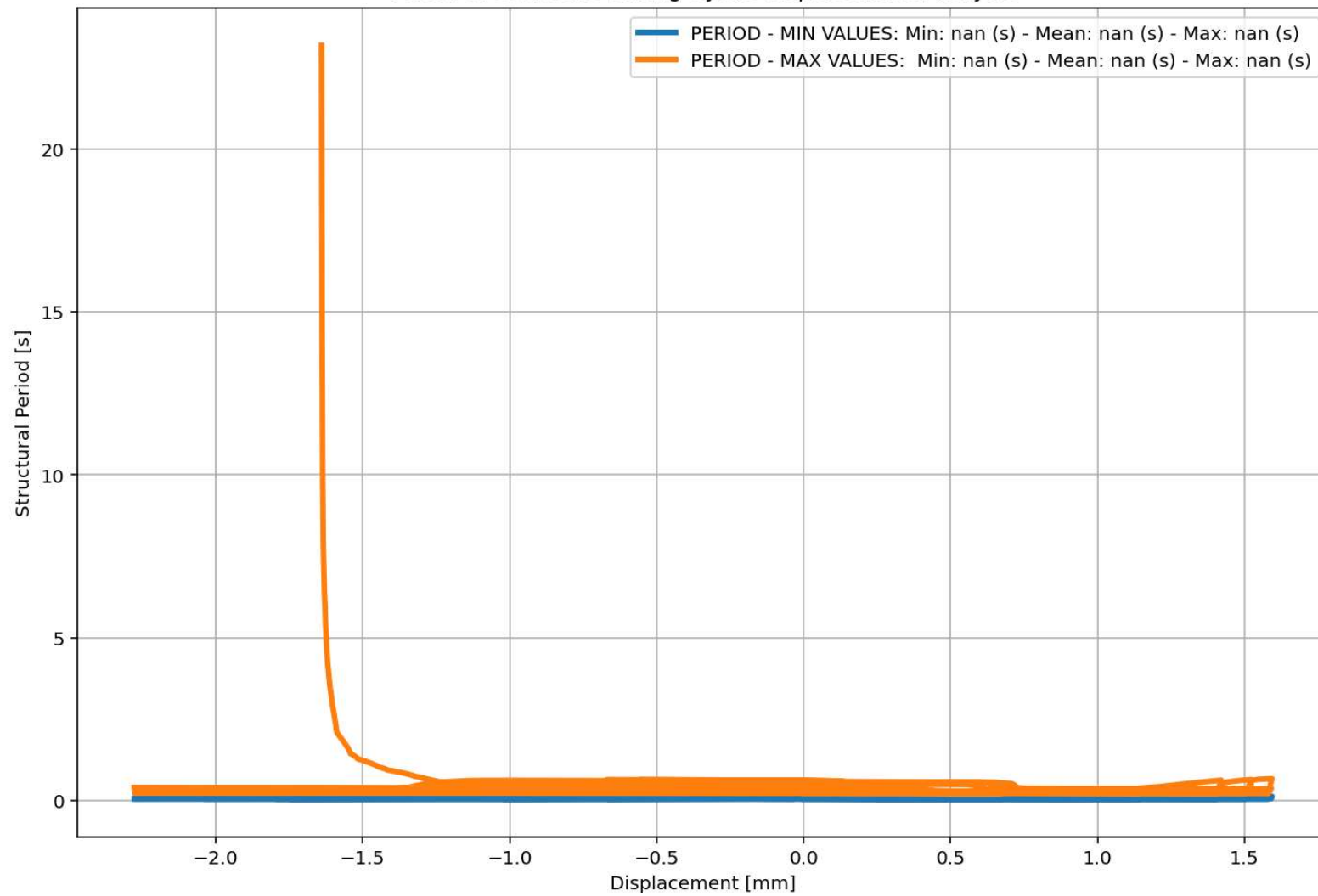


Base Shear Reaction and Displacement of Structure During Cyclic-Displacement Analysis

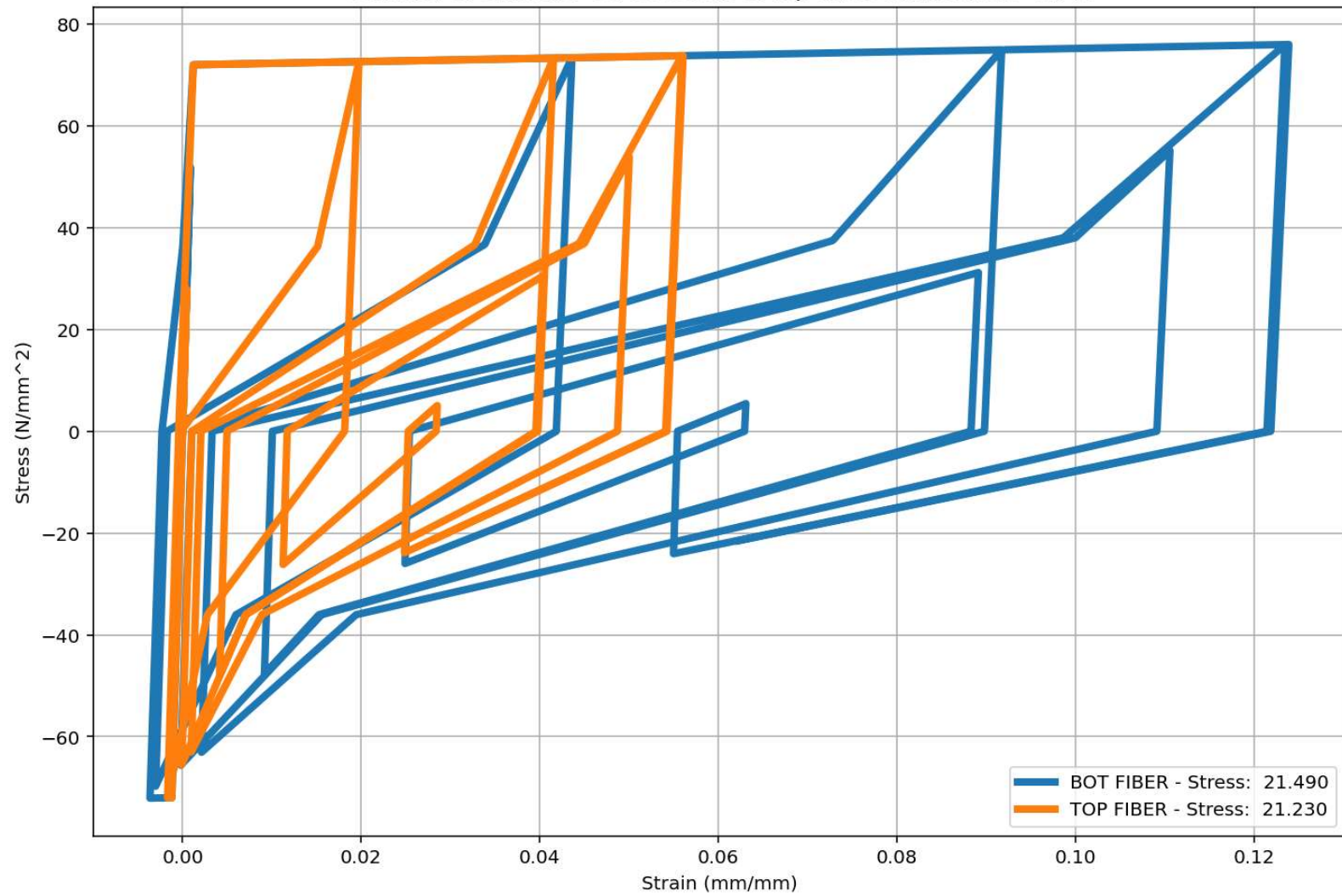


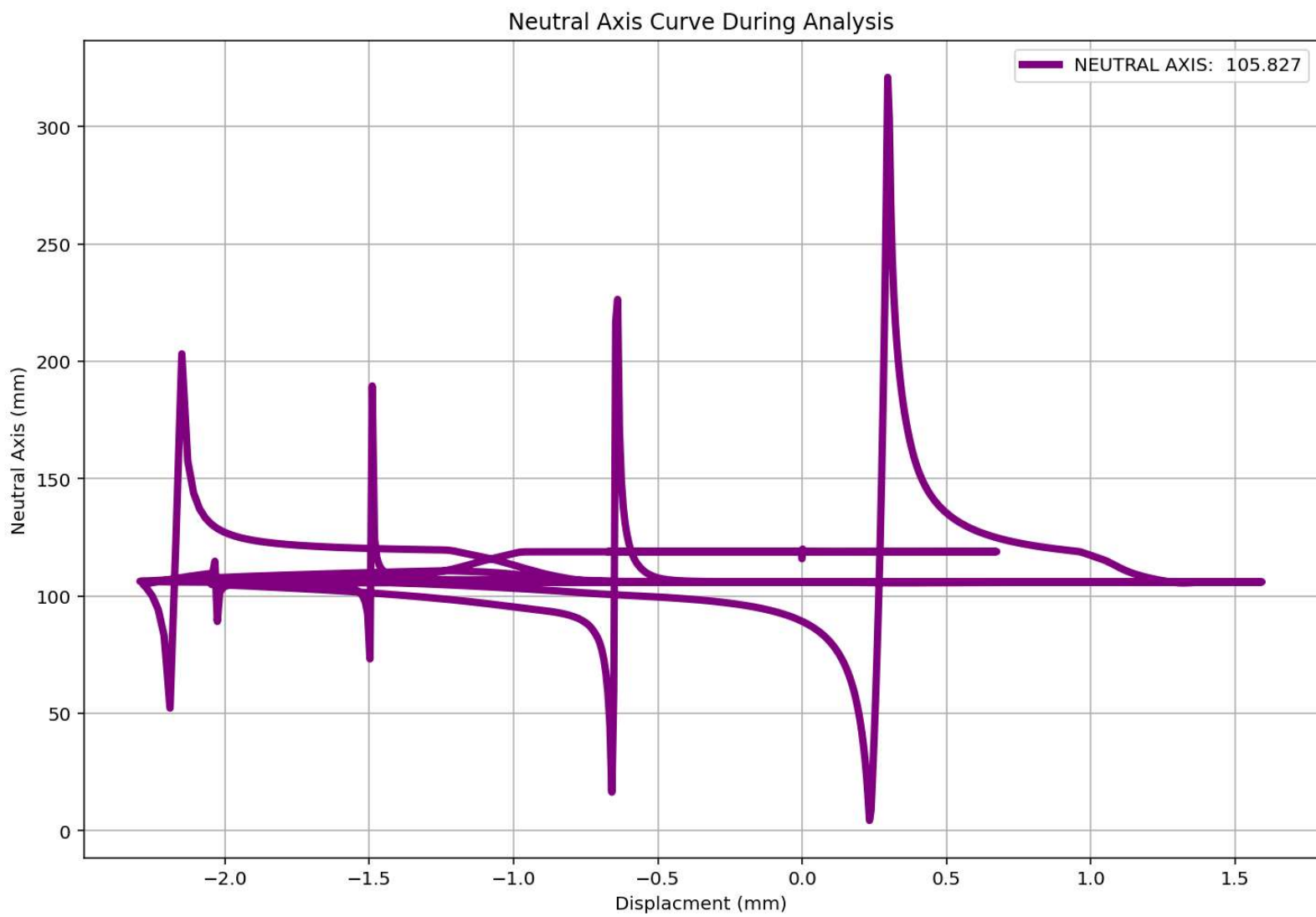


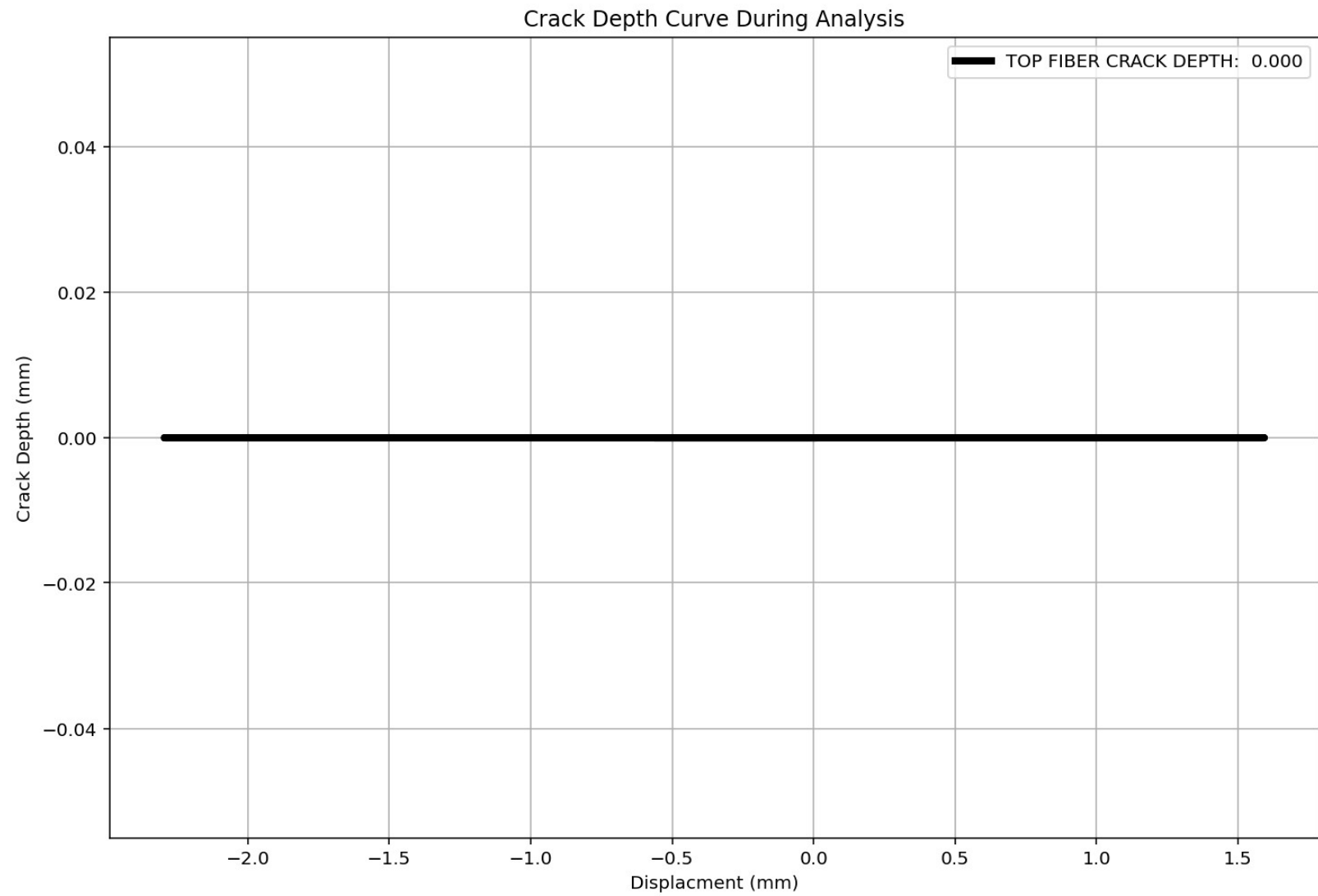
Period of Structure During Cyclic Displacement Analysis



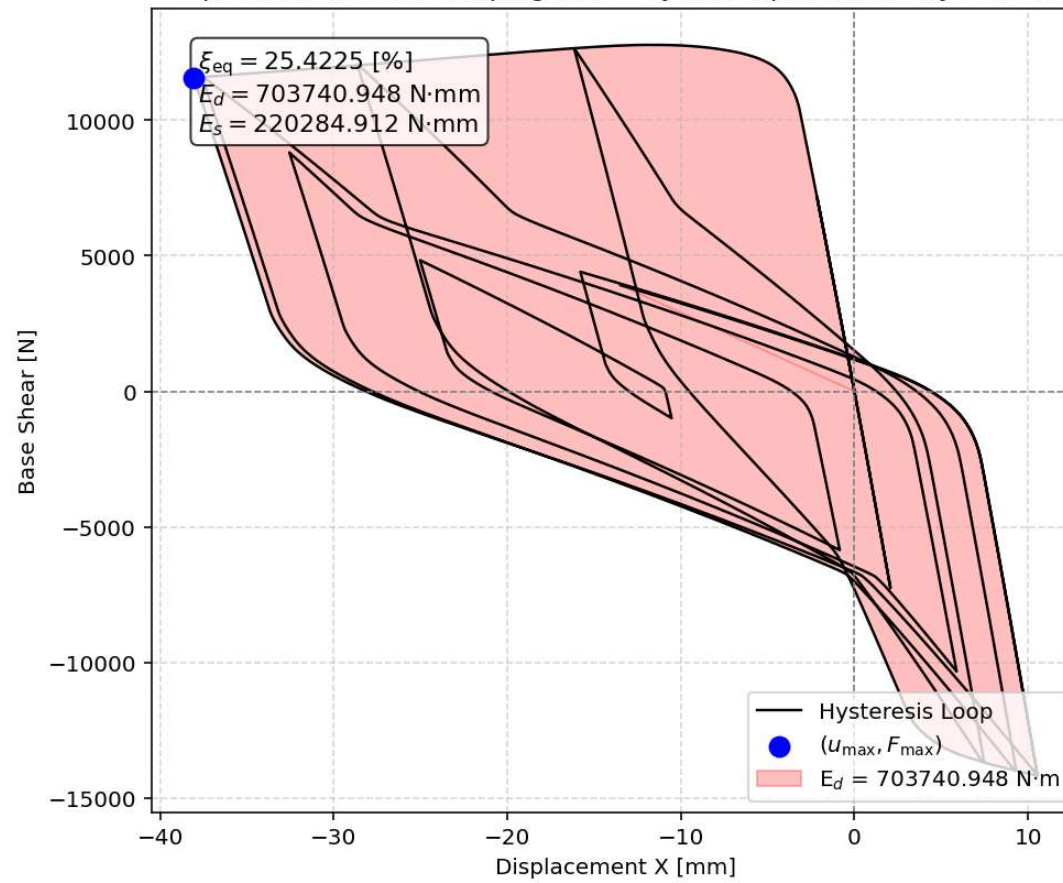
Behavior of element - Stress-strain of Top and Bottom Fibers Curve



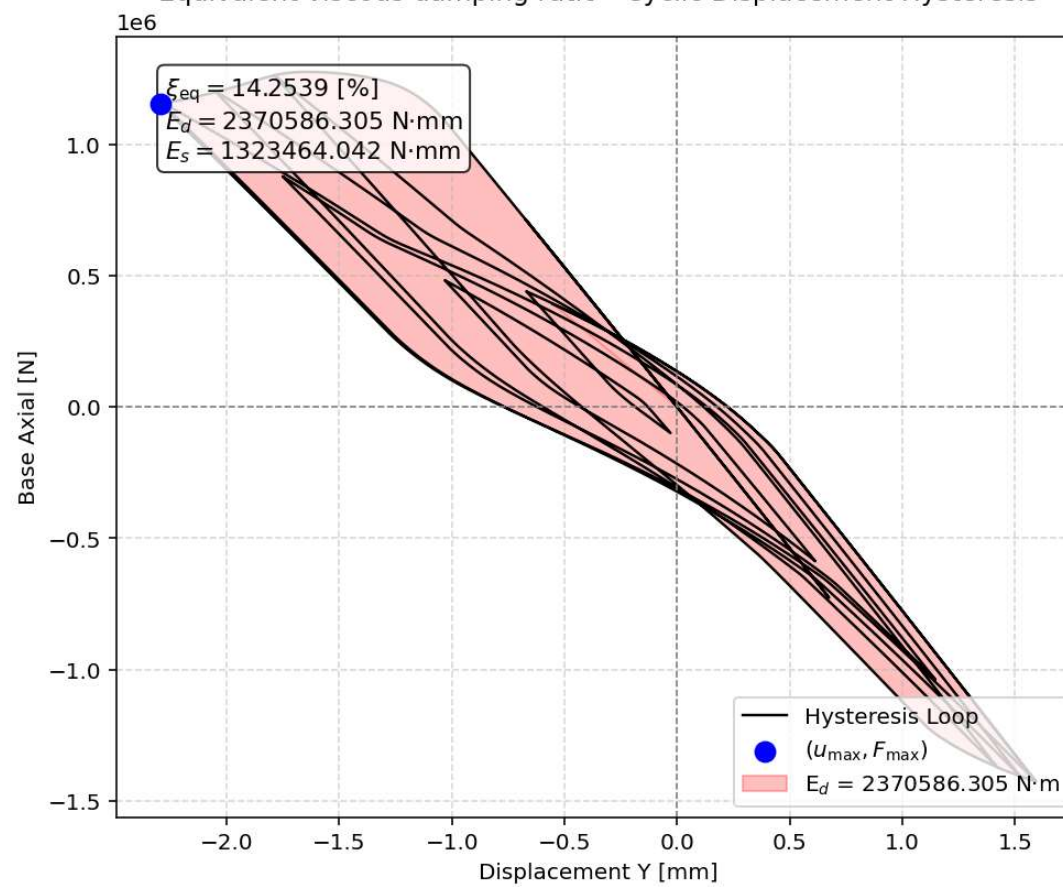


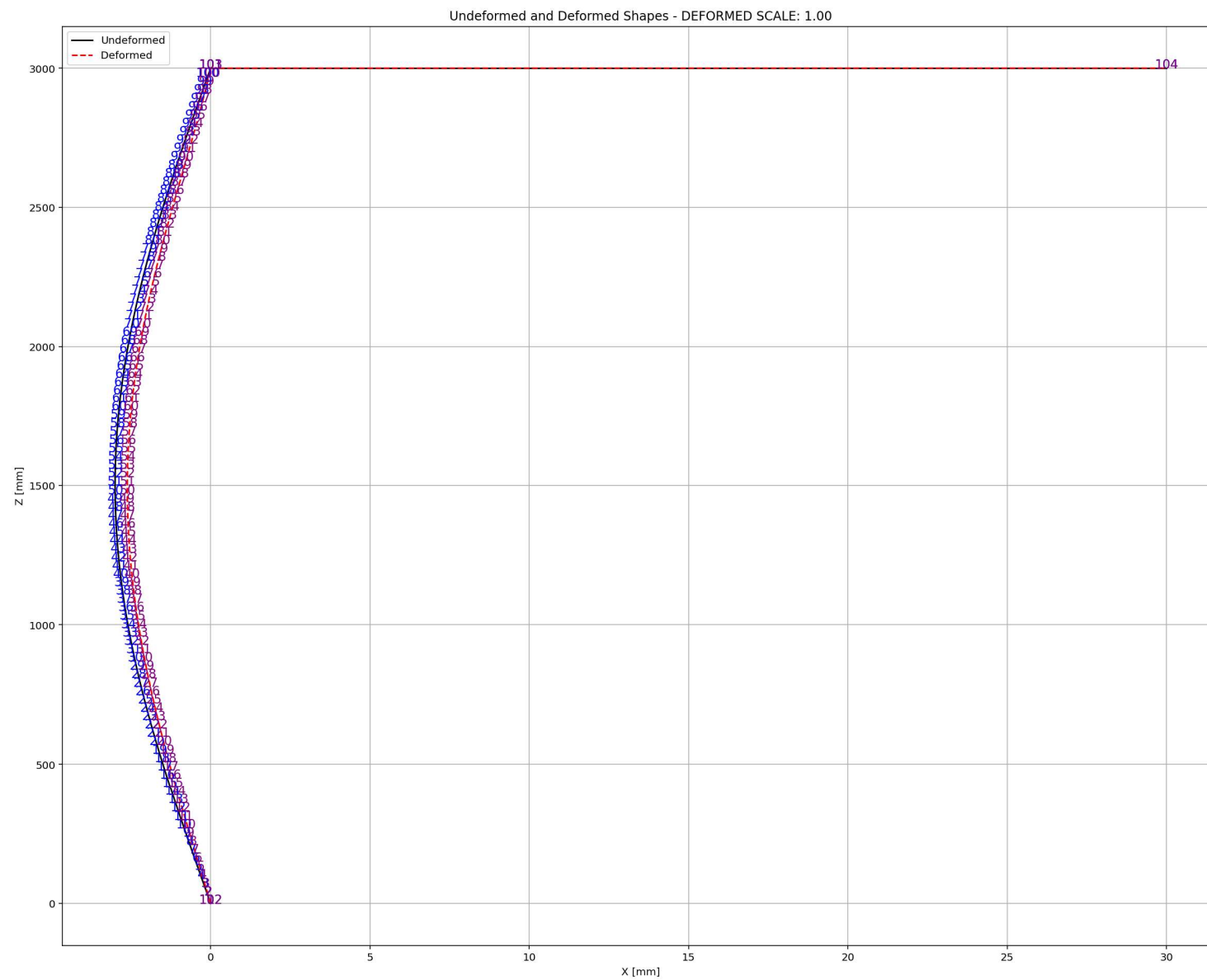


Equivalent viscous damping ratio - Cyclic Displacement Hysteresis



Equivalent viscous damping ratio - Cyclic Displacement Hysteresis





STATIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...)_AM_STEEL_CONCRETE_COLUMN_POST-BUCKLING_8_ANA.py

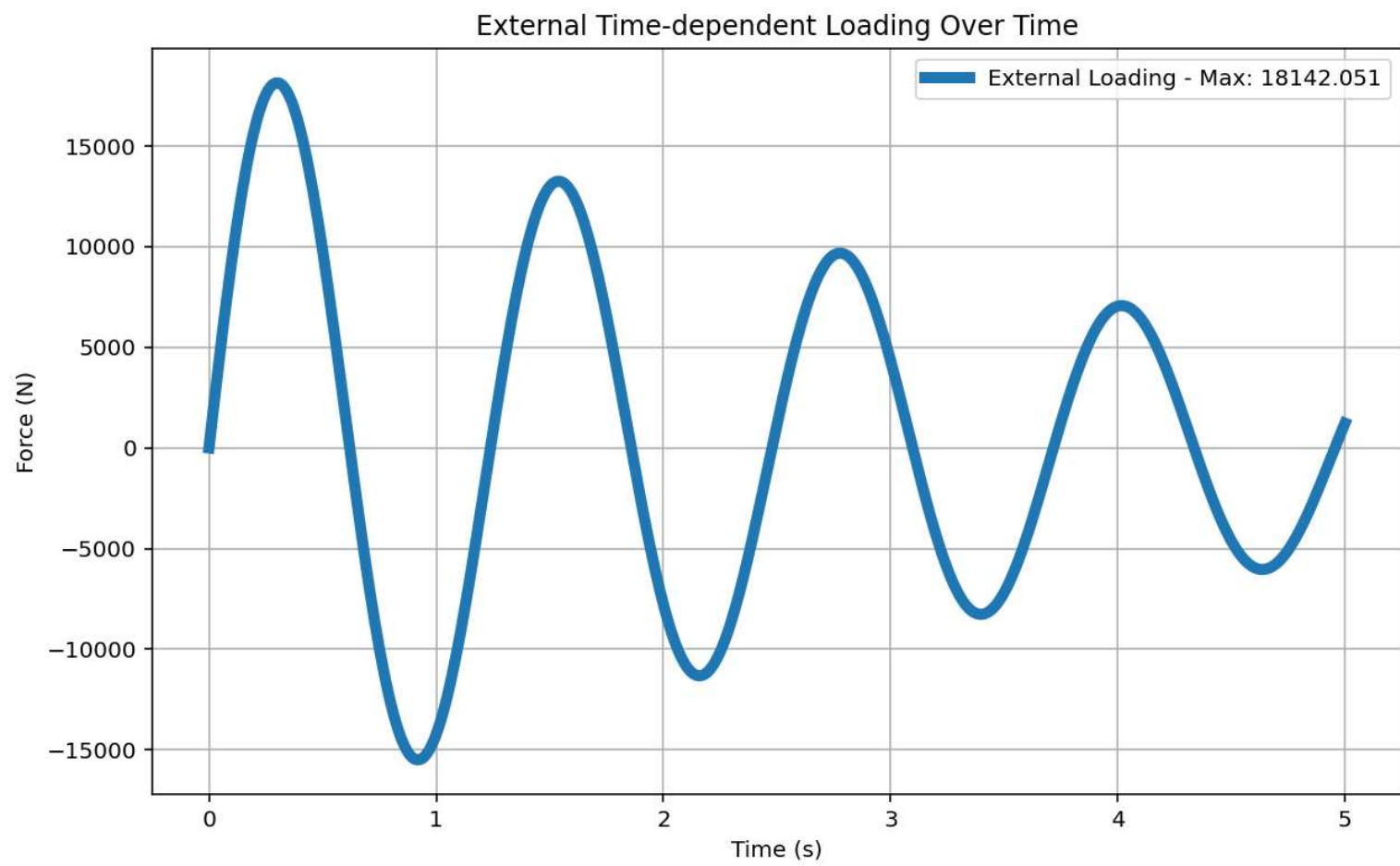
P(t)_AM_STEEL_CONC...-BUCKLING_8_ANA.py X AFTER_FIRE_01_STEEL_SECTION.py X

```
1346 # EXTERNAL TIME-DEPENDENT LOADING ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1347 #ELE_TYPE = 'elasticBeamColumn' # MATERIAL LINEARITY
1348 ELE_TYPE = 'nonlinearBeamColumn' # MATERIAL AND GEOMETRIC NONLINEARITIE
1349 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1350 ANAL_TYPE = 'STATIC_EXTERNAL_TIME-DEPENDENT_LOADING'
1351
1352 DATA = POST_BUCKLING_COLUMN(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MA
1353
1354 (reactionX_ETDLS, reactionY_ETDLS, reactionZ_ETDLS, disp_mid_ETDLS,
1355 ele_axialforce_ETDLS, ele_shearforce_ETDLS, ele_momentforce_ETDLS,
1356 node_displacementsX_ETDLS, node_displacementsY_ETDLS,
1357 dispX_ETDLS, dispY_ETDLS,
1358 PERIOD_MIN_ETDLS, PERIOD_MAX_ETDLS,
1359 STRESSb_ETDLS, STRAINb_ETDLS, STRESSt_ETDLS, STRAINt_ETDLS, CURVATURE_ETDL
1360
1361
1362 XDATA = disp_mid_ETDLS
1363 YDATA = reactionY_ETDLS
1364 XLABEL = 'Displacement [mm]'
1365 YLABEL = 'Base Reaction [N]'
1366 TITLE = 'Base Reaction and Displacement of Structure During Static External
1367 COLOR = 'black'
1368 SEMIOLOGY = False
1369 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMIOLOGY)
1370
1371 # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1372 plt.figure(0, figsize=(12, 8))
1373 plt.plot(disp_mid_ETDLS, PERIOD_MIN_ETDLS, linewidth=3)
1374 plt.plot(disp_mid_ETDLS, PERIOD_MAX_ETDLS, linewidth=3)
1375 plt.title('Period of Structure During Static External Time-dependent Loadin
1376 plt.ylabel('Structural Period [s]')
1377 plt.xlabel('Displacement [mm]')
1378 #plt.semilogy()
1379 plt.grid()
```

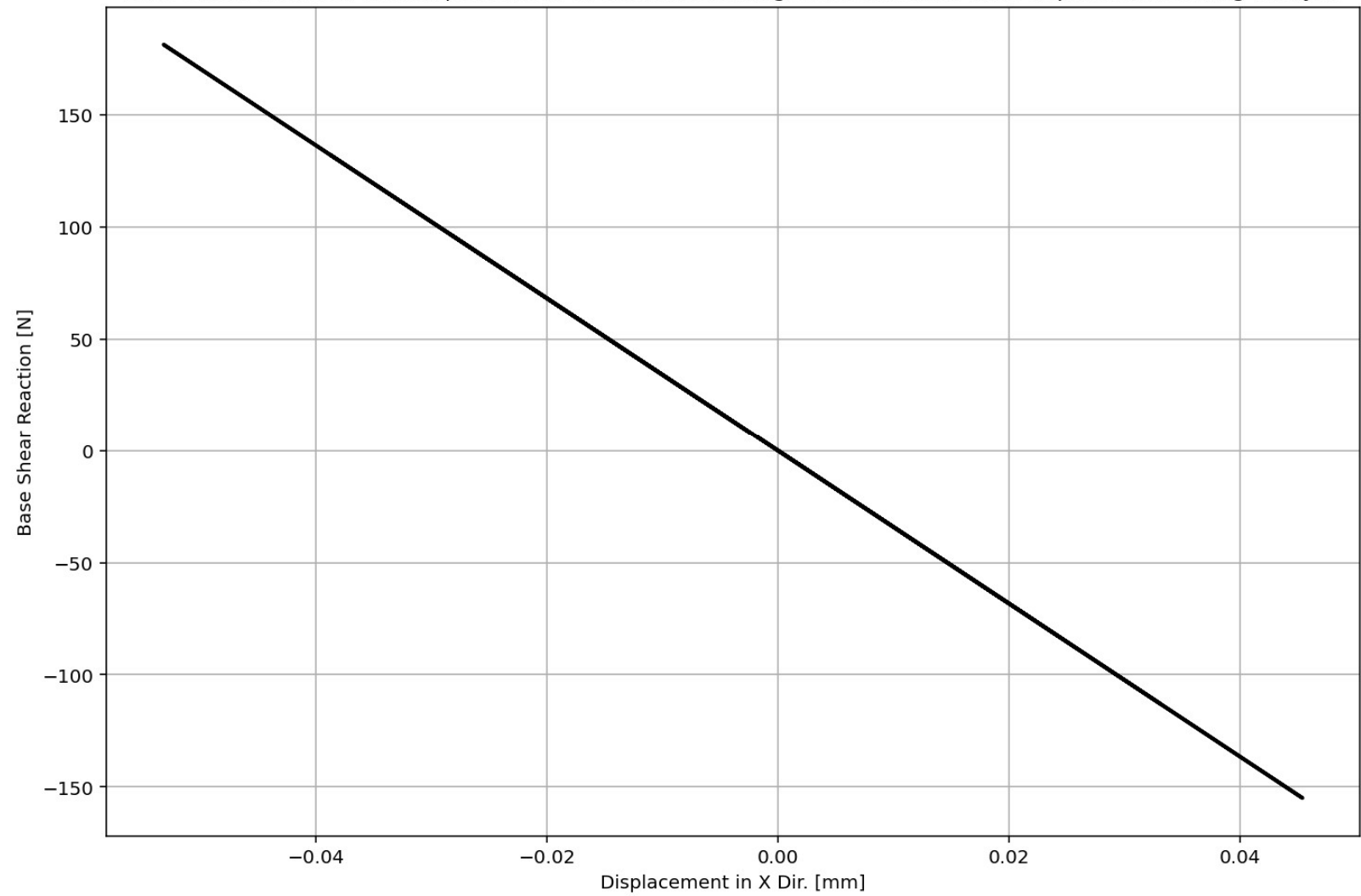
Base Reaction and Displacement of Structure During Static External Time-dependent Loading Analysis

IPython Console Files Help Variable Explorer Debugger Plots History

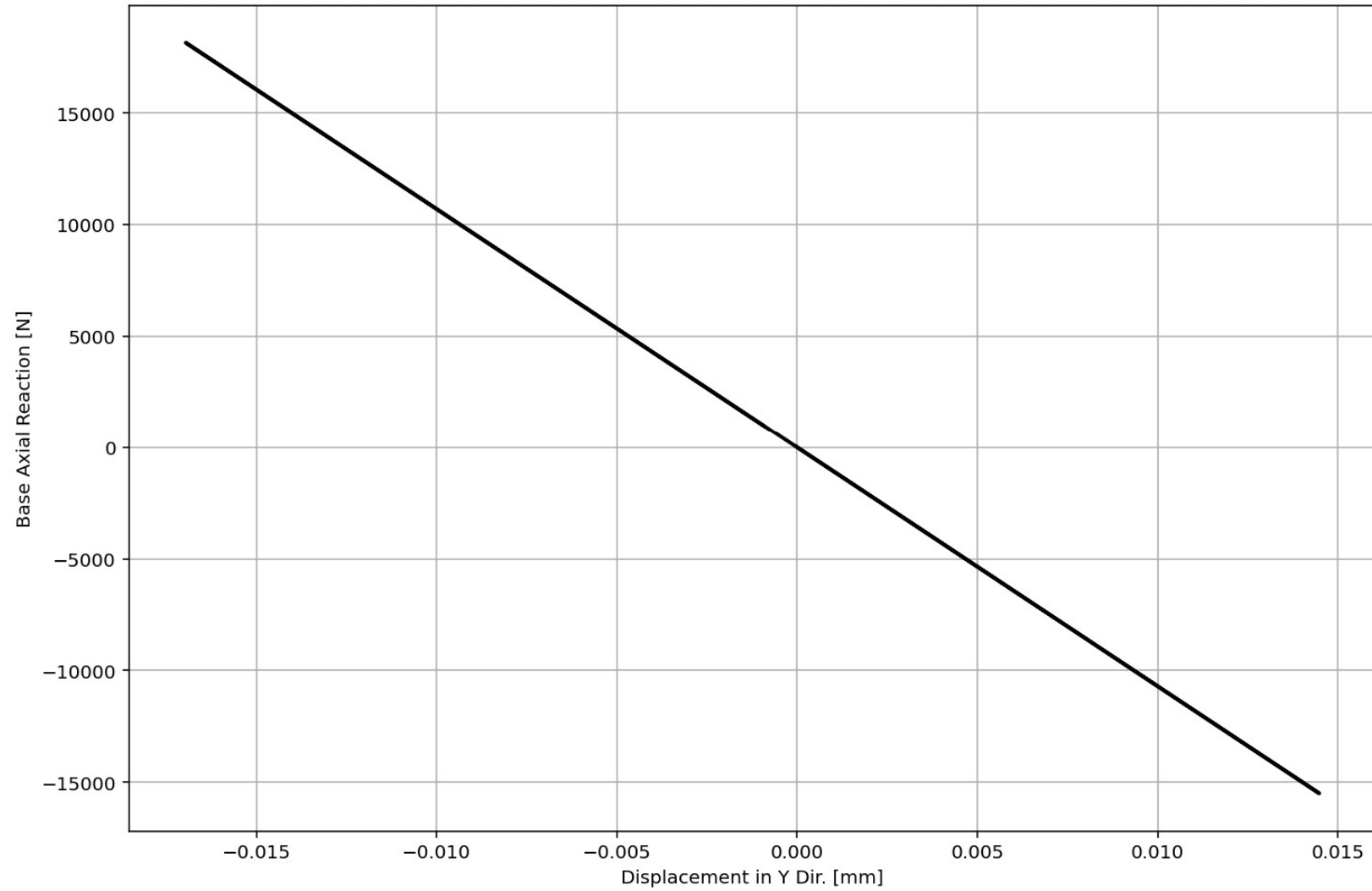
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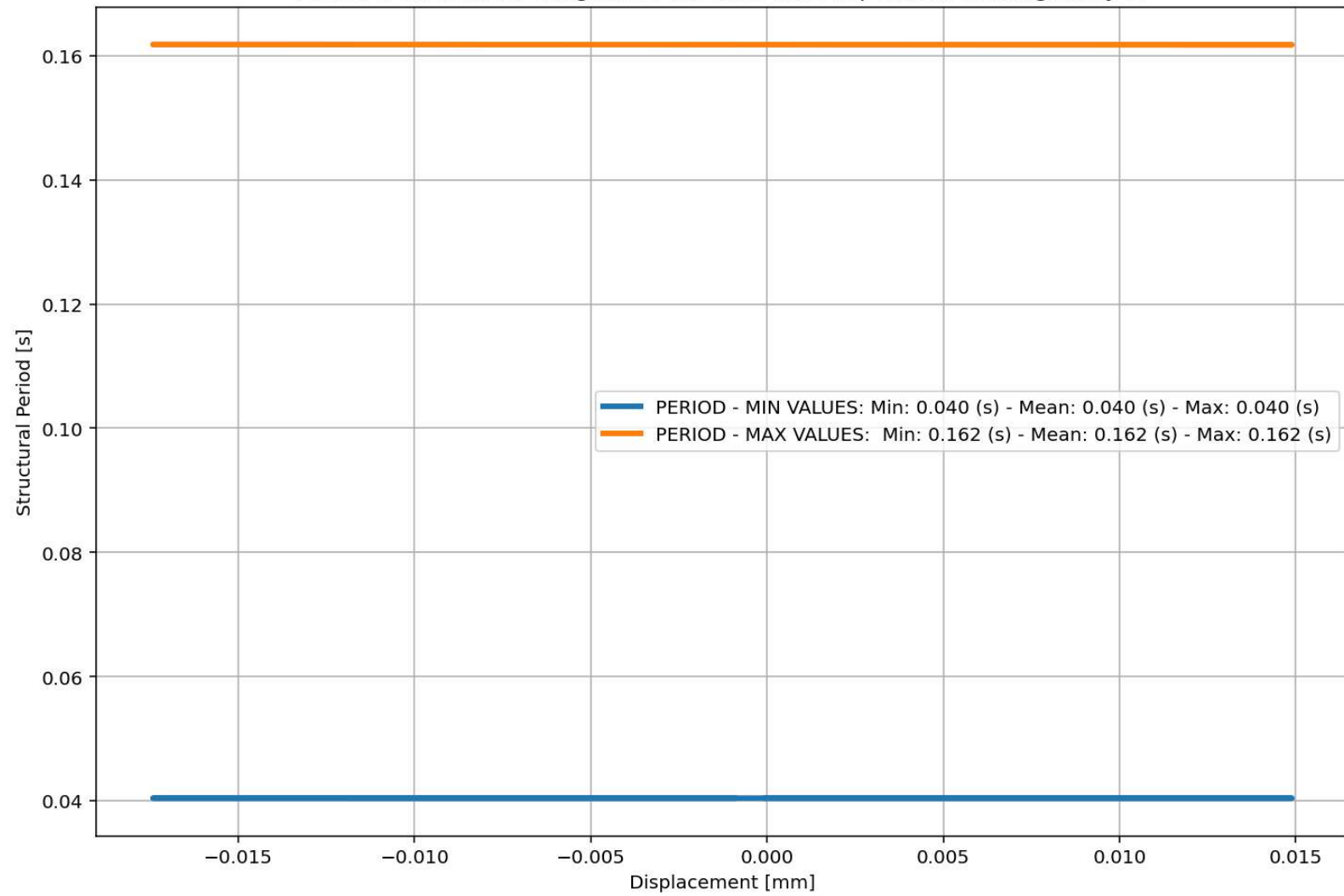
Base Shear Reaction and Displacement of Structure During Static External Time-dependent Loading Analysis



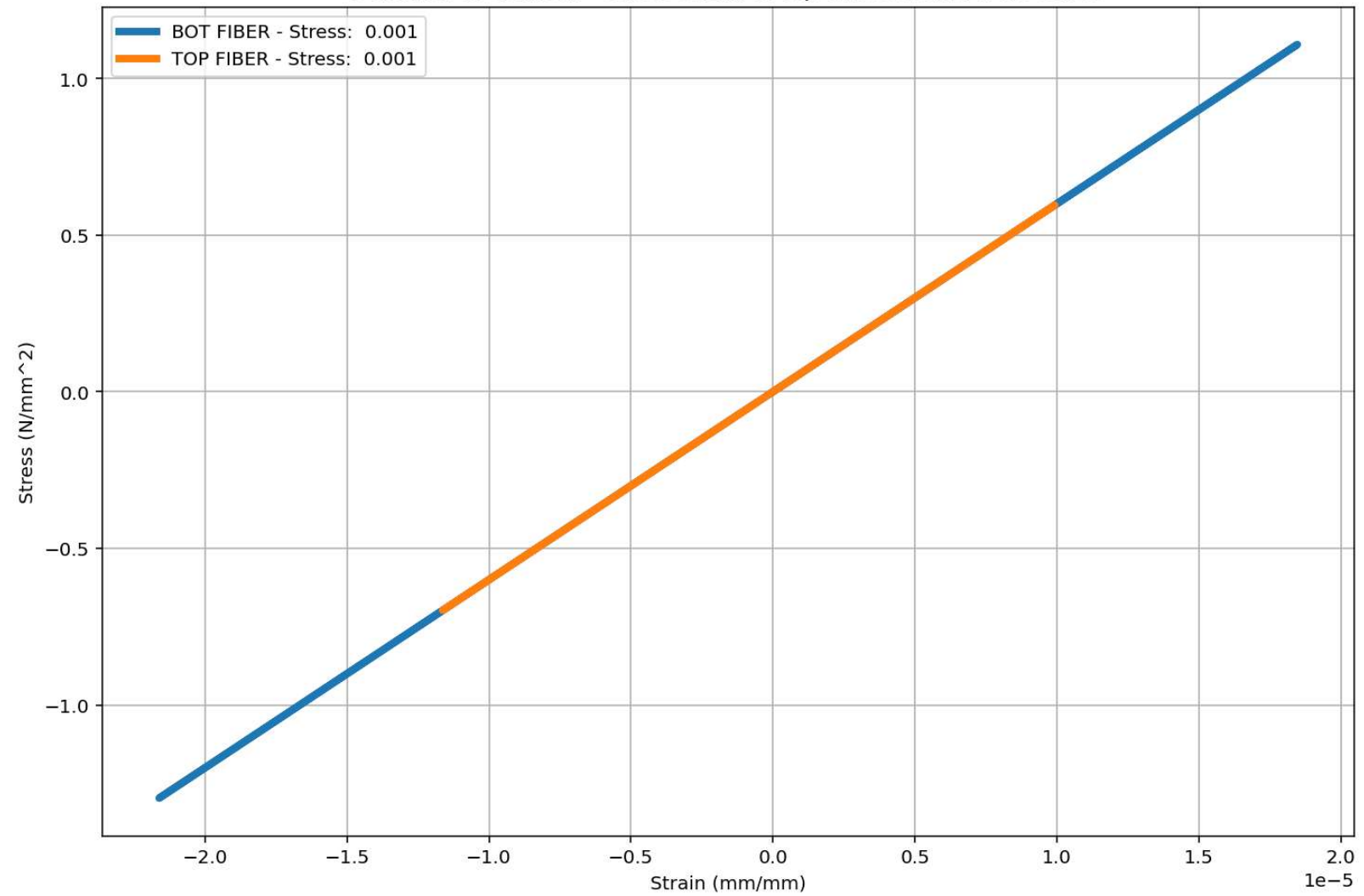
Base Axial Reaction and Displacement of Structure During Static External Time-dependent Loading Analysis

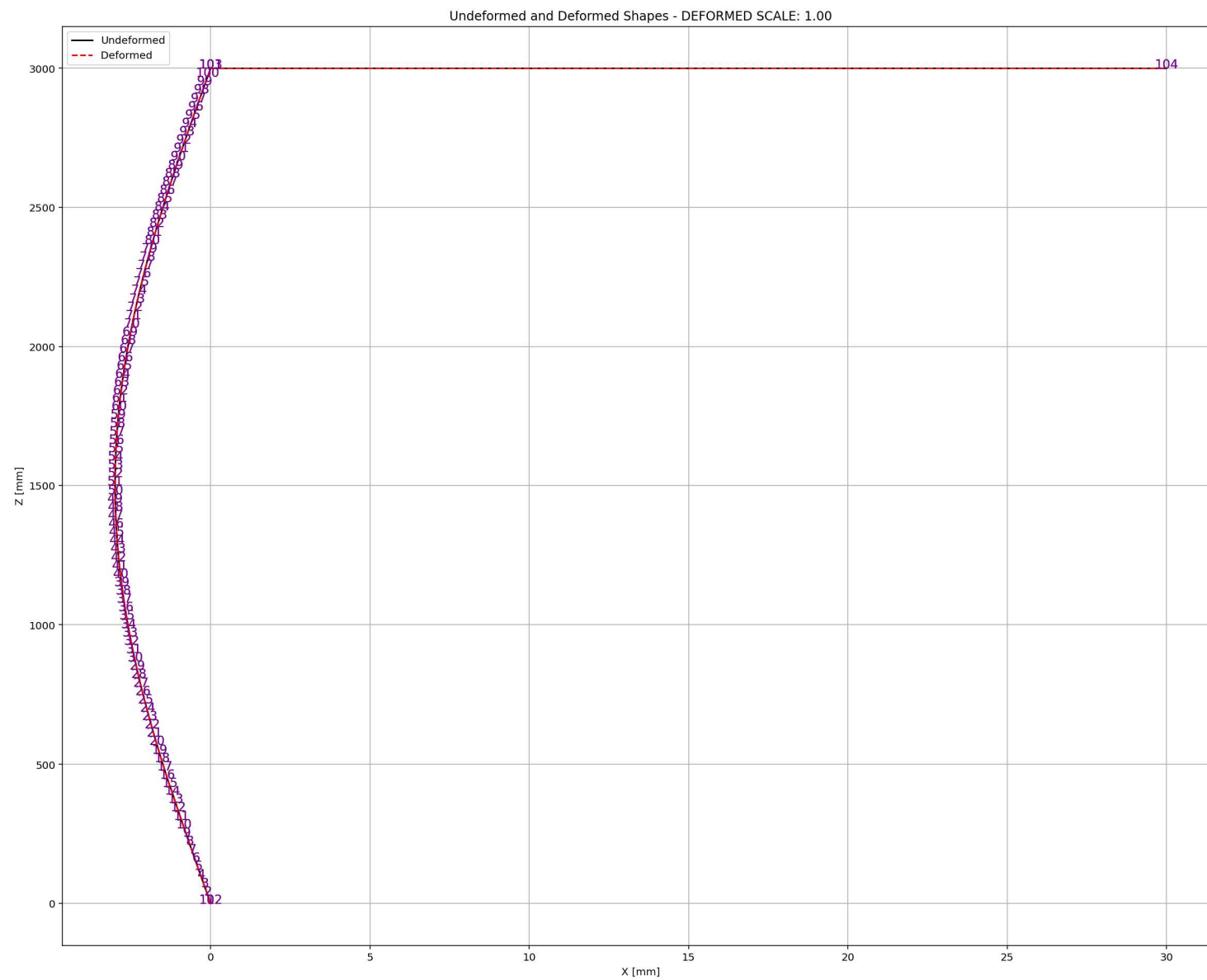


Period of Structure During Static External Time-dependent Loading Analysis

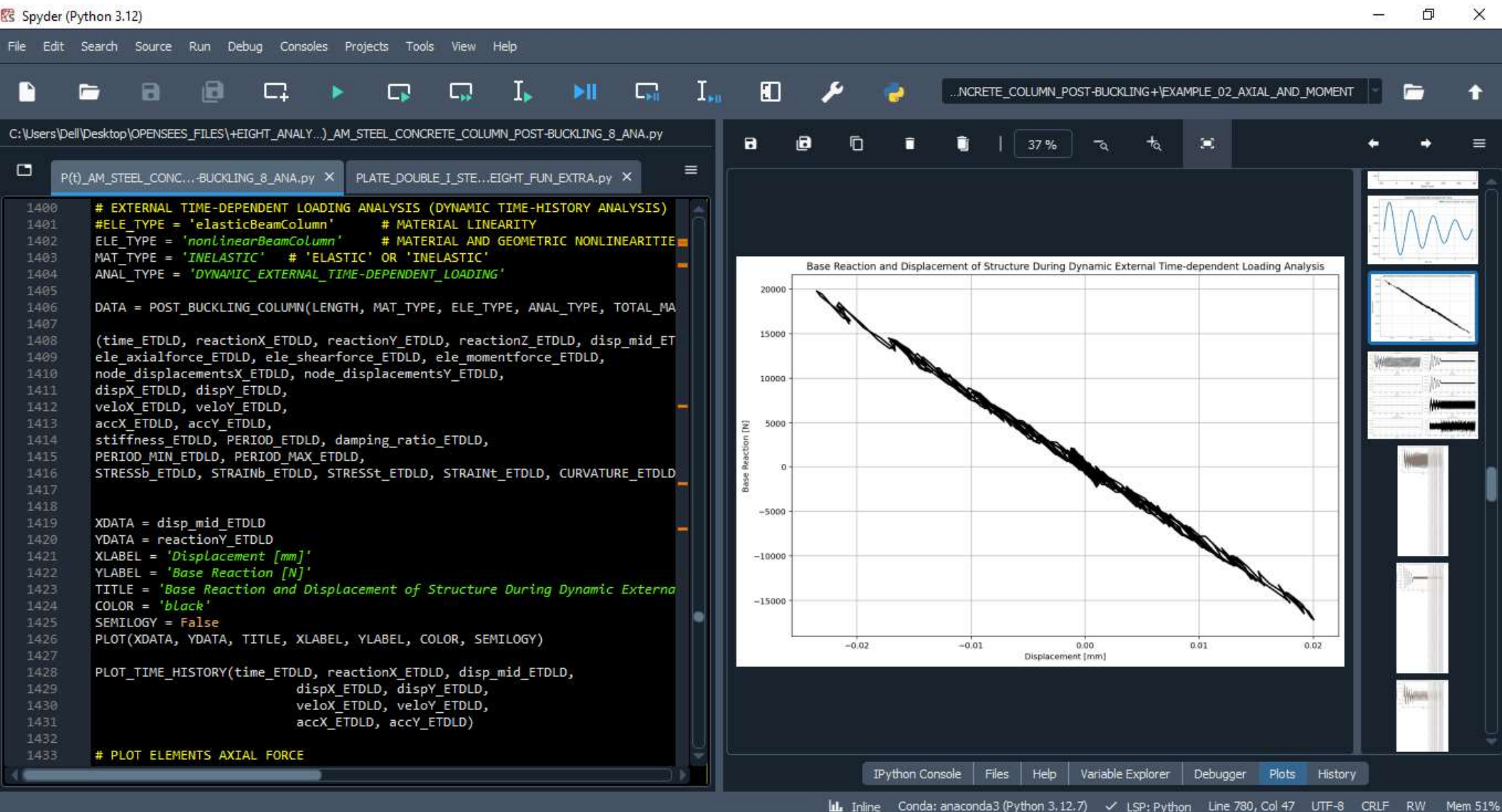


Behavior of element - Stress-strain of Top and Bottom Fibers Curve

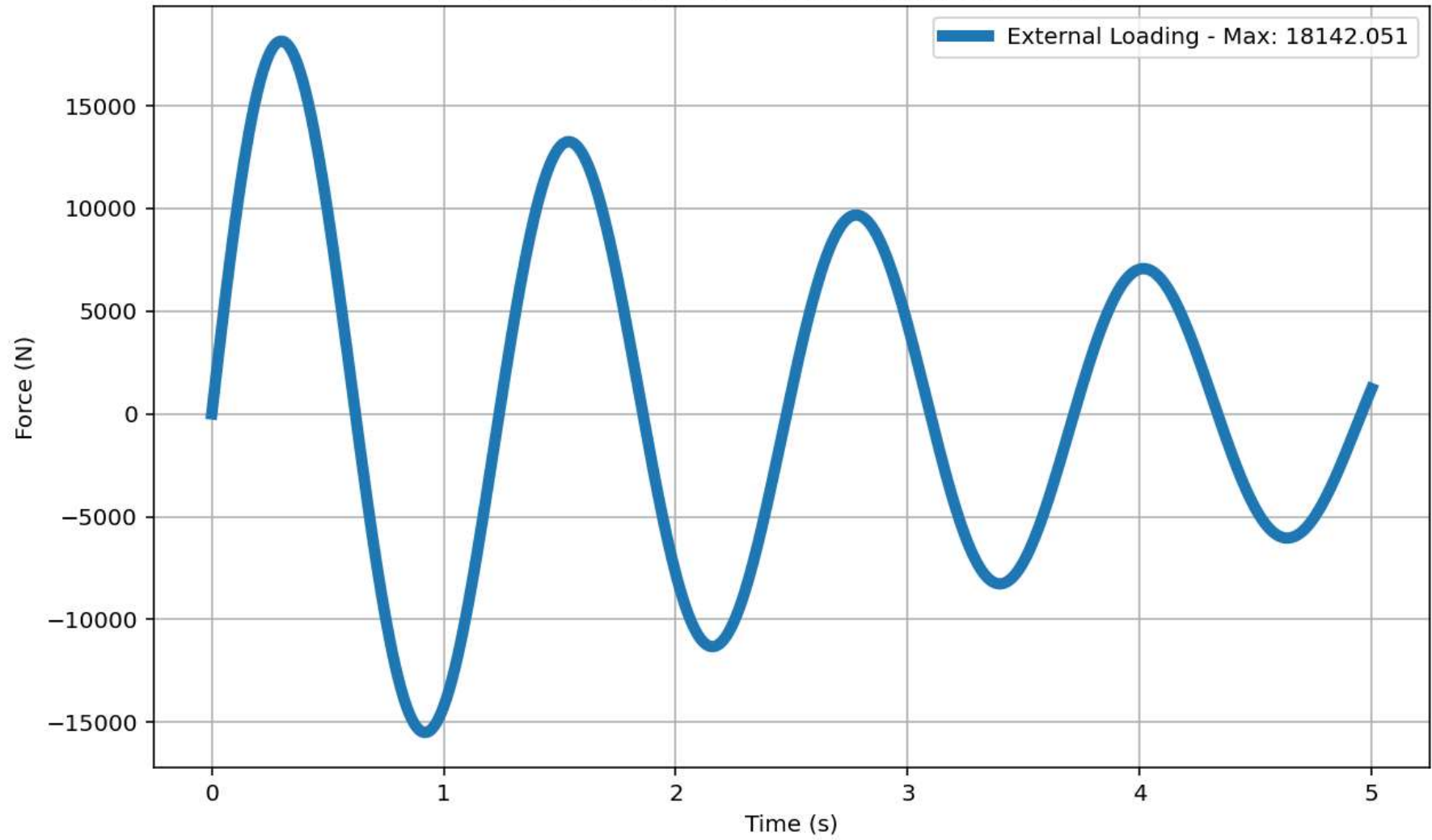




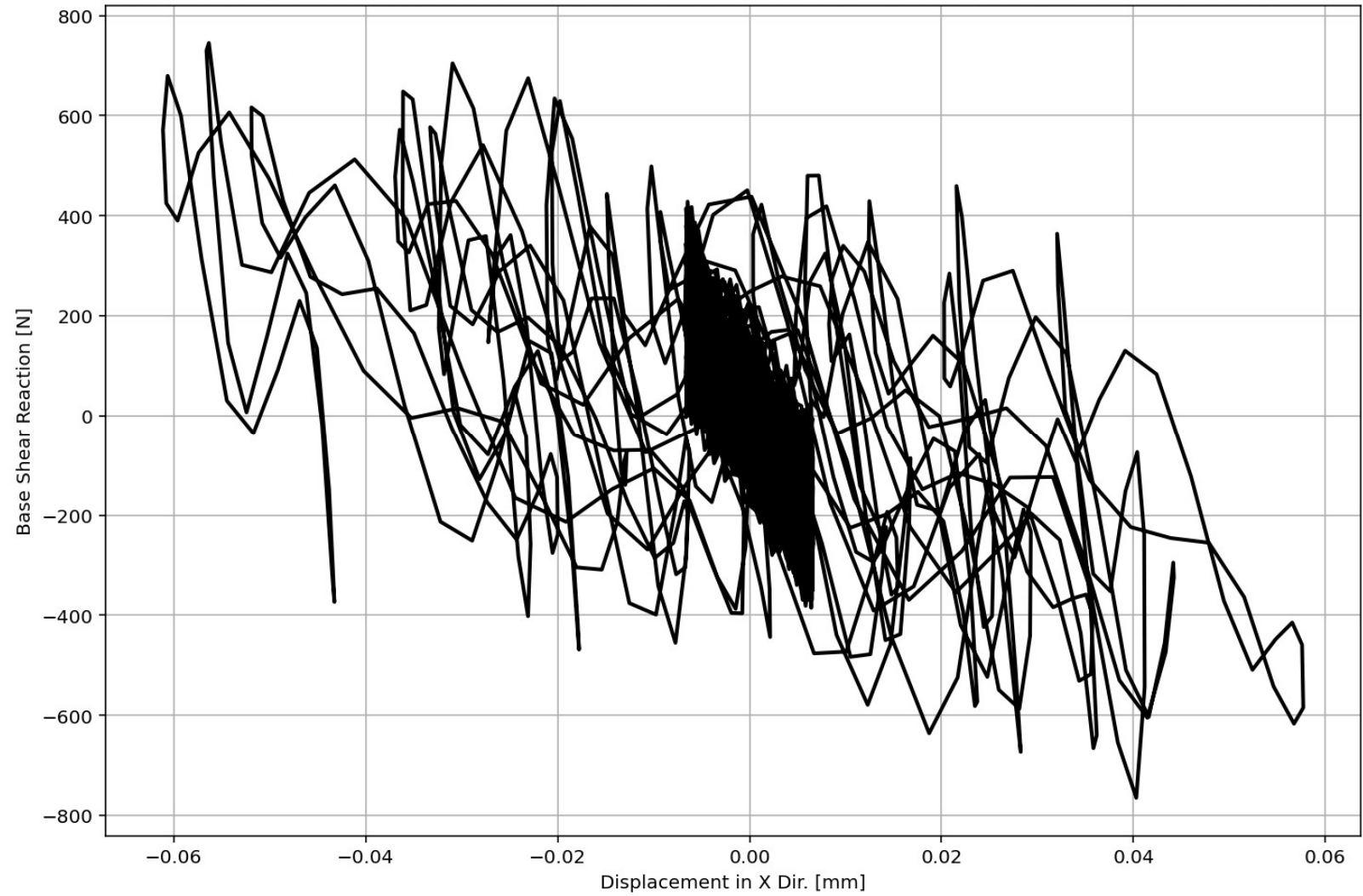
DYNAMIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS



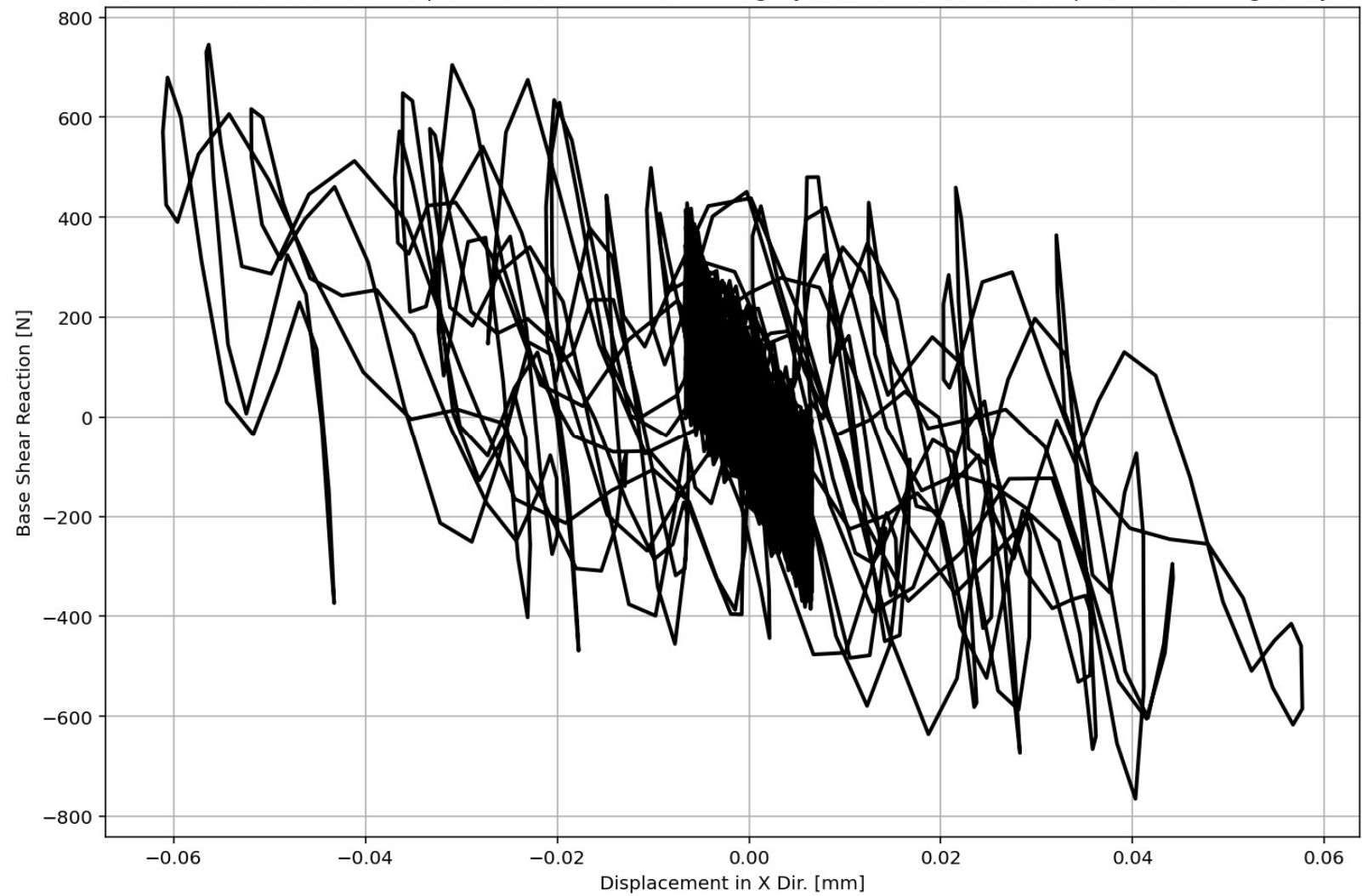
External Time-dependent Loading Over Time

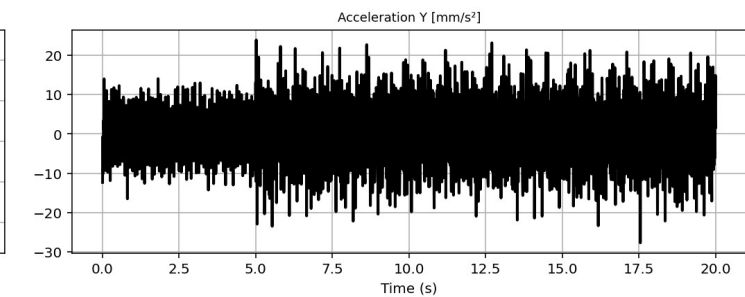
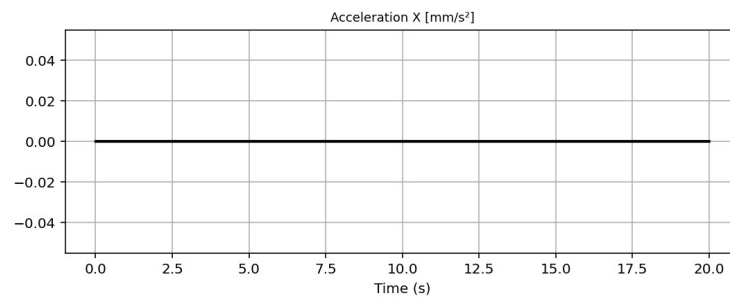
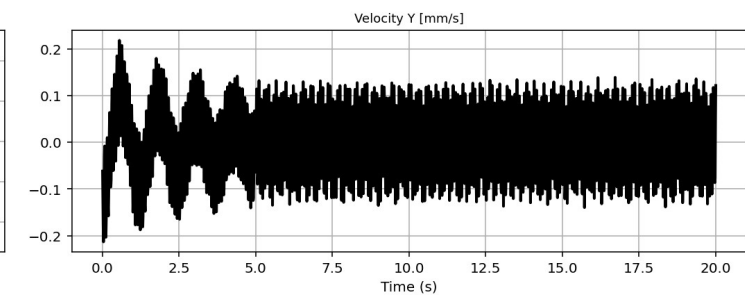
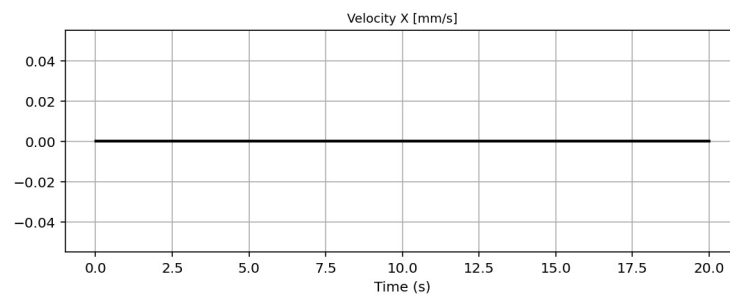
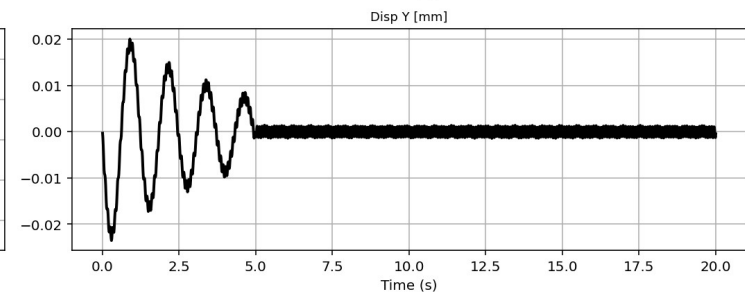
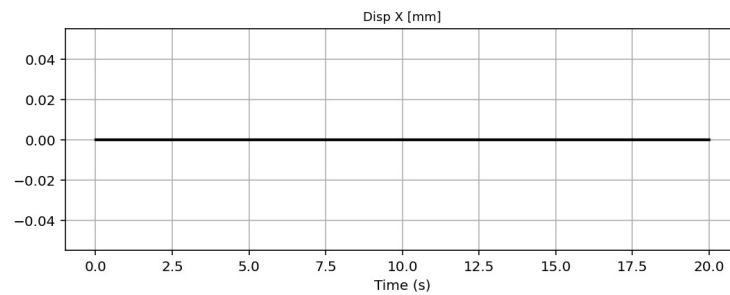
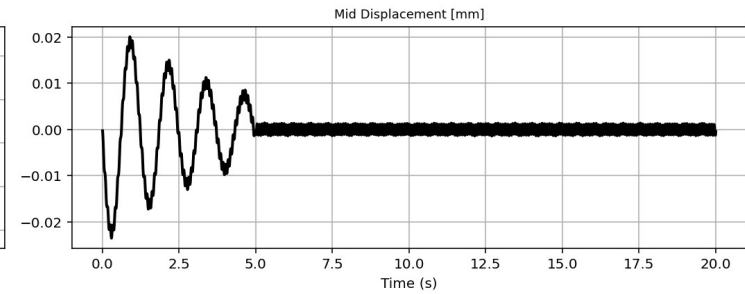
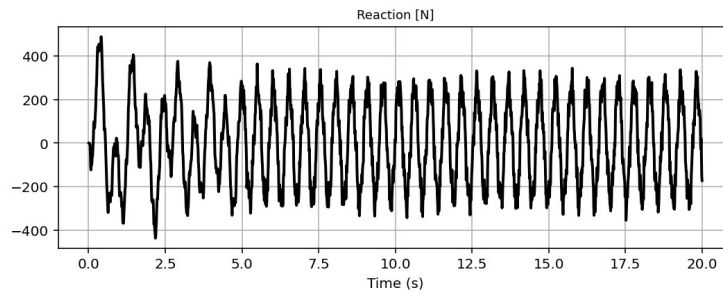


Base Shear Reaction and Displacement of Structure During Dynamic External Time-dependent Loading Analysis



Base Shear Reaction and Displacement of Structure During Dynamic External Time-dependent Loading Analysis

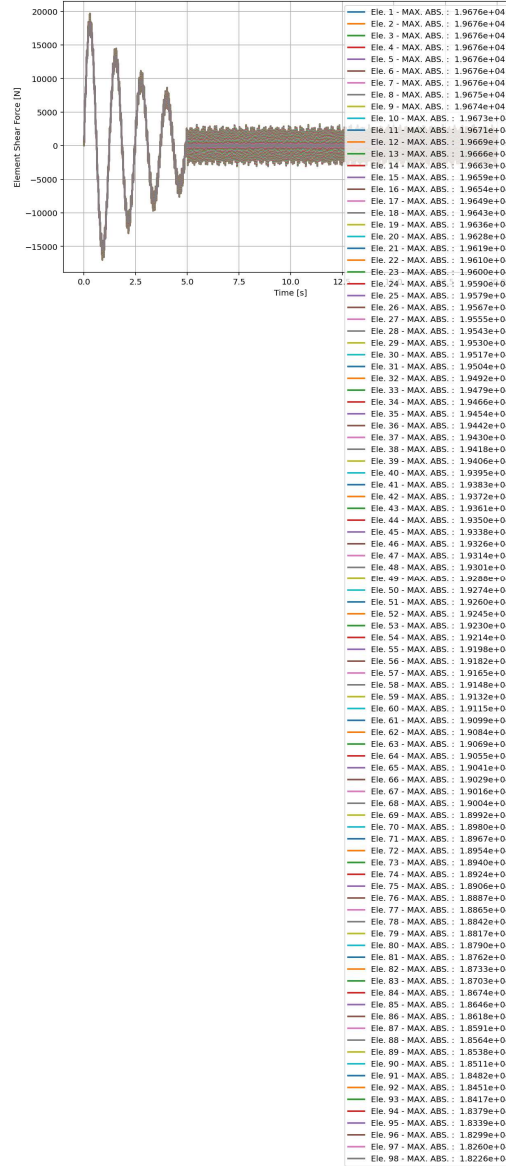




elements force in X Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis



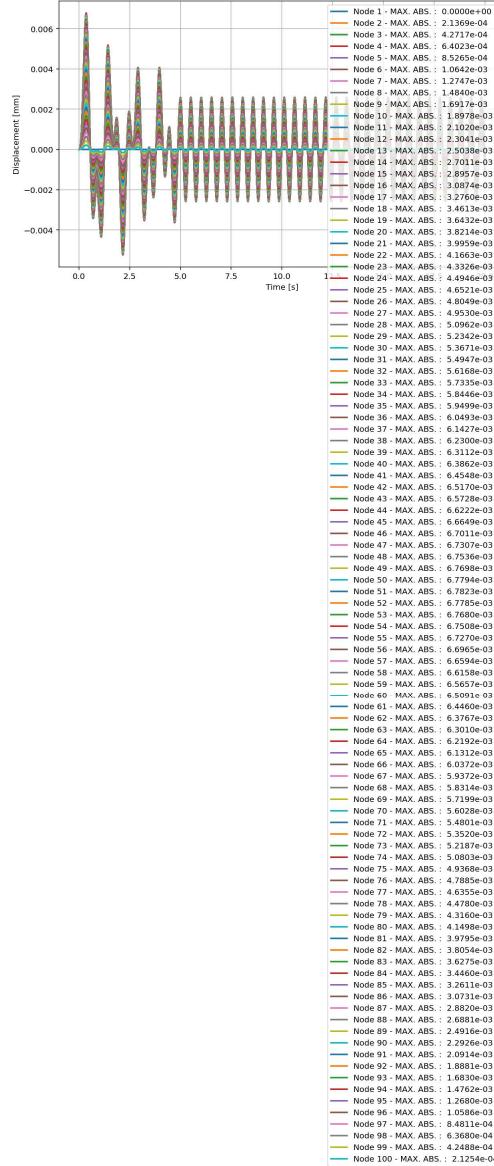
elements force in Y Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis



elements force in Z Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis



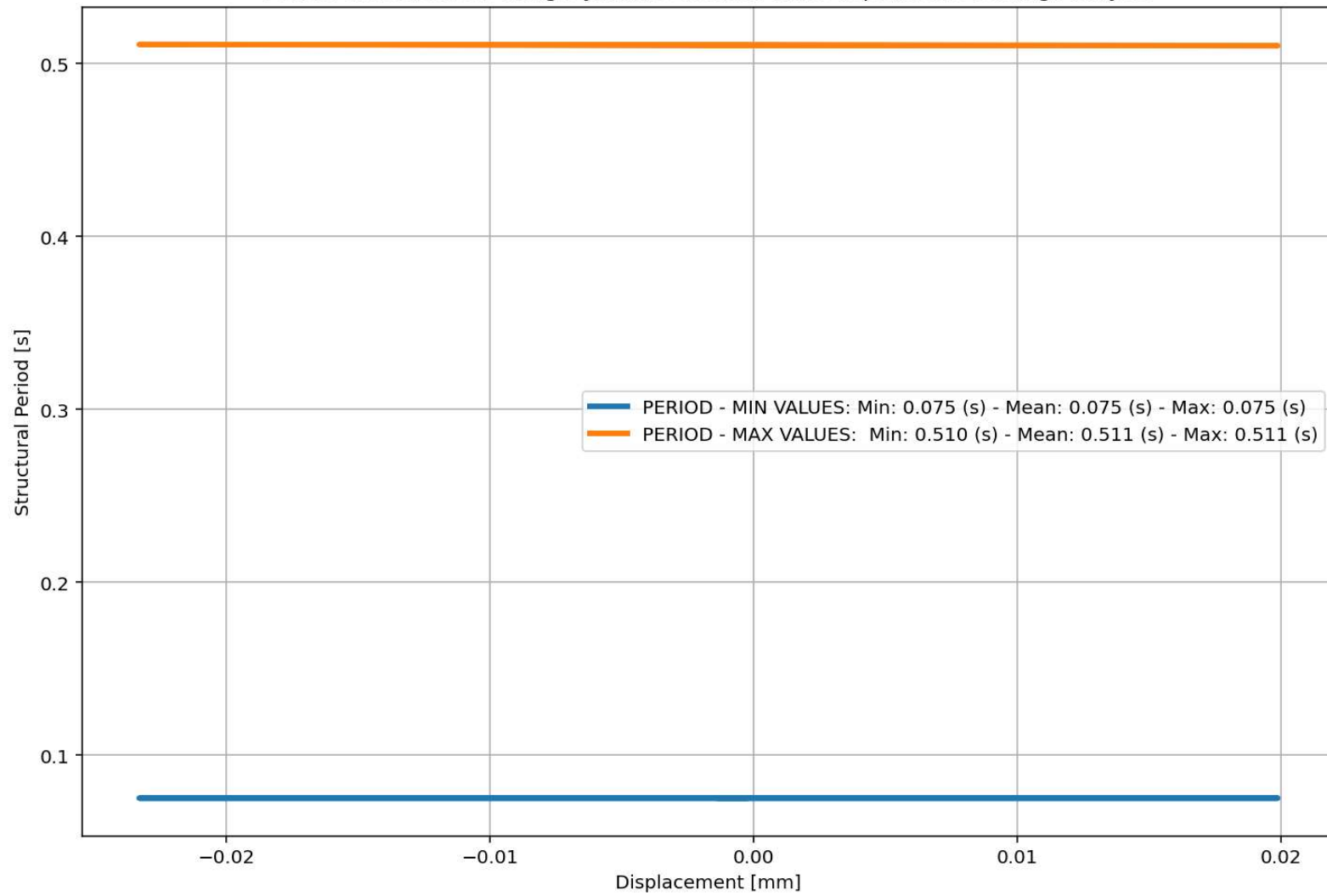
Node Displacements in X Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis

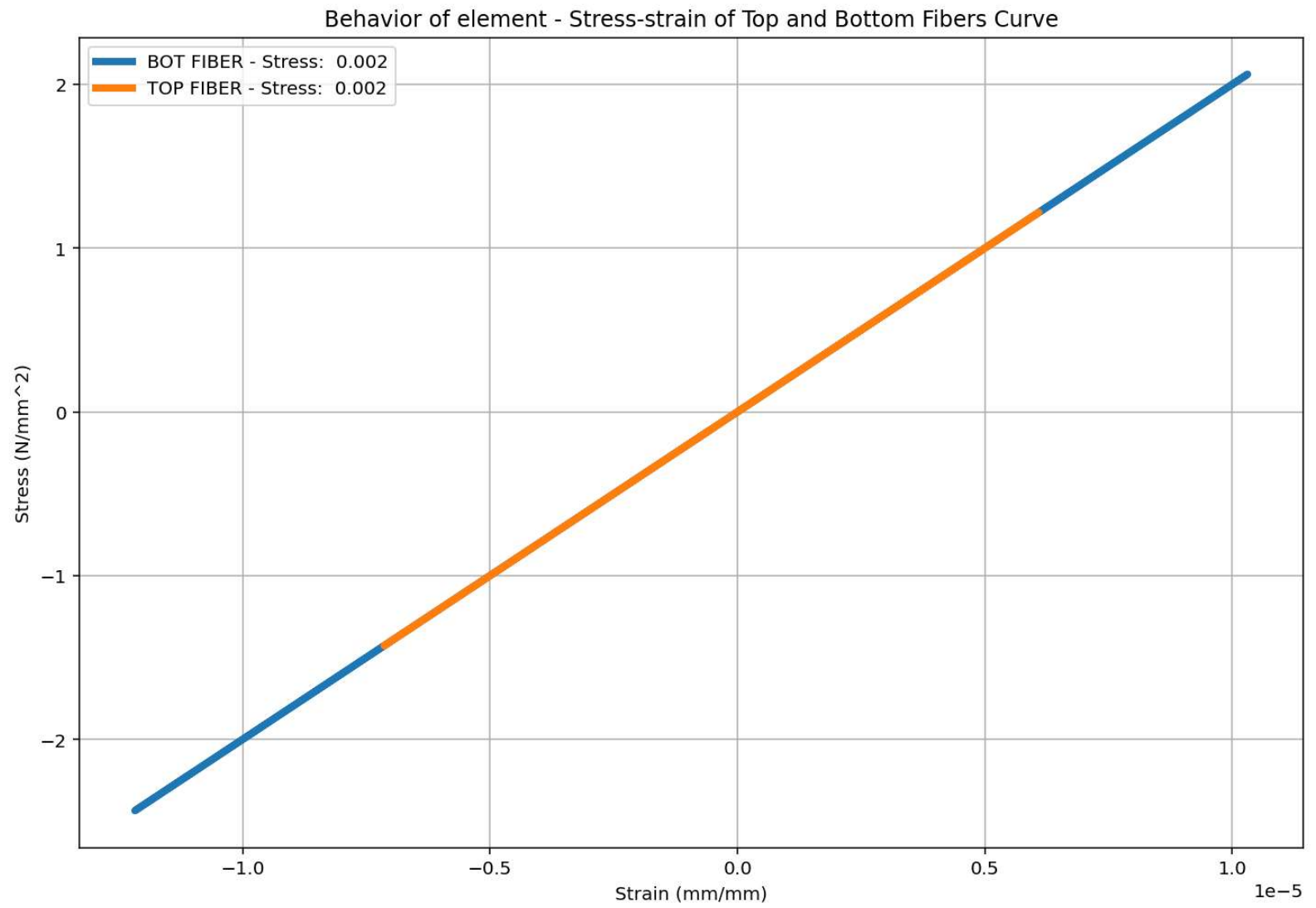


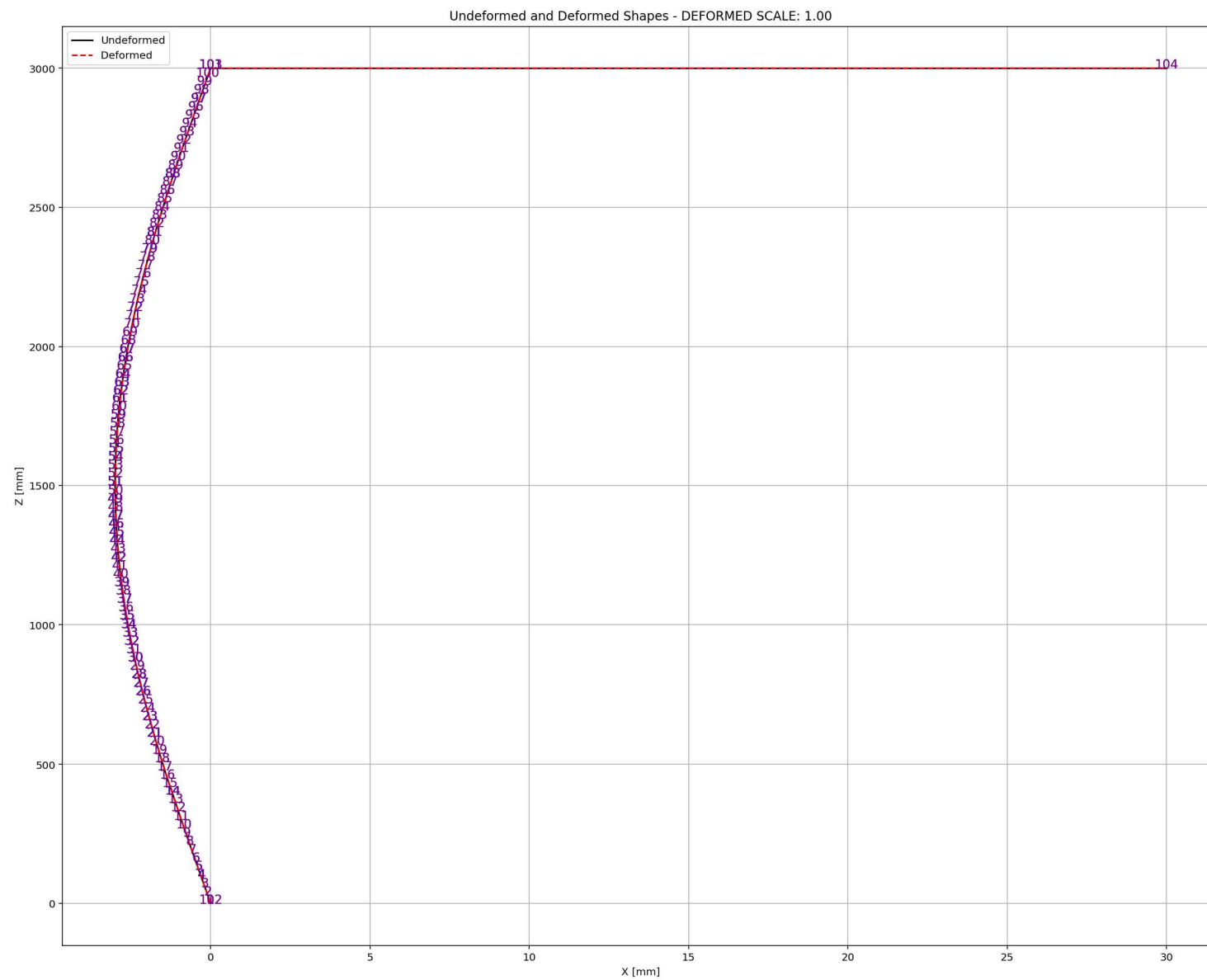
Node Displacements in Y Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis



Period of Structure During Dynamic External Time-dependent Loading Analysis







FREE-VIBRATION ANALYSIS

Spyder (Python 3.12)

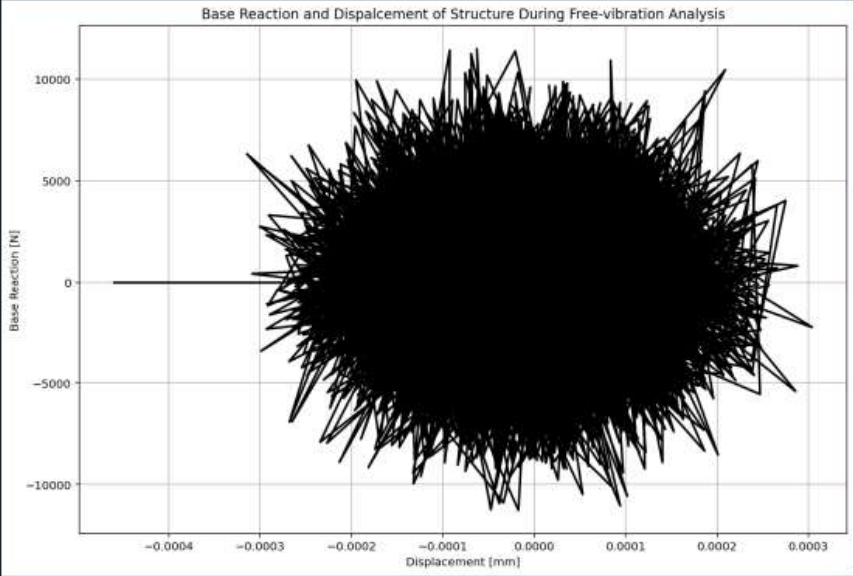
File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\ DELL\Desktop\OPENSEES_FILES\+EIGHT_ANALY...) _AM_STEEL_CONCRETE_COLUMN_POST-BUCKLING_8_ANA.py

P(t)_AM_STEEL_CONC...-BUCKLING_8_ANA.py X PLATE_DOUBLE_I_STE...EIGHT_FUN_EXTRA.py X

```
1478 # FREE-VIBRATION ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
1479 #ELE_TYPE = 'elasticBeamColumn' # MATERIAL LINEARITY
1480 ELE_TYPE = 'nonlinearBeamColumn' # MATERIAL AND GEOMETRIC NONLINEARITIE
1481 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1482 ANAL_TYPE = 'FREE-VIBRATION'
1483 DATA = POST_BUCKLING_COLUMN(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MA
1484
1485 (time_FV, reactionX_FV, reactionY_FV, reactionZ_FV, disp_mid_FV,
1486 ele_axialforce_FV, ele_shearforce_FV, ele_momentforce_FV,
1487 node_displacementsX_FV, node_displacementsY_FV,
1488 dispX_FV, dispY_FV,
1489 veloX_FV, veloY_FV,
1490 accX_FV, accY_FV,
1491 stiffness_FV, PERIOD_FV, damping_ratio_FV,
1492 PERIOD_MIN_FV, PERIOD_MAX_FV,
1493 STRESSb_FV, STRAINb_FV, STRESSt_FV, STRAINt_FV, CURVATURE_FV, NA_FV, CD_FV)
1494
1495 XDATA = disp_mid_FV
1496 YDATA = reactionY_FV
1497 XLABEL = 'Displacement [mm]'
1498 YLABEL = 'Base Reaction [N]'
1499 TITLE = 'Base Reaction and Dispalcement of Structure During Free-vibration
1500 COLOR = 'black'
1501 SEMIOLOGY = False
1502 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMIOLOGY)
1503
1504 PLOT_TIME_HISTORY(time_FV, reactionX_FV, disp_mid_FV,
1505 dispX_FV, dispY_FV,
1506 veloX_FV, veloY_FV,
1507 accX_FV, accY_FV)
1508
1509 # PLOT ELEMENTS AXIAL FORCE
1510 YLABEL = 'Element Axial Force [N]'
1511 TITLE = "elements force in X Dir. vs Time for Beam Element During Free-vib
```

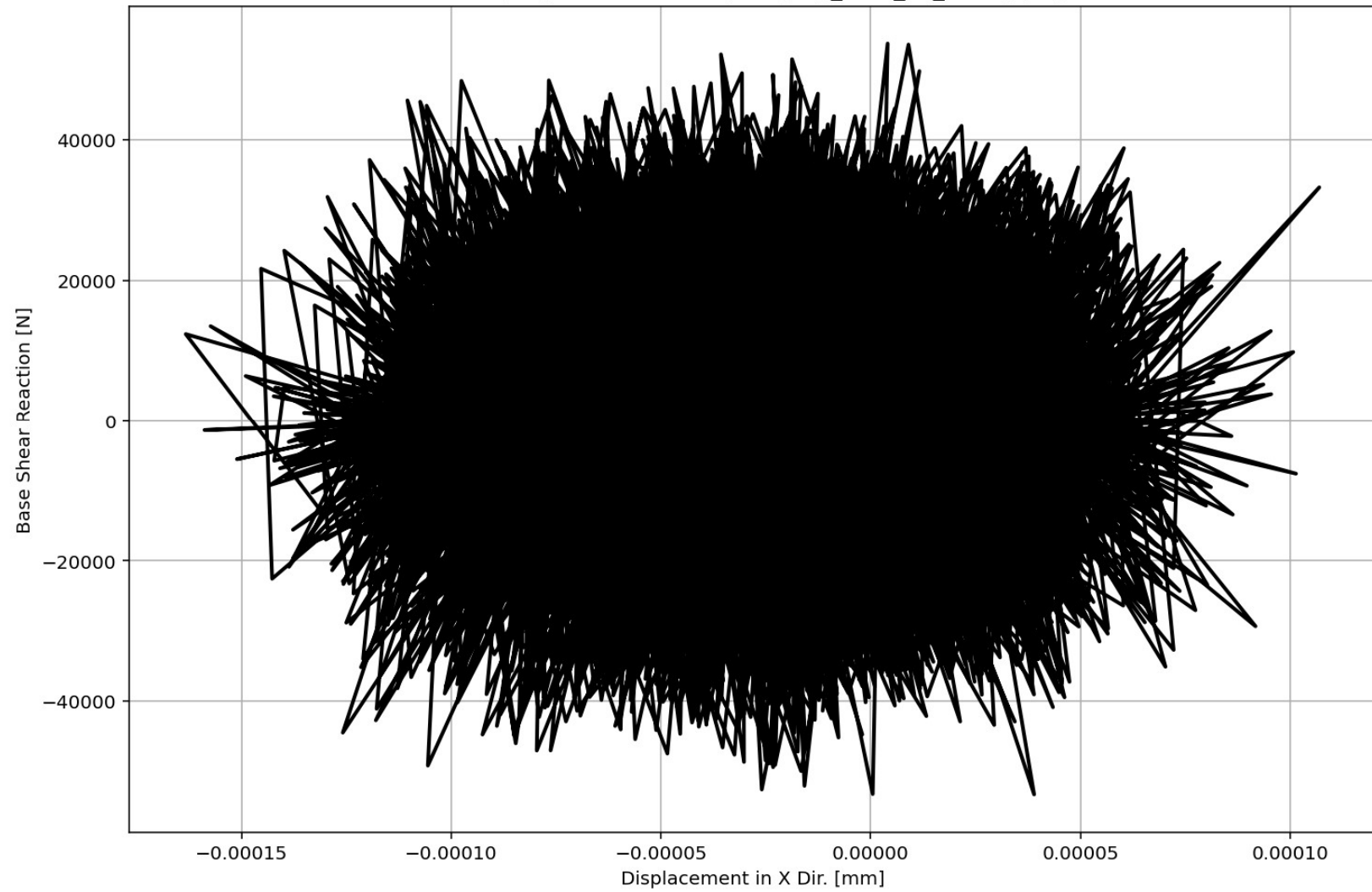
Base Reaction and Dispalcement of Structure During Free-vibration Analysis



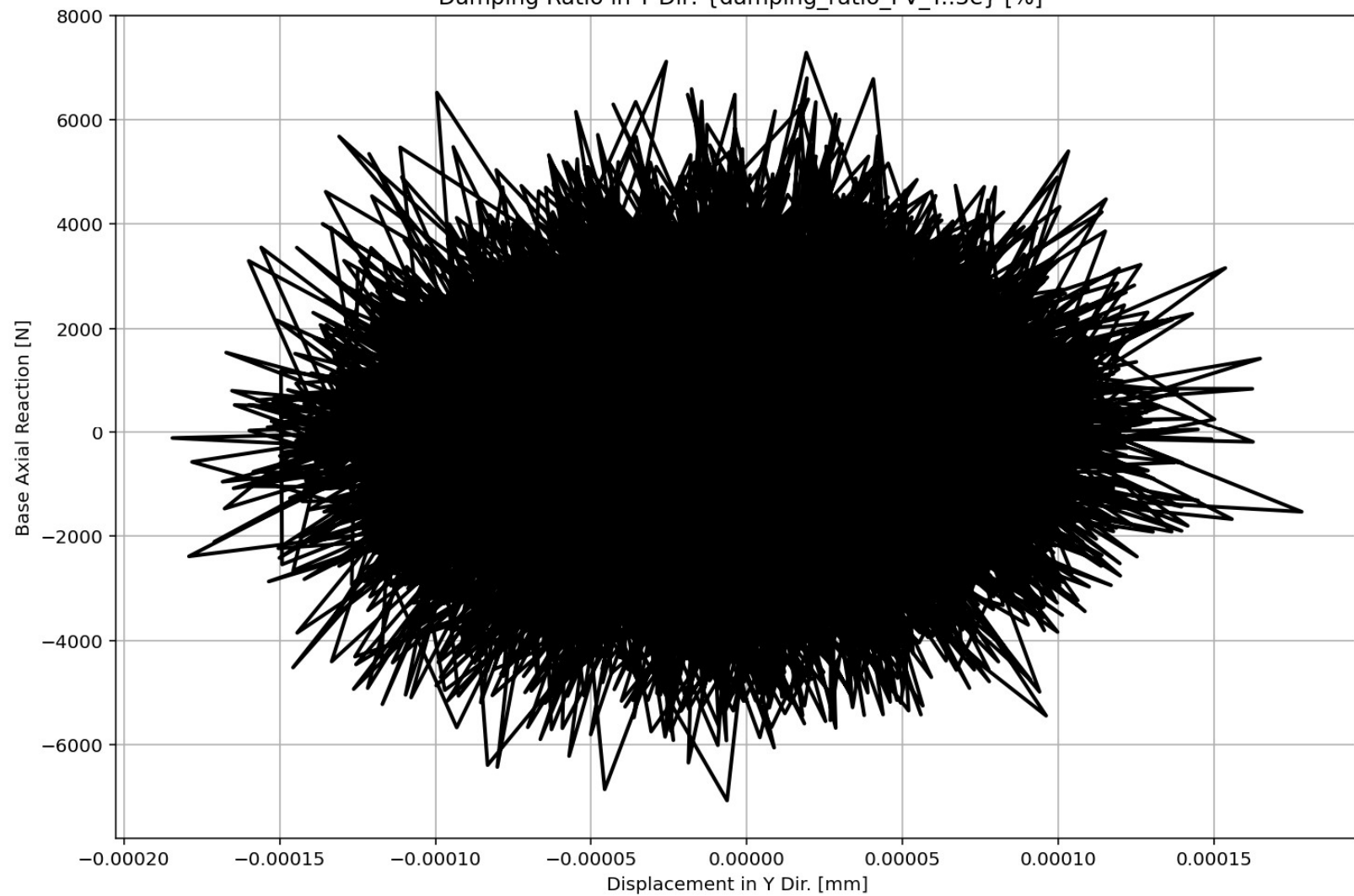
IPython Console Files Help Variable Explorer Debugger Plots History

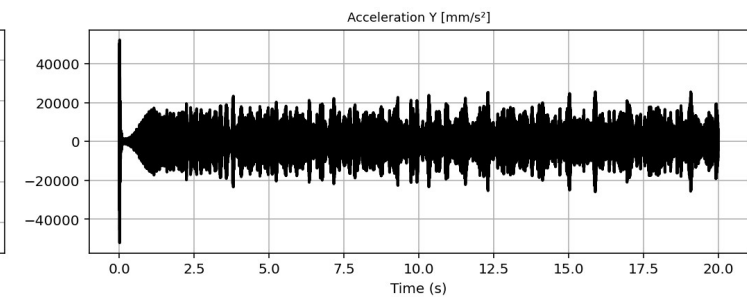
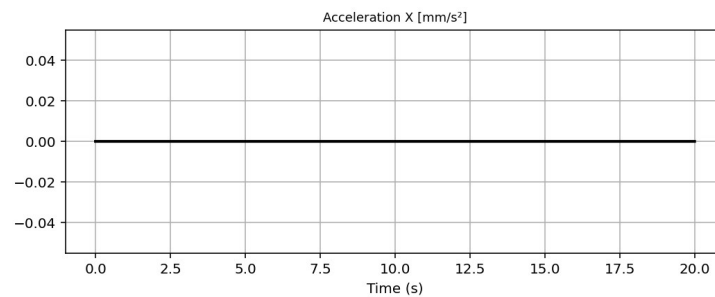
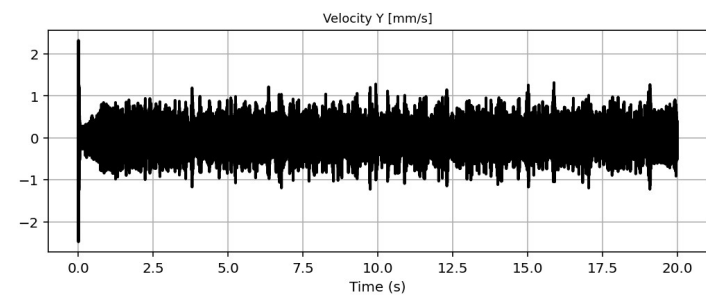
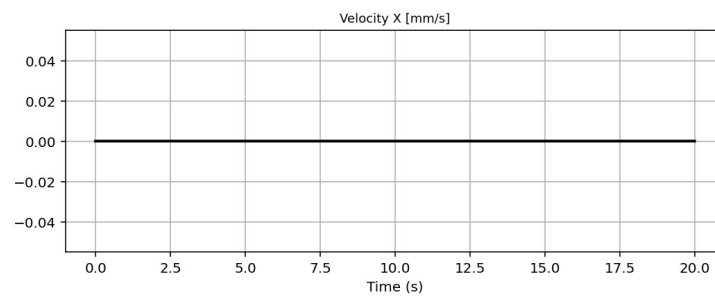
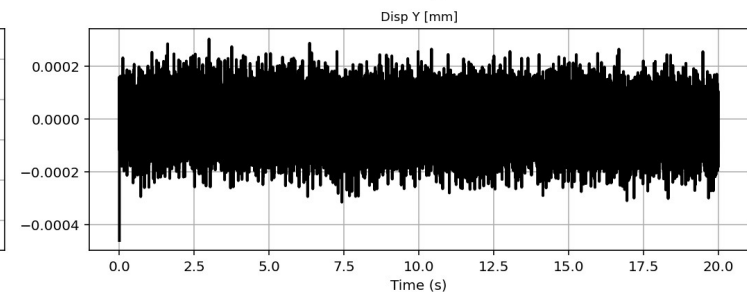
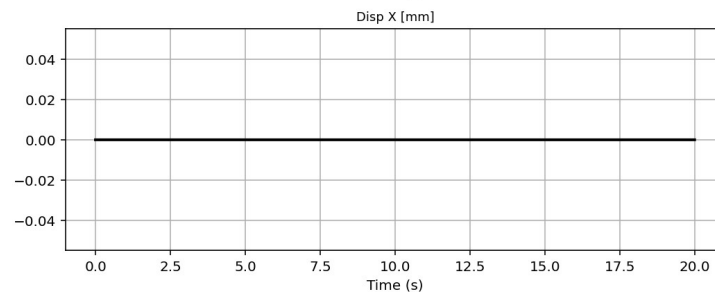
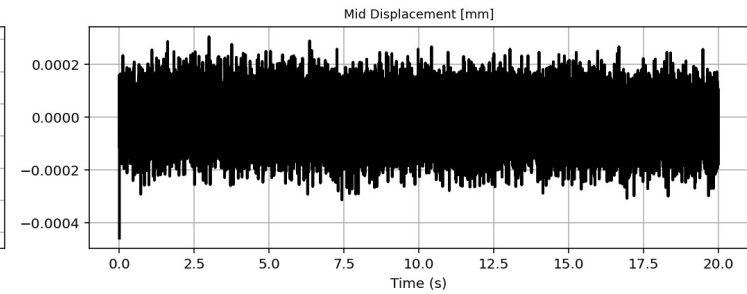
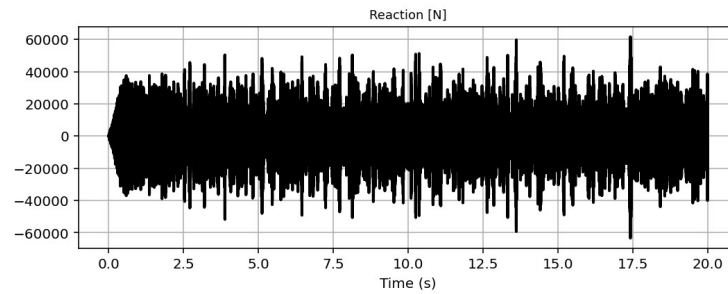
Int. Inline Conda: anaconda3 (Python 3.12.7) ✓ LSP: Python Line 780, Col 47 UTF-8 CR LF RW Mem 52%

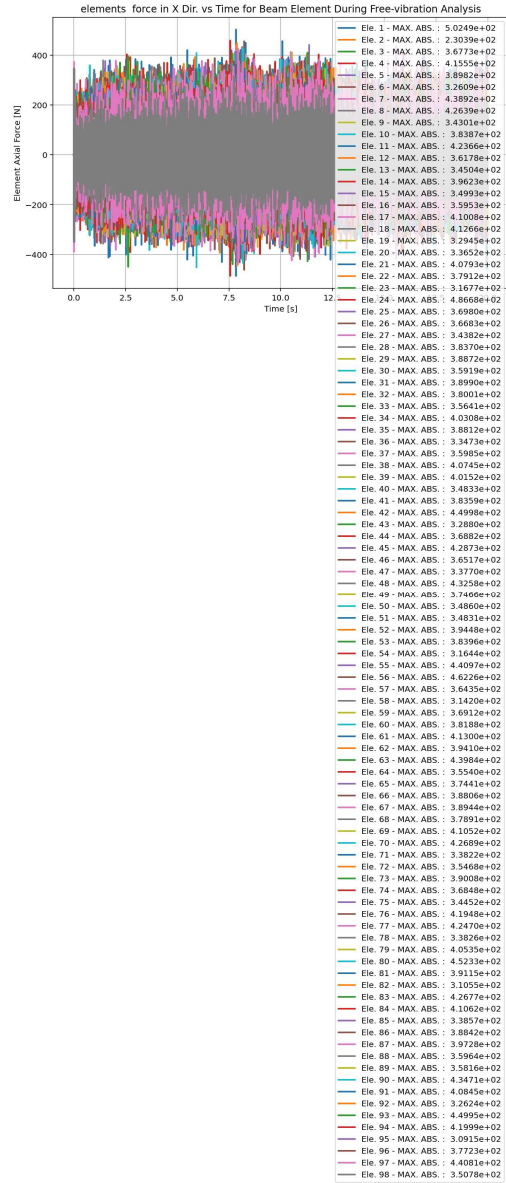
Base Shear Reaction and Displacement of Structure During Free-vibration Analysis
Damping Ratio in X Dir: {damping_ratio_FV_X:.3e} [%]



Base Axial Reaction and Dispalcement of Structure During Free-vibration Analysis
Damping Ratio in Y Dir: {damping_ratio_FV_Y:.3e} [%]

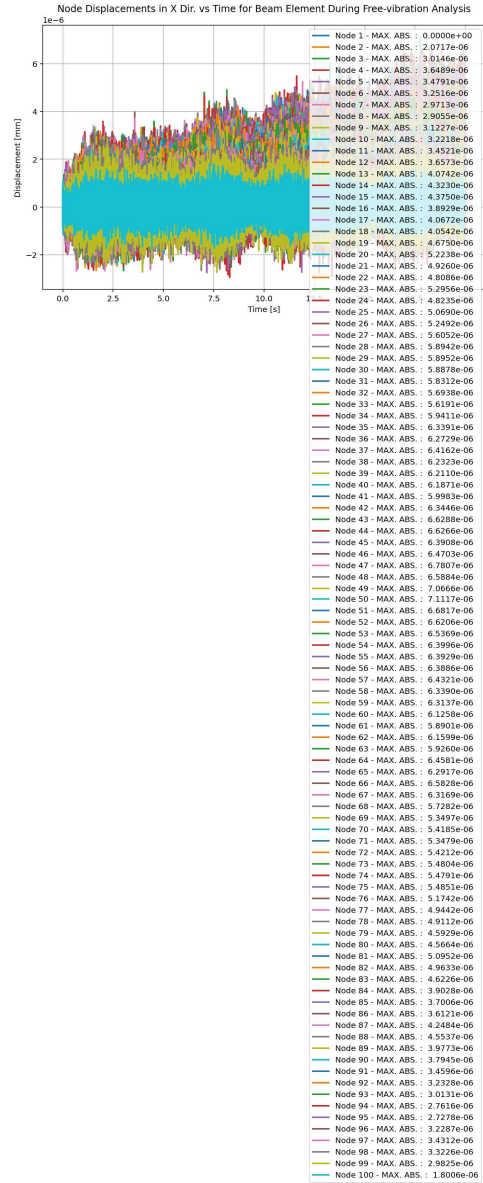


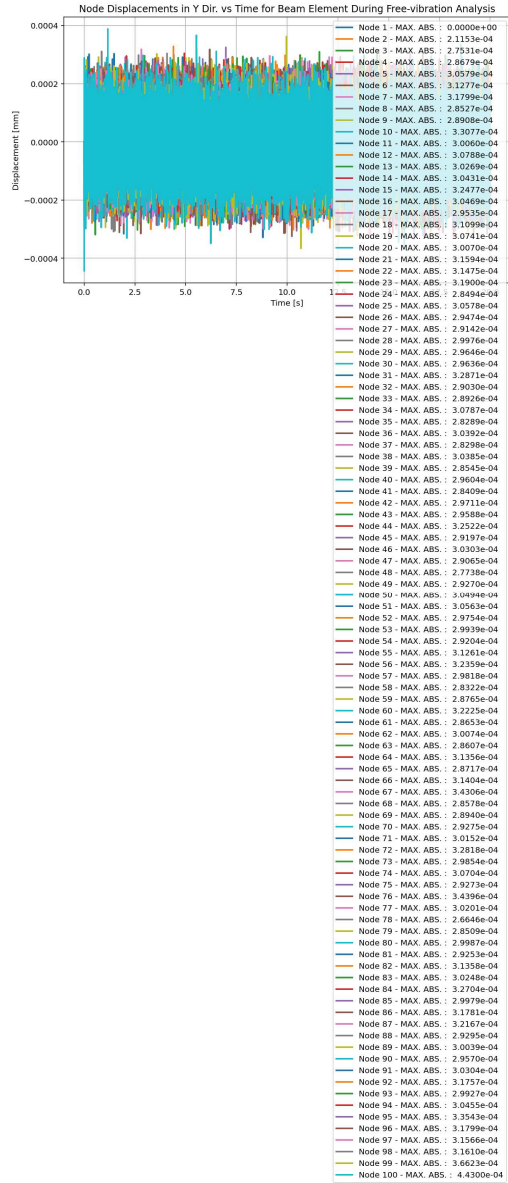




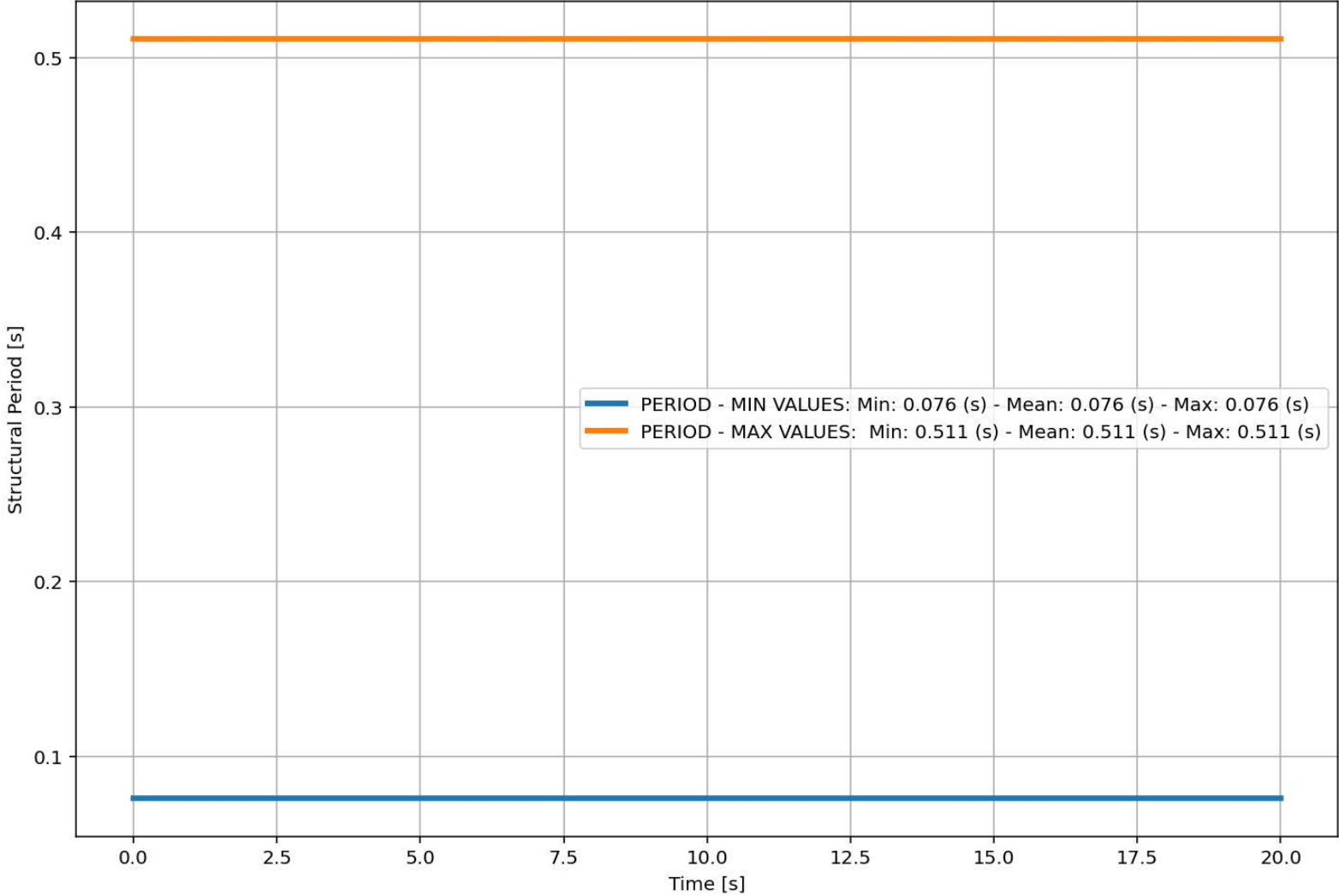




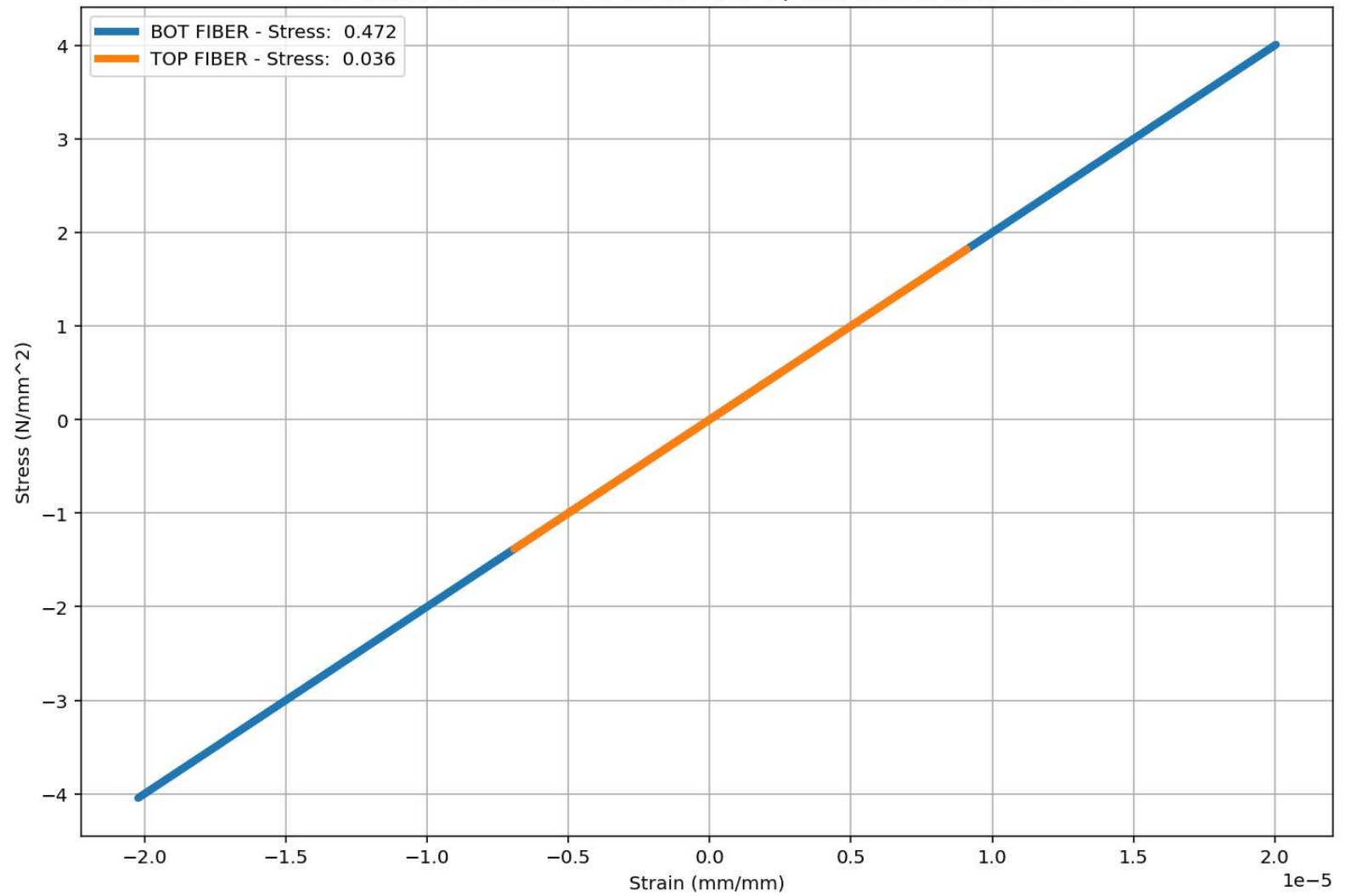


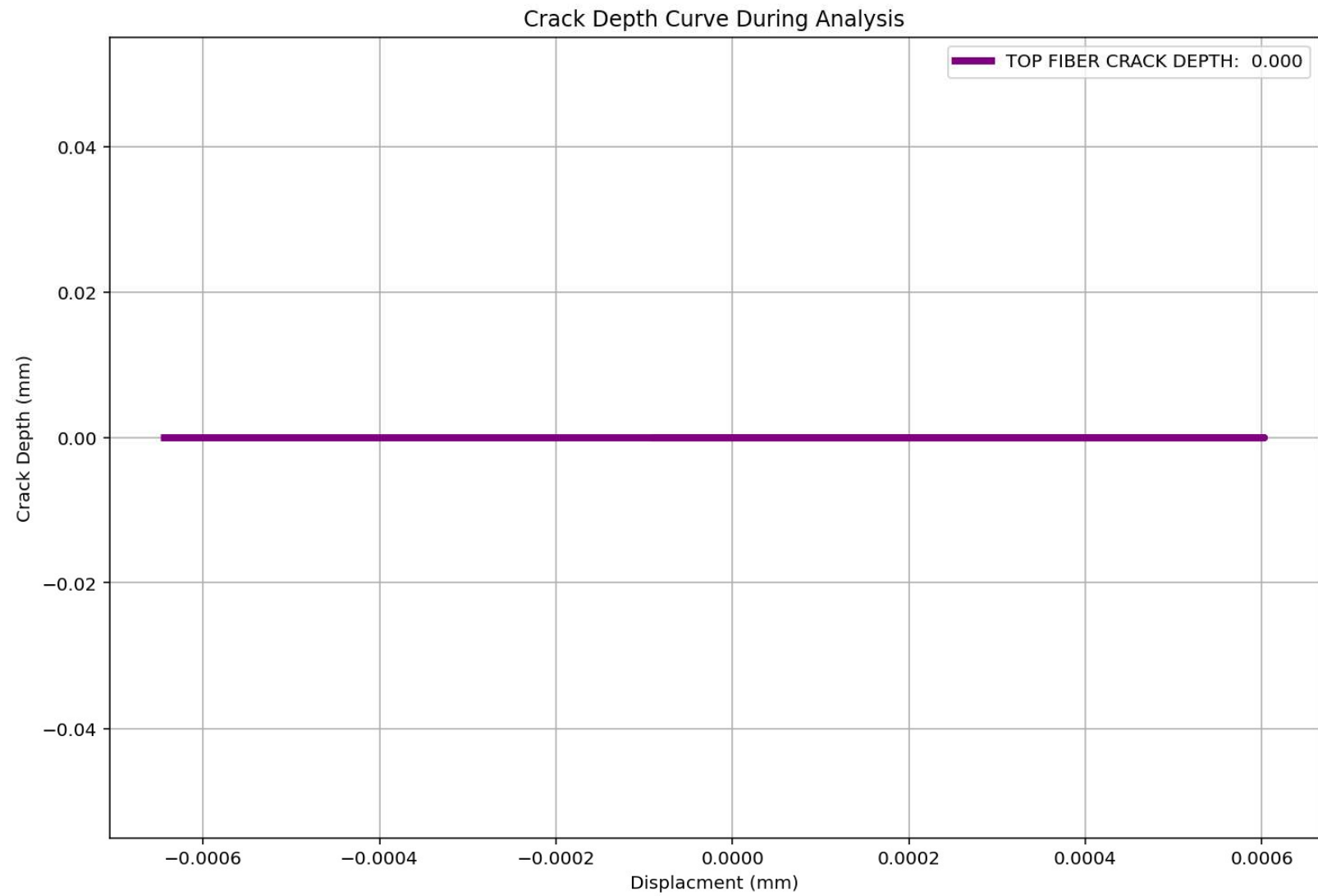


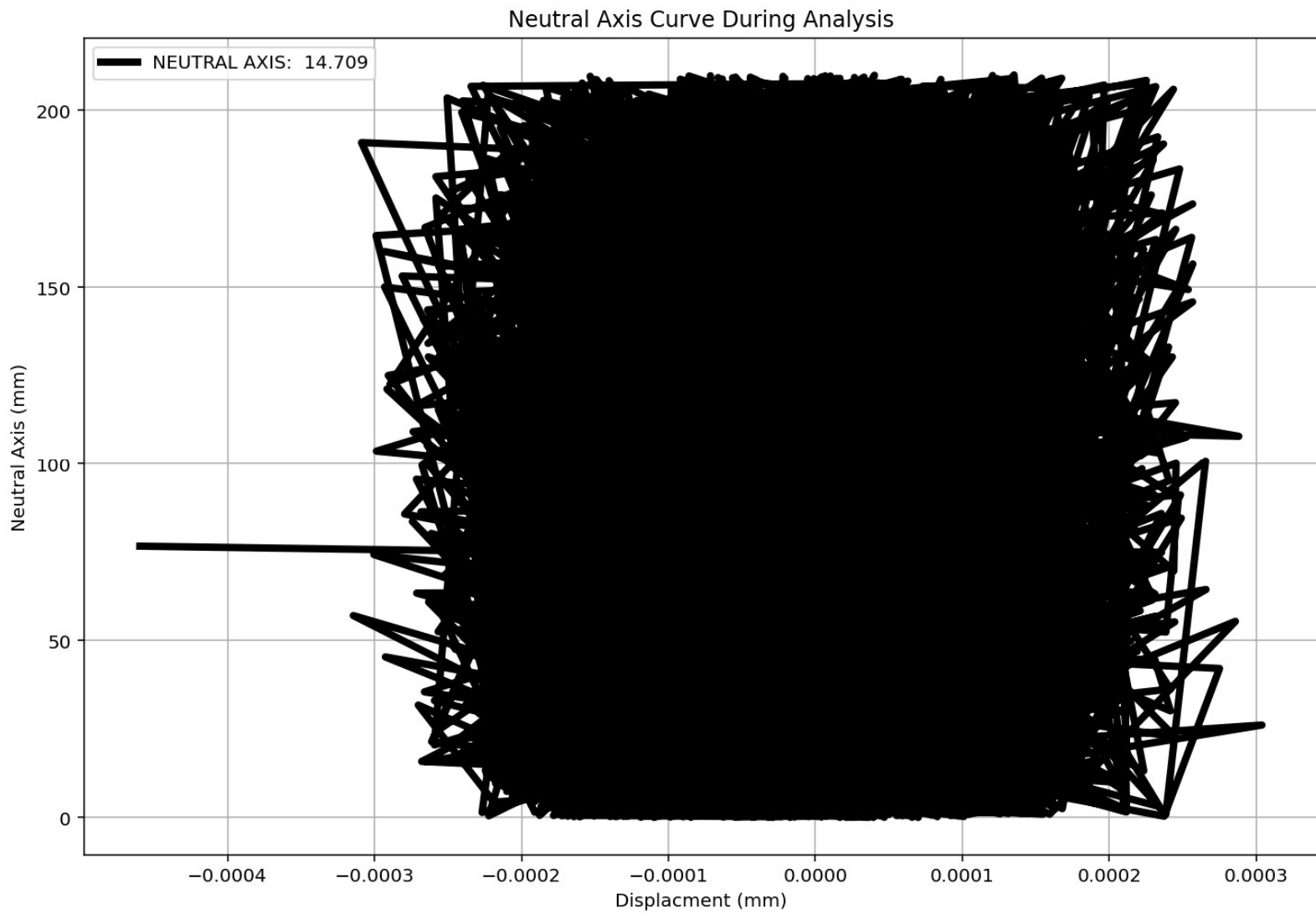
Period of Structure During Free-vibration Analysis



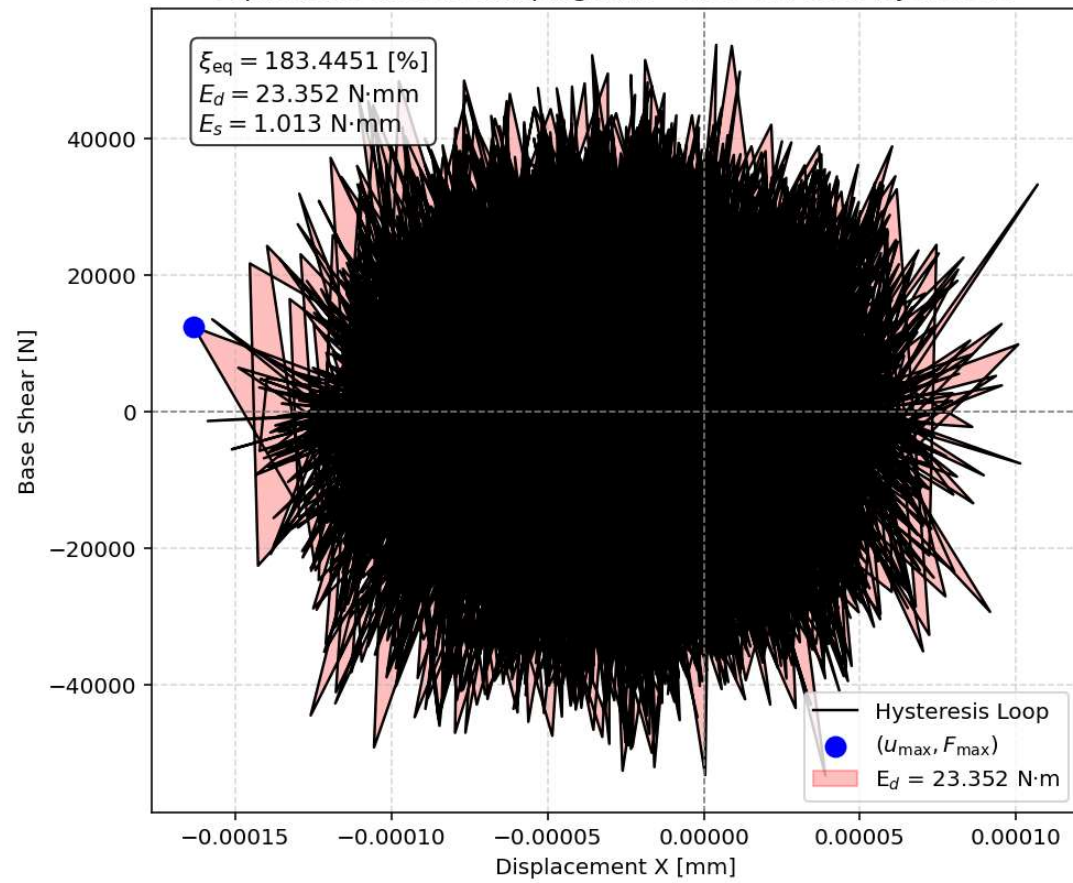
Behavior of element - Stress-strain of Top and Bottom Fibers Curve

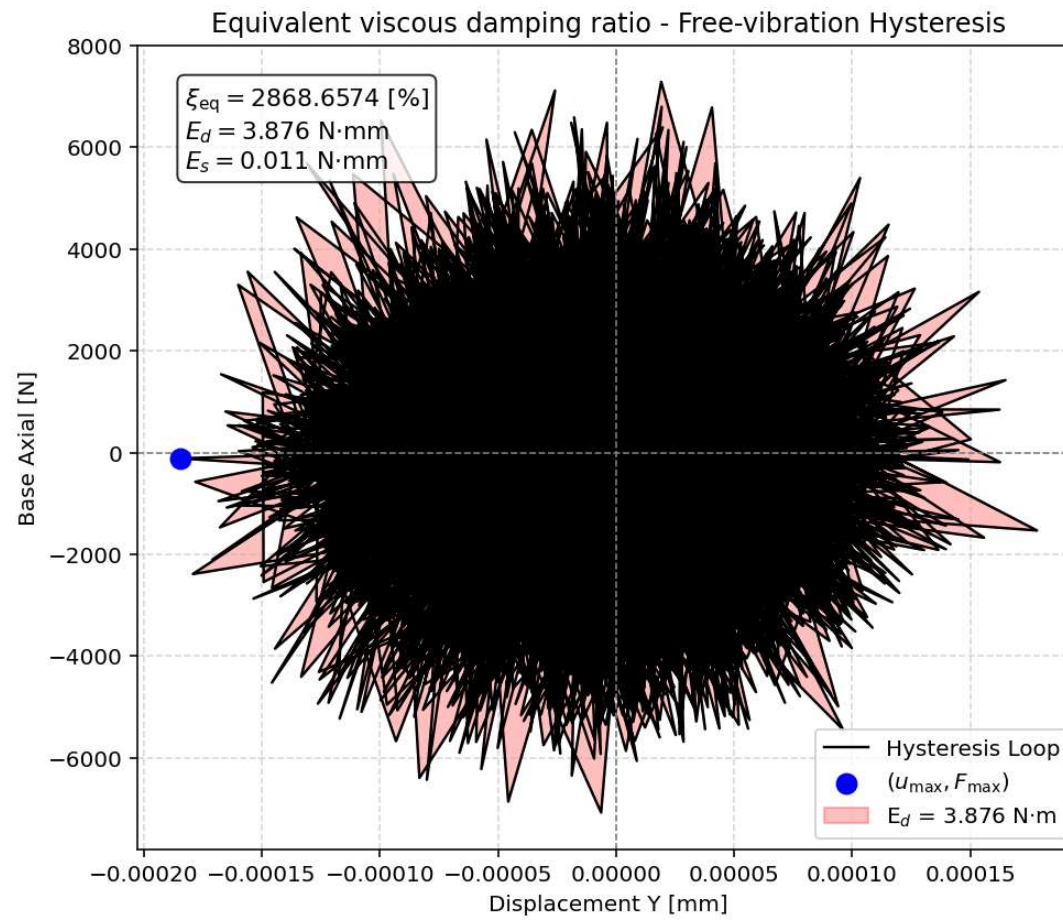


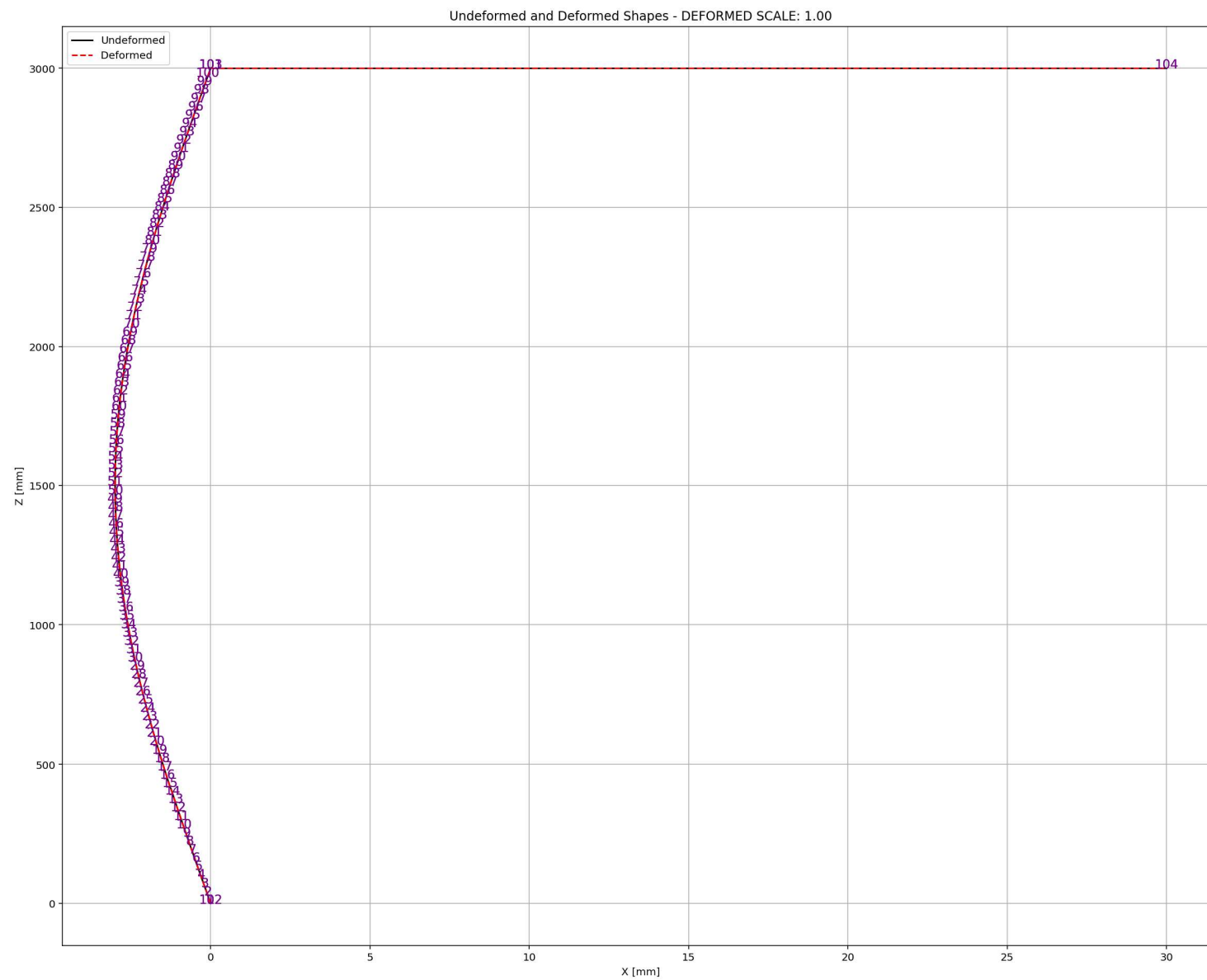




Equivalent viscous damping ratio - Free-vibration Hysteresis







SEISMIC ANALYSIS

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...)_AM_STEEL_CONCRETE_COLUMN_POST-BUCKLING_8_ANA.py

P(t)_AM_STEEL_CONC...-BUCKLING_8_ANA.py X

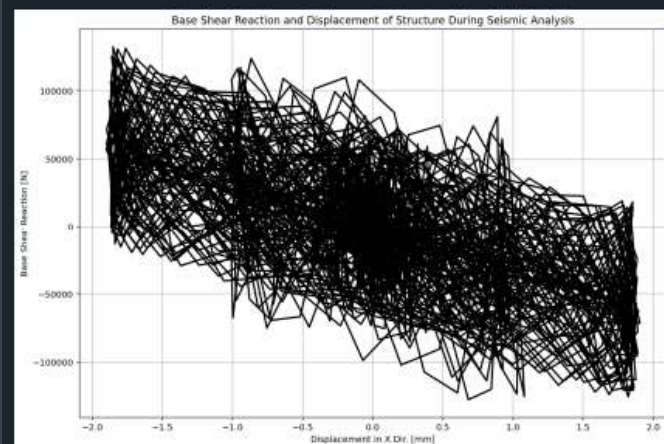
```

955 if ANAL_TYPE == 'SEISMIC': # DYNAMIC TIME-HISTORY ANALYSIS
956     %% DEFINE PARAMETERS FOR SEISMIC ANALYSIS
957     duration = 20.0           # [s] Analysis duration
958     dt = 0.01                # [s] Time step
959     GMfact = 9810            # [mm/s^2] standard acceleration of gravity or stand
960     SSF_X = 2.50             # Seismic Acceleration Scale Factor in X Direction
961     SSF_Y = 2.50             # Seismic Acceleration Scale Factor in Y Direction
962     iv0_X = 0.0005           # [mm/s] Initial velocity applied to the node in
963     iv0_Y = 0.0005           # [mm/s] Initial velocity applied to the node in
964     st_iv0 = 0.0             # [s] Initial velocity applied starting time
965     SEI = 'Y'                # Seismic Direction
966     DR = 0.03                # Damping ratio
967
968     # Define time series for input motion (Acceleration time history)
969     if SEI == 'X':
970         SEISMIC_TAG_01 = 100
971         ops.timeSeries('Path', SEISMIC_TAG_01, '-dt', dt, '-filePath', 'Ground_Accelerati
972         # Define load patterns
973         # pattern UniformExcitation $patternTag $dof -accel $stsTag <-vel0 $vel0> <-fact $
974         ops.pattern('UniformExcitation', SEISMIC_TAG_01, 1, '-accel', SEISMIC_TAG_01, '-v
975     if SEI == 'Y':
976         SEISMIC_TAG_02 = 200
977         ops.timeSeries('Path', SEISMIC_TAG_02, '-dt', dt, '-filePath', 'Ground_Accelerati
978         ops.pattern('UniformExcitation', SEISMIC_TAG_02, 2, '-accel', SEISMIC_TAG_02, '-v
979     if SEI == 'XY':
980         SEISMIC_TAG_01 = 100
981         ops.timeSeries('Path', SEISMIC_TAG_01, '-dt', dt, '-filePath', 'Ground_Accelerati
982         # Define load patterns
983         # pattern UniformExcitation $patternTag $dof -accel $stsTag <-vel0 $vel0> <-fact $
984         ops.pattern('UniformExcitation', SEISMIC_TAG_01, 1, '-accel', SEISMIC_TAG_01, '-v
985         SEISMIC_TAG_02 = 200
986         ops.timeSeries('Path', SEISMIC_TAG_02, '-dt', dt, '-filePath', 'Ground_Accelerati
987         ops.pattern('UniformExcitation', SEISMIC_TAG_02, 2, '-accel', SEISMIC_TAG_02, '-v
988     print('Seismic Defined Done '\

```

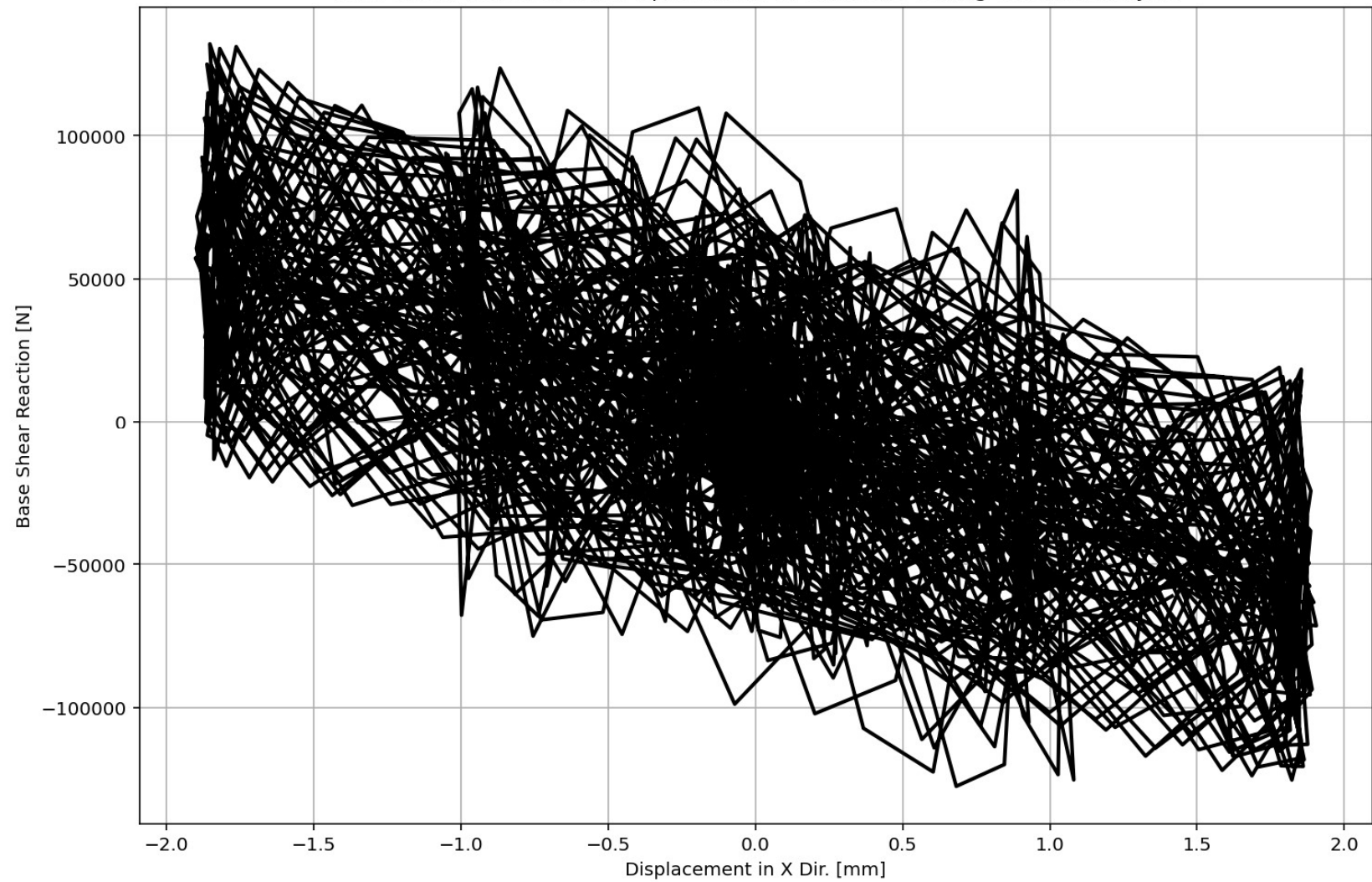
...DISSIPATION_CAPACITY_INDEX\57_POST_FIRE+EXAMPLE_01_COLUMN

28 %

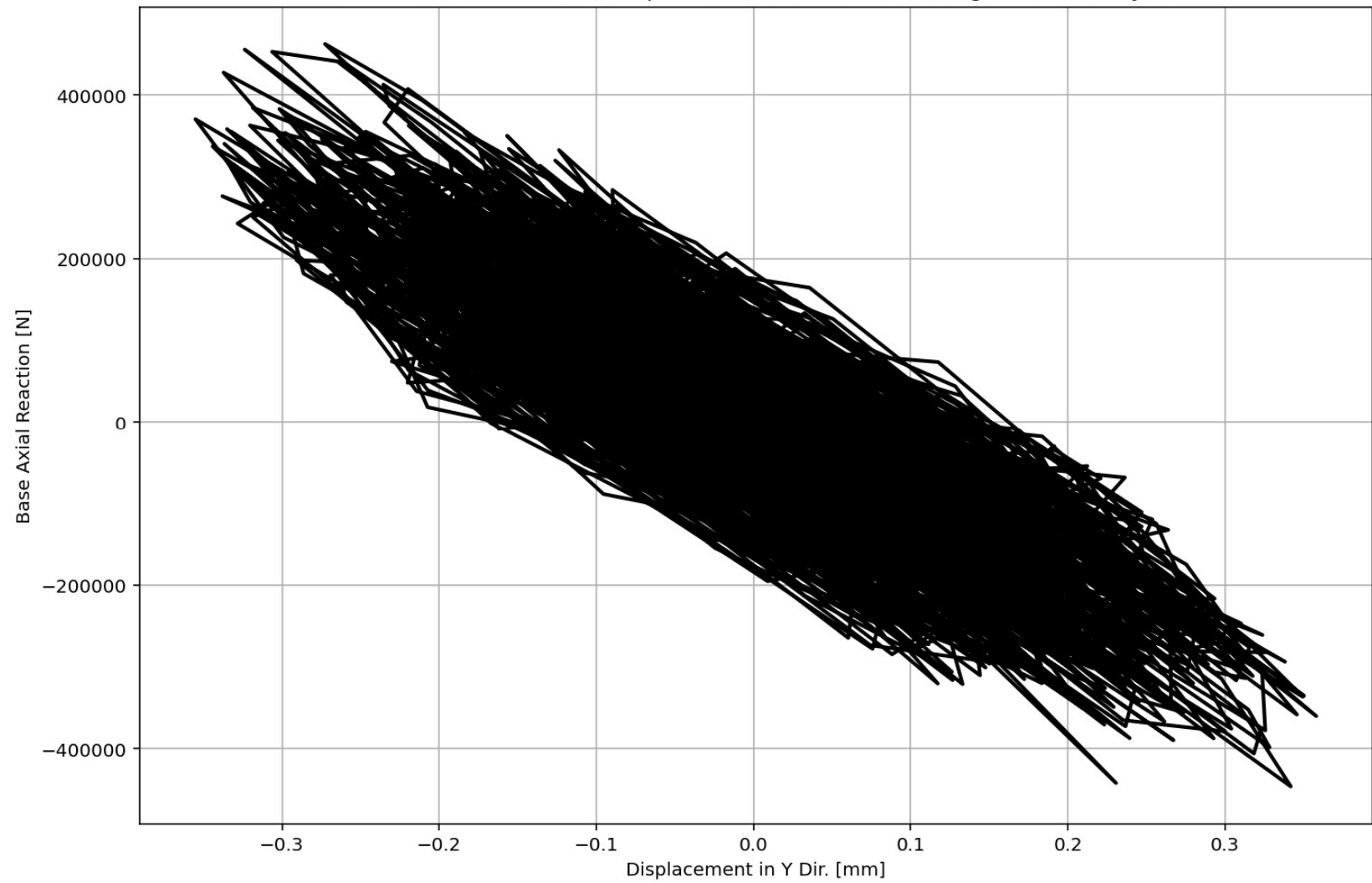


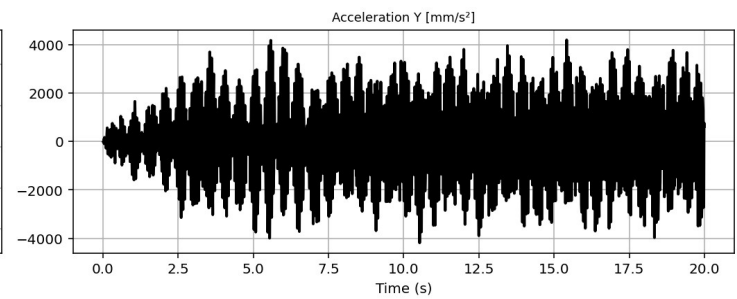
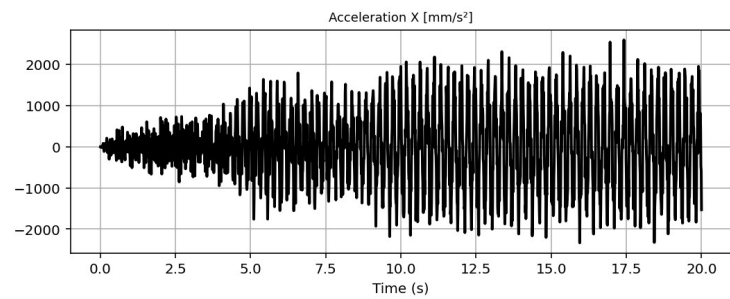
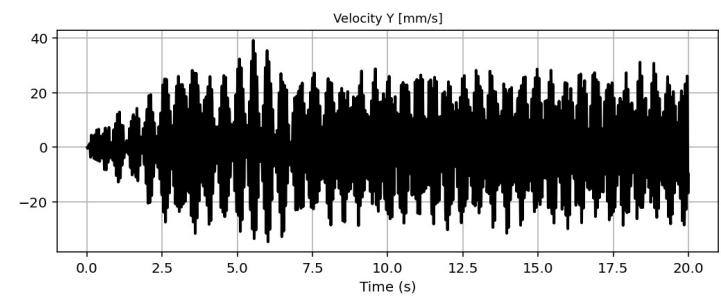
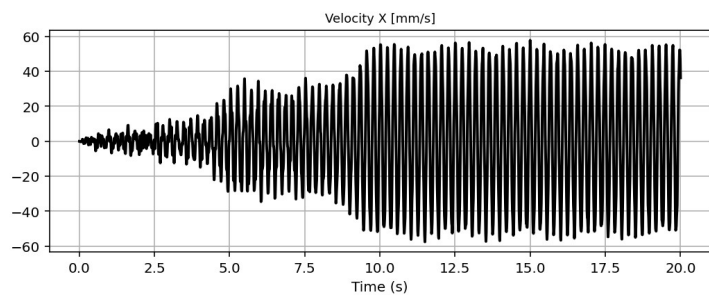
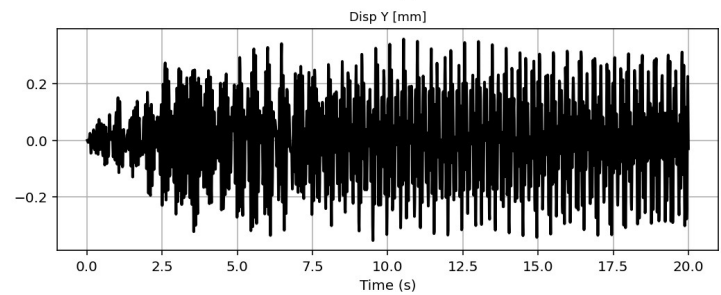
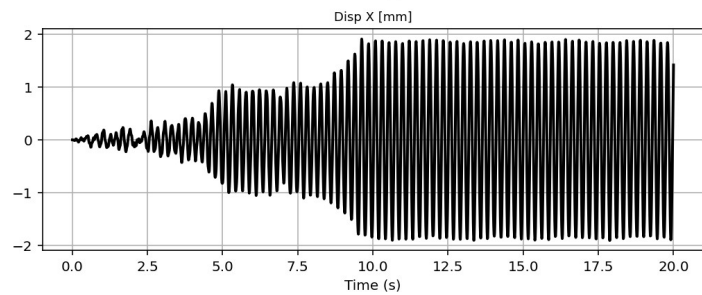
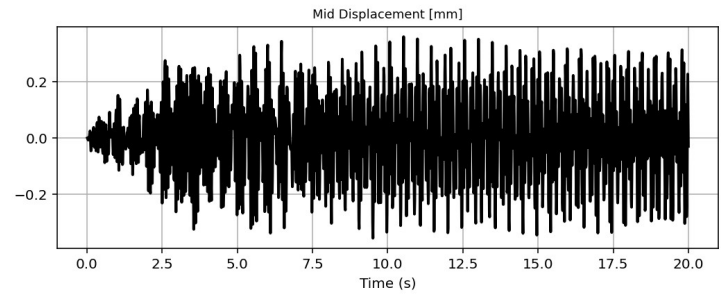
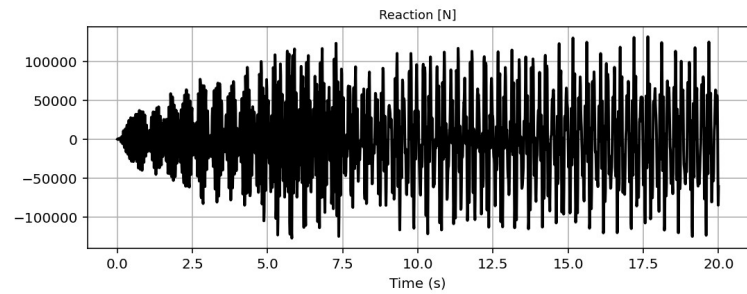
IPython Console Plots Variable Explorer Help Debugger History Files

Base Shear Reaction and Displacement of Structure During Seismic Analysis

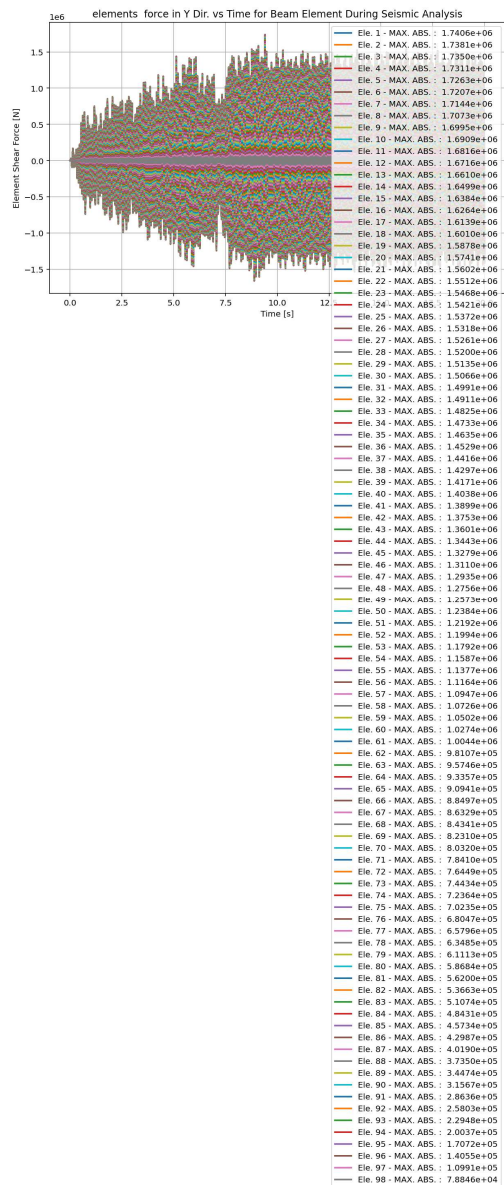


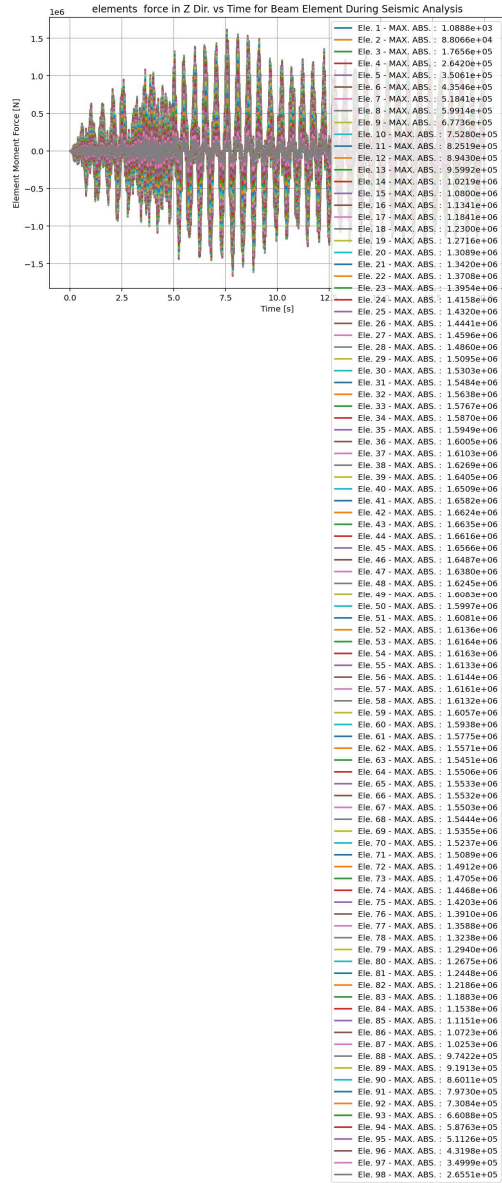
Base Axial Reaction and Displacment of Structure During Seismic Analysis

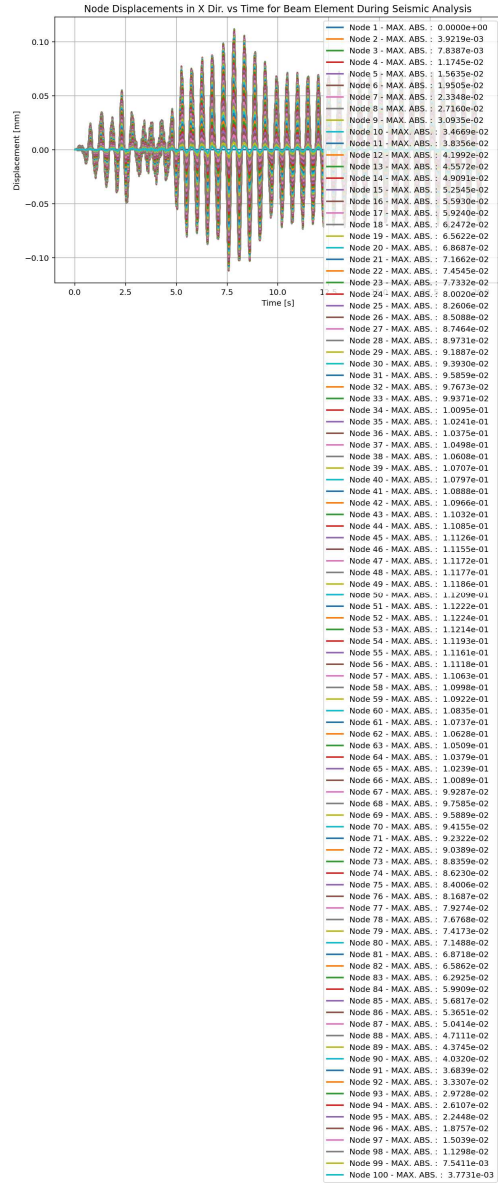


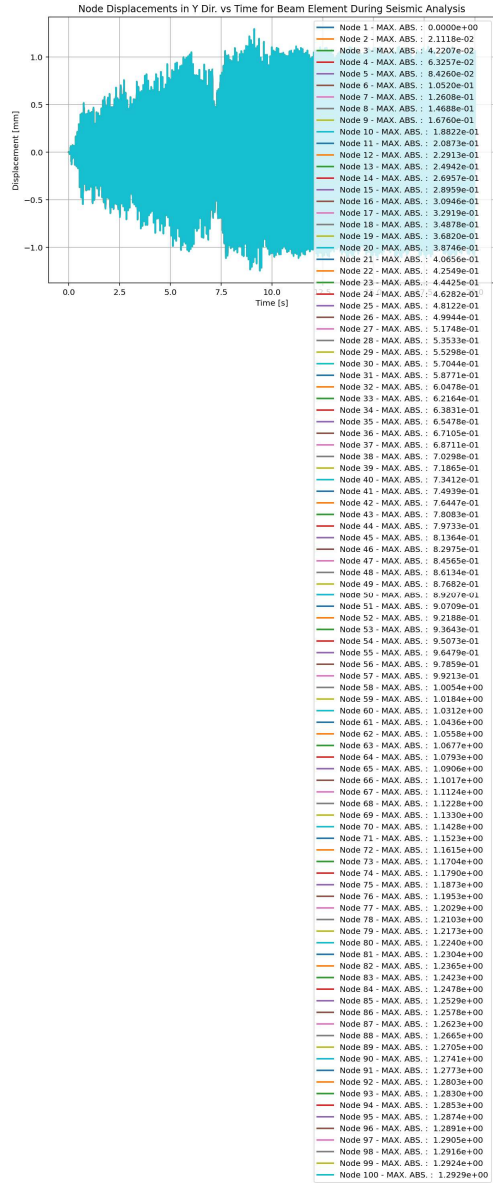




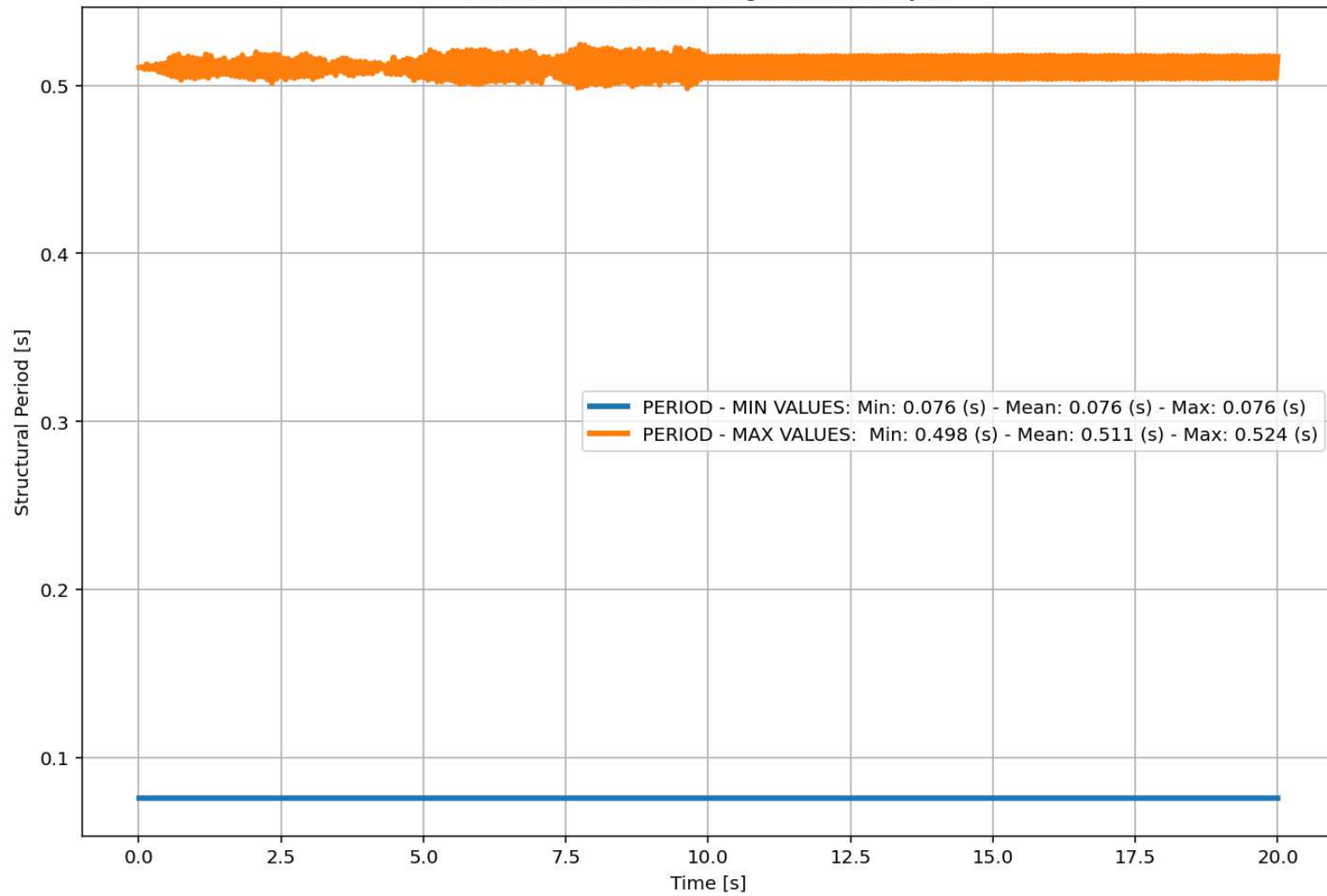


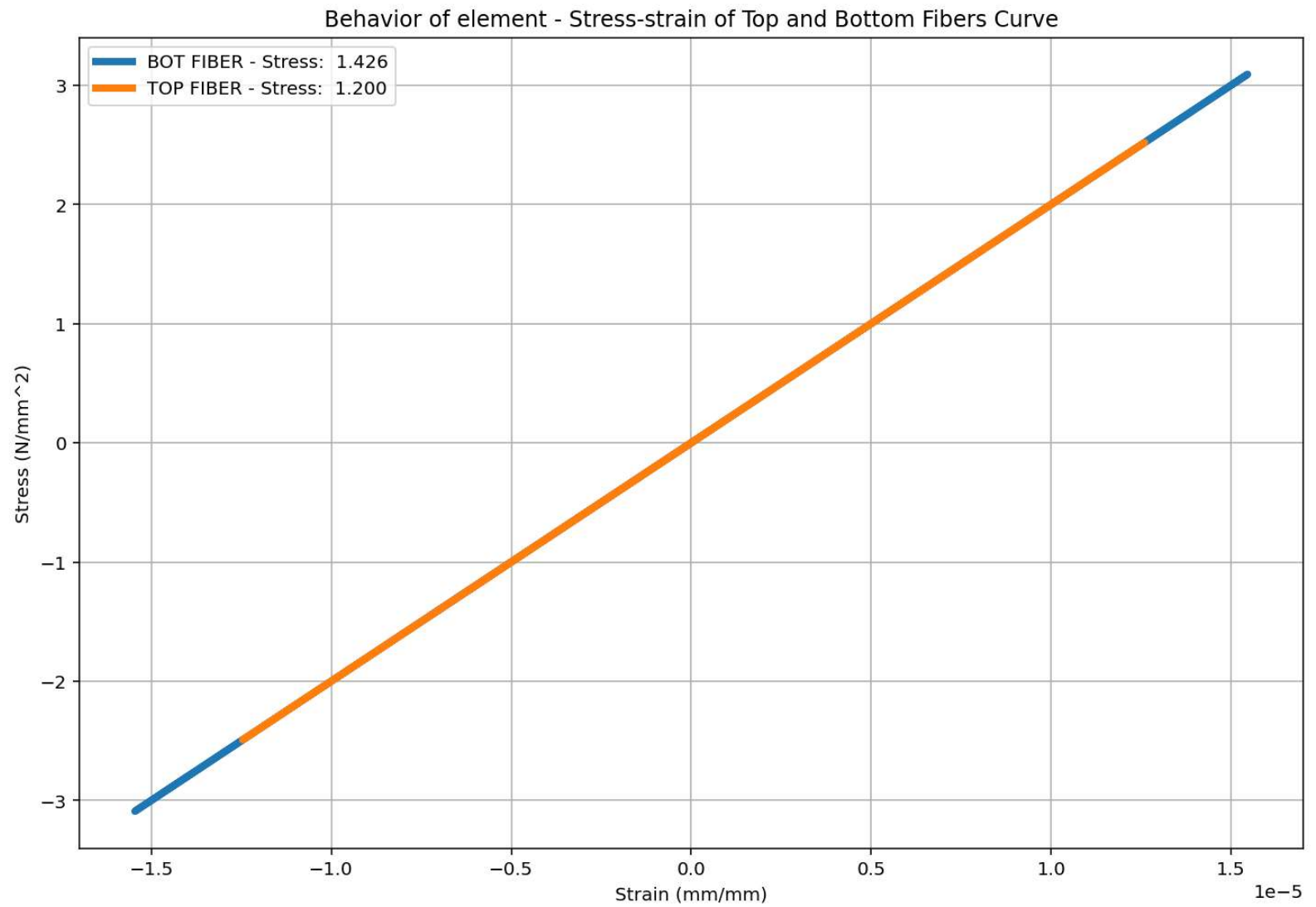


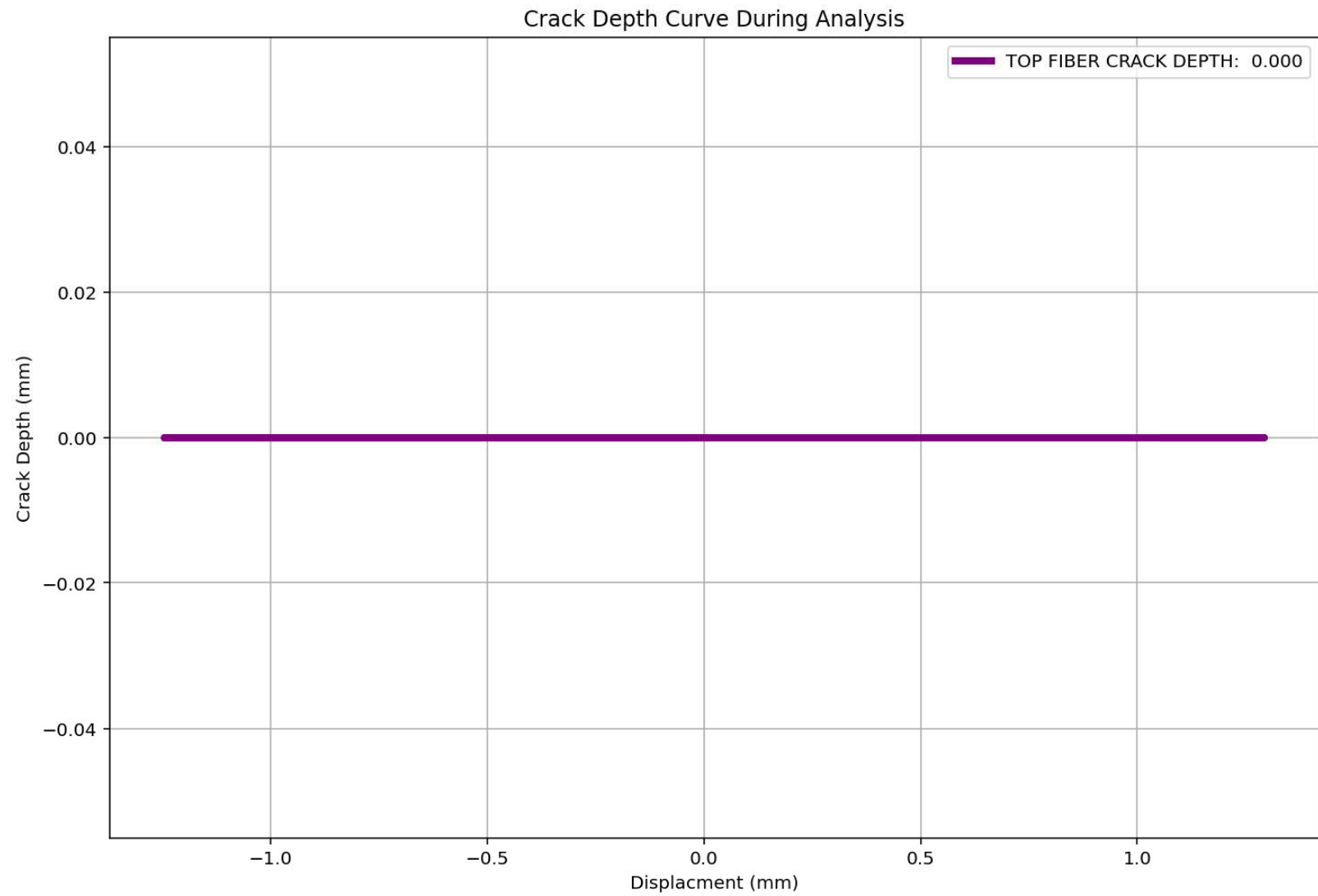


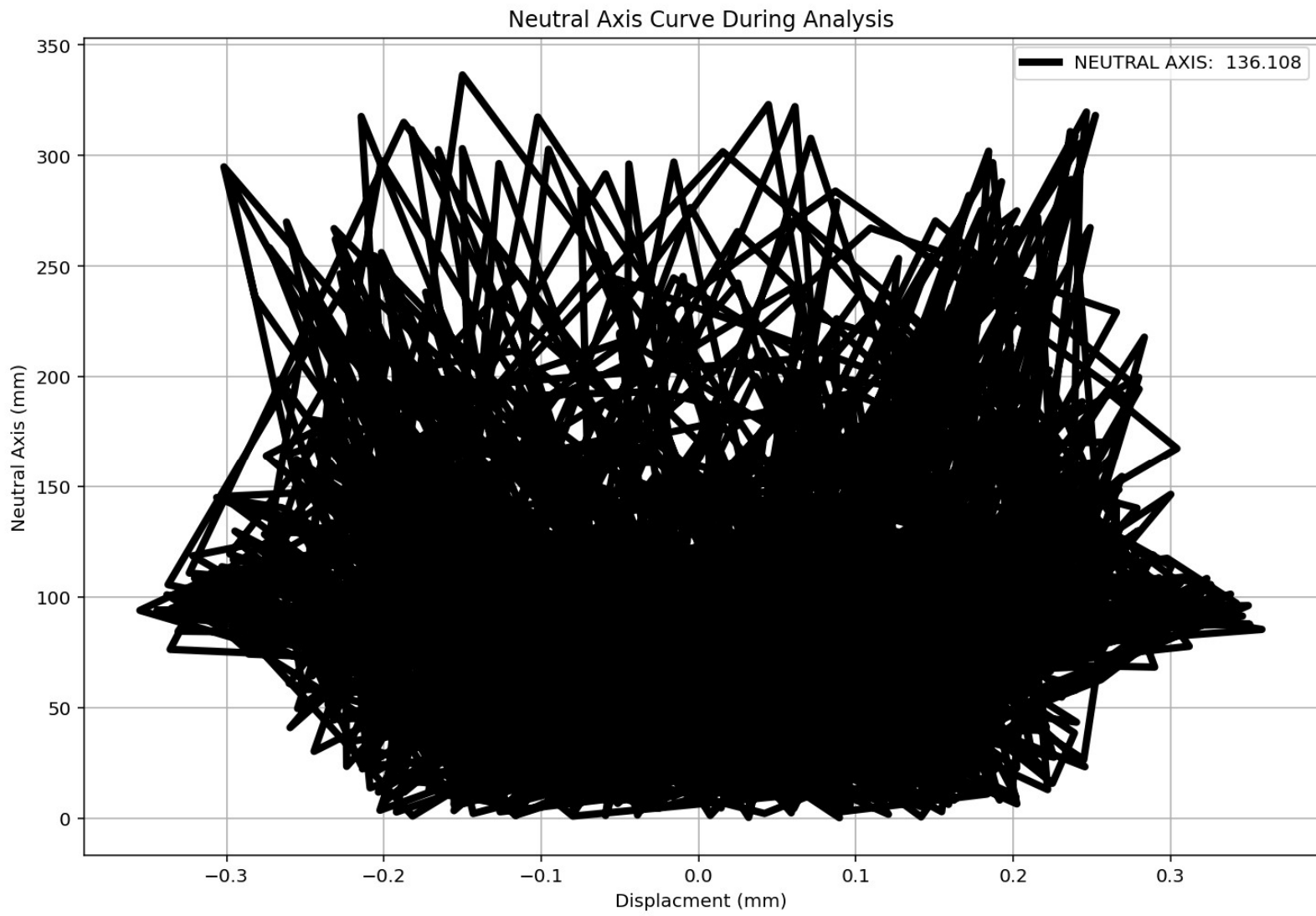


Period of Structure During Seismic Analysis

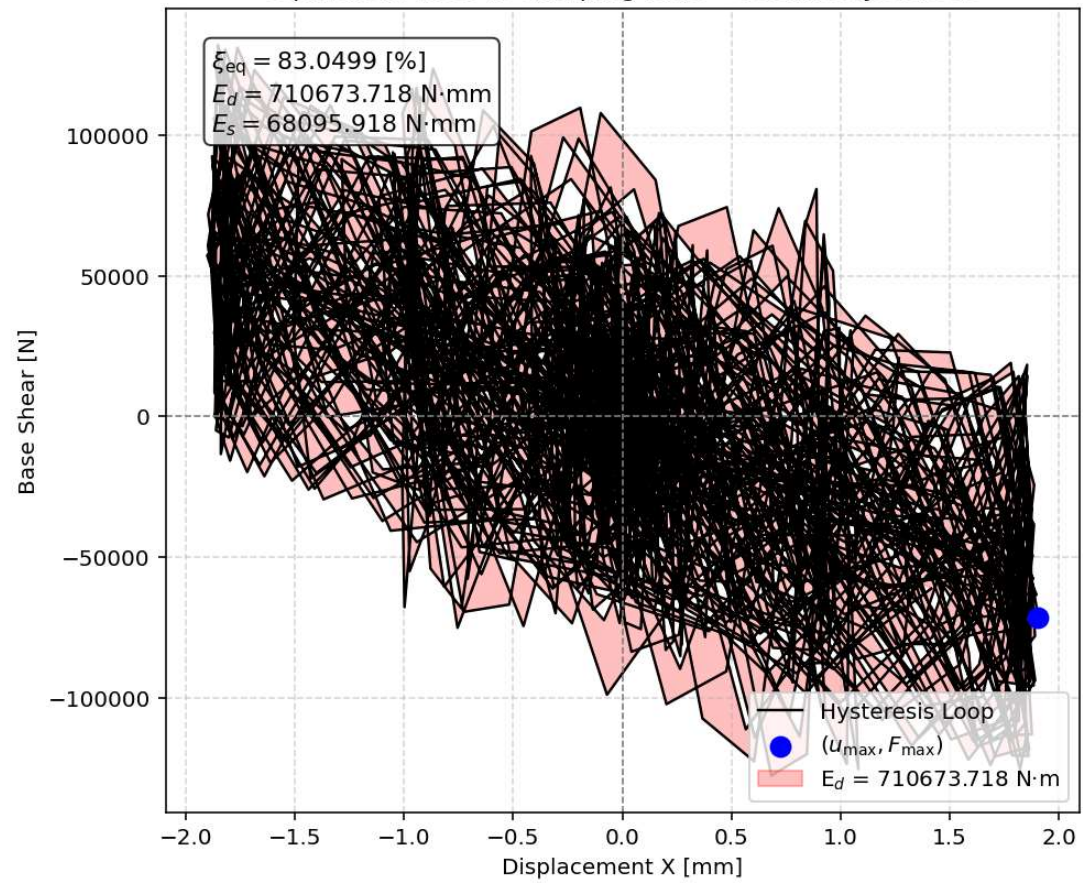




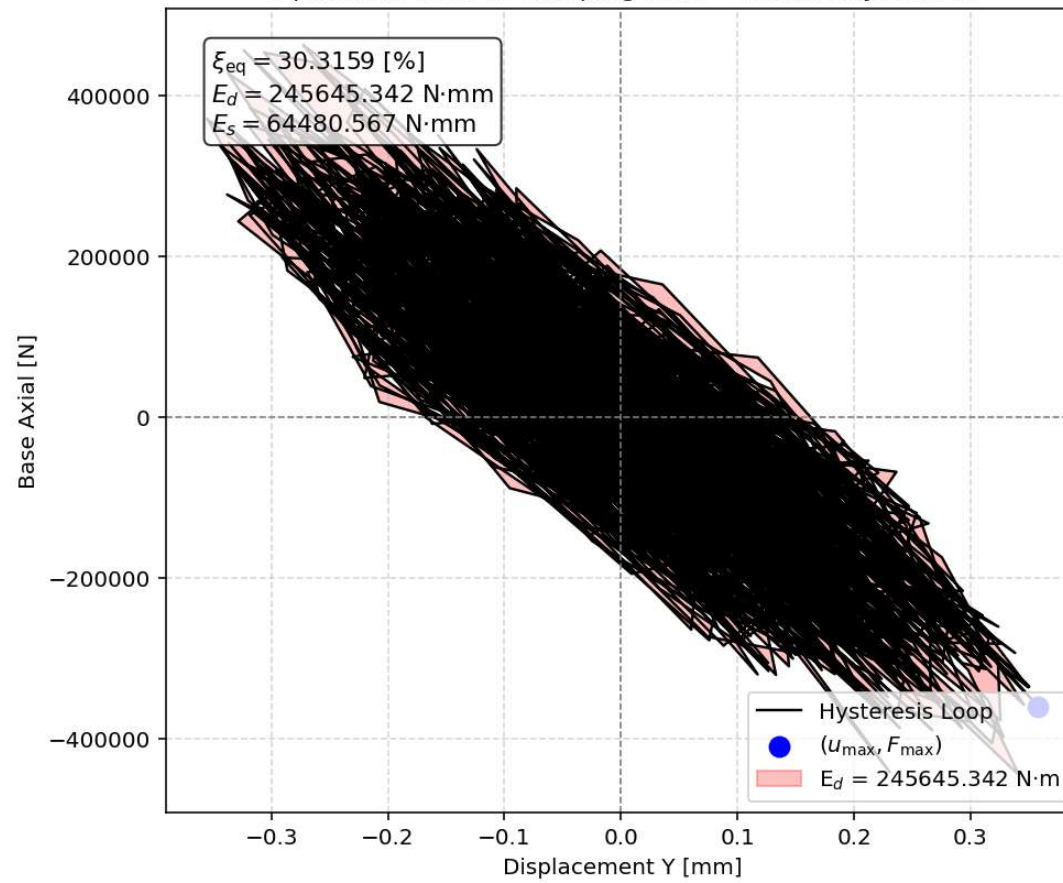


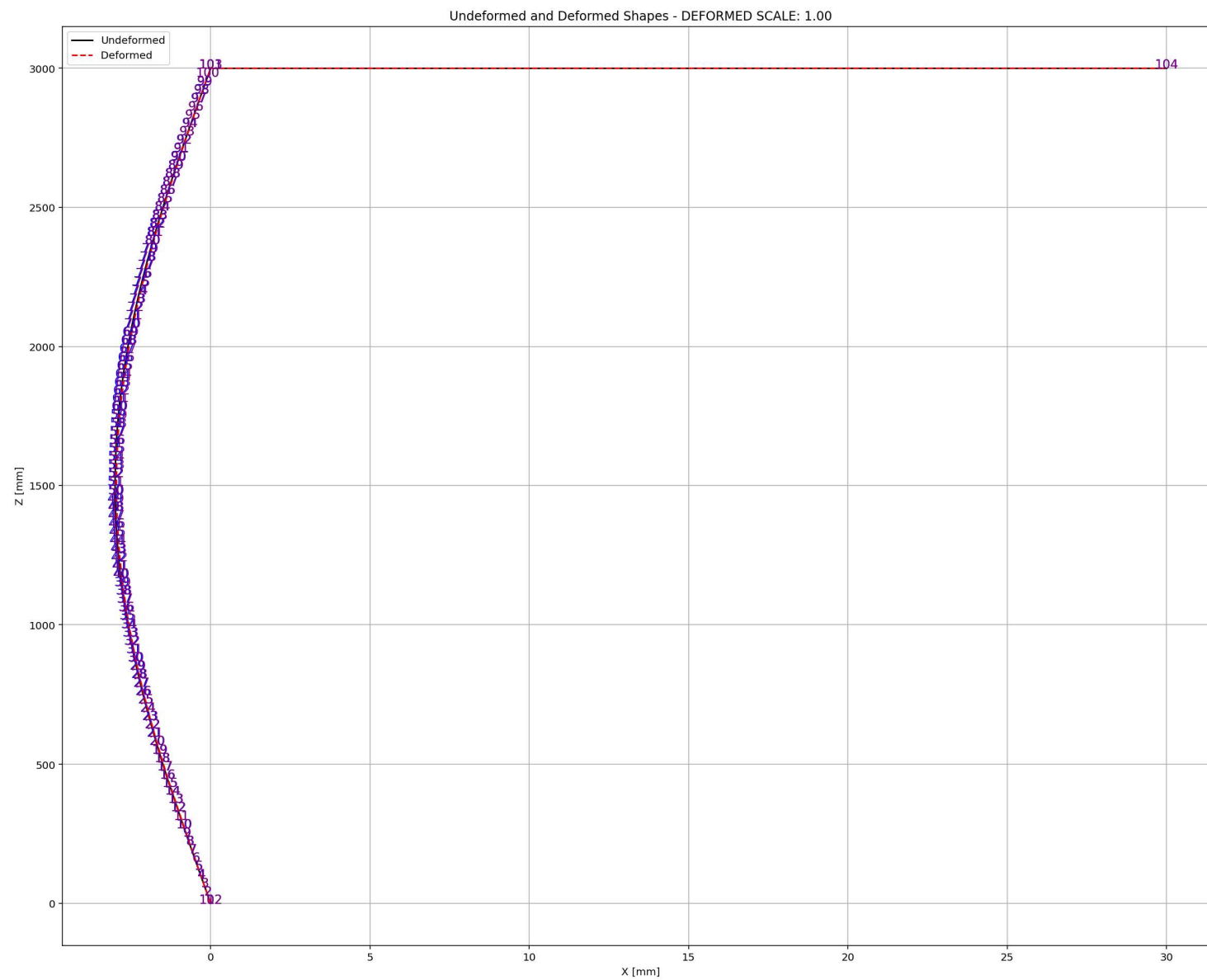


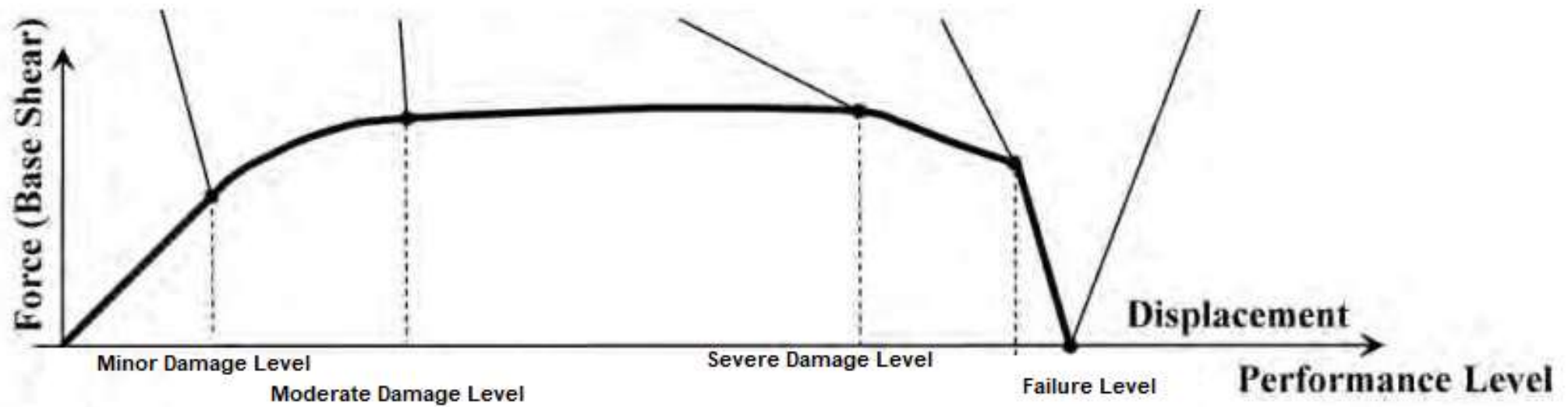
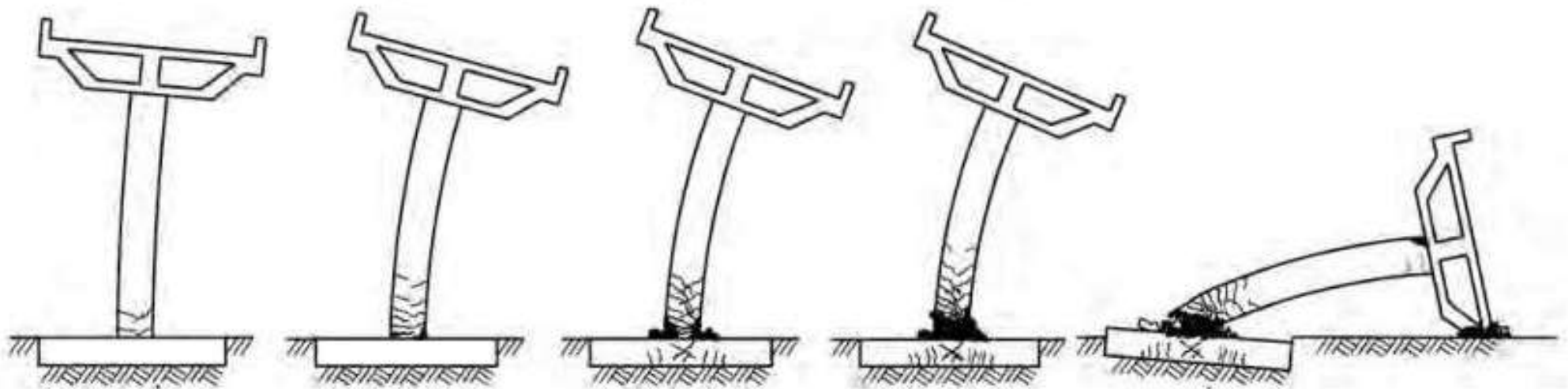
Equivalent viscous damping ratio - Seismic Hysteresis

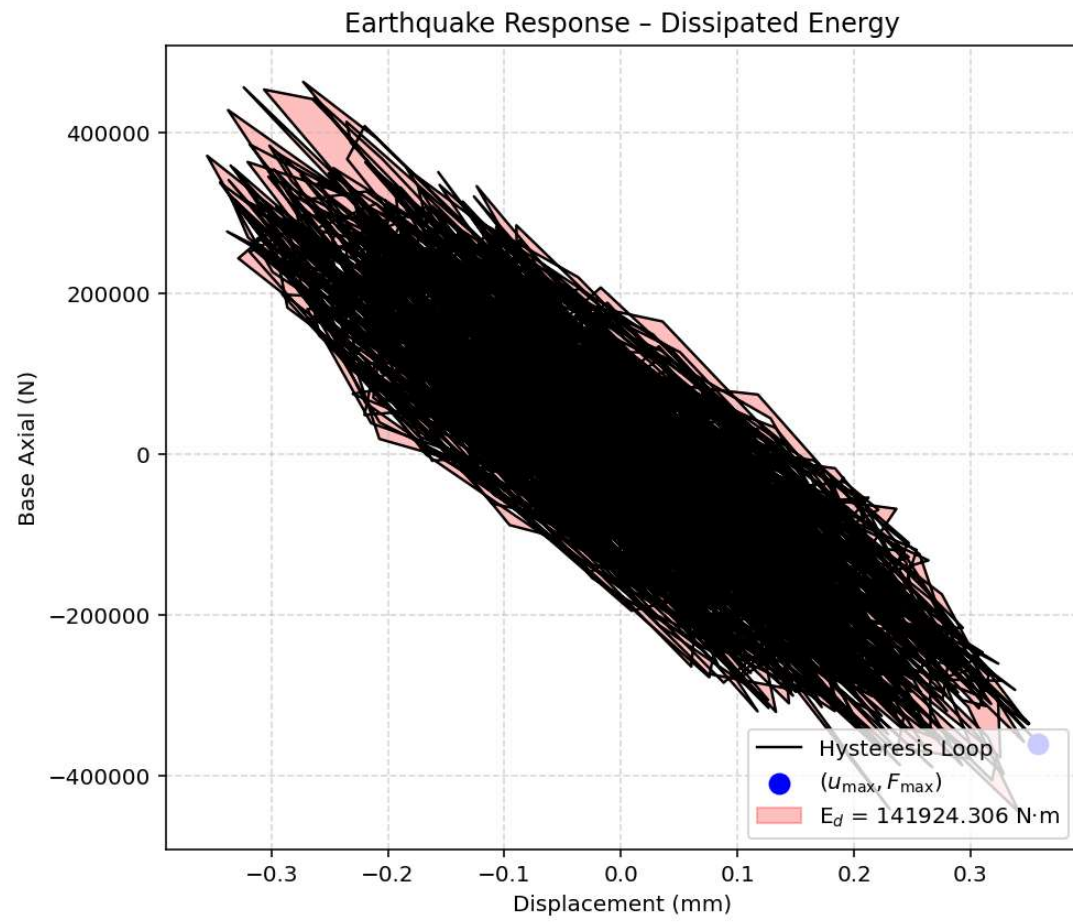


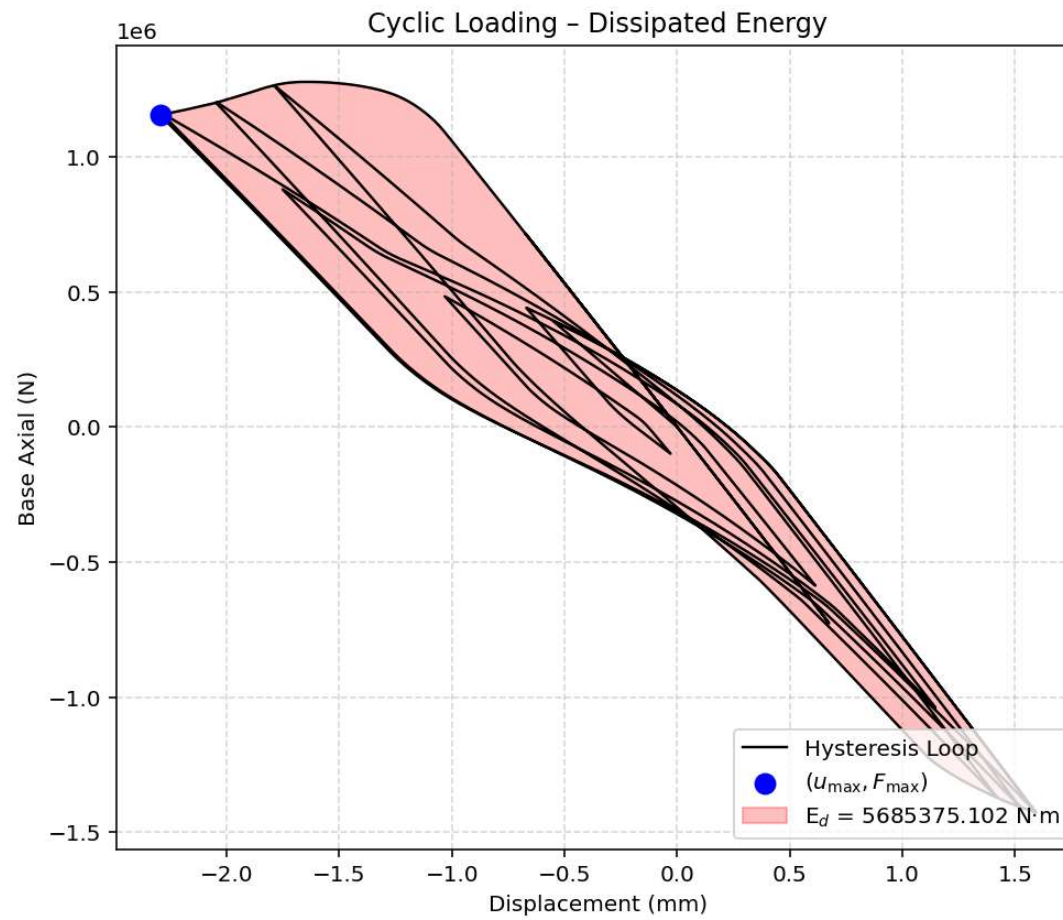
Equivalent viscous damping ratio - Seismic Hysteresis







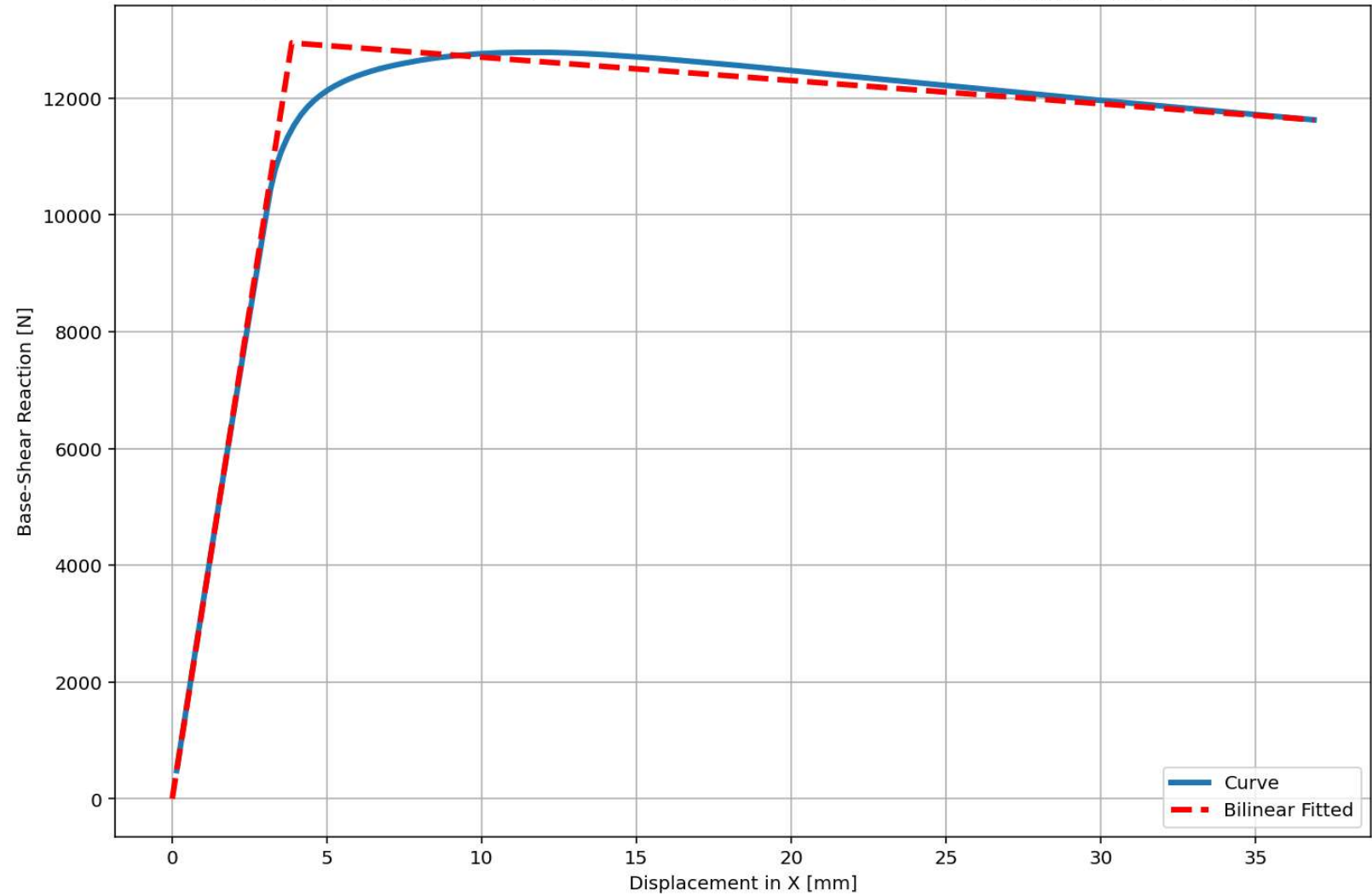




Dissipated Energy from Earthquake= 141924.31 N·mm
Dissipated Energy from Cyclic Displacement= 5685375.10 N·m

DISSIPATED ENERGY CAPACITY INDEX: 2.496 [%]
ZONE 1: VERY MINOR DAMAGE

Base Shear-Displacement Analysis - Ductility Ratio: 9.5498 - Over Strength Factor: 0.8982



+=====+

= Analysis curve fitted =

Disp Base Shear

[[0.00000000e+00 0.00000000e+00]

[3.86386277e+00 1.29458814e+04]

[3.68991417e+01 1.16281984e+04]]

+=====+

+-----+

Structure Elastic Stiffness : 3350.50

Structure Plastic Stiffness : 315.13

Structure Tangent Stiffness : -39.89

Structure Ductility Ratio : 9.55

Structure Over Strength Factor: 0.90

+-----+

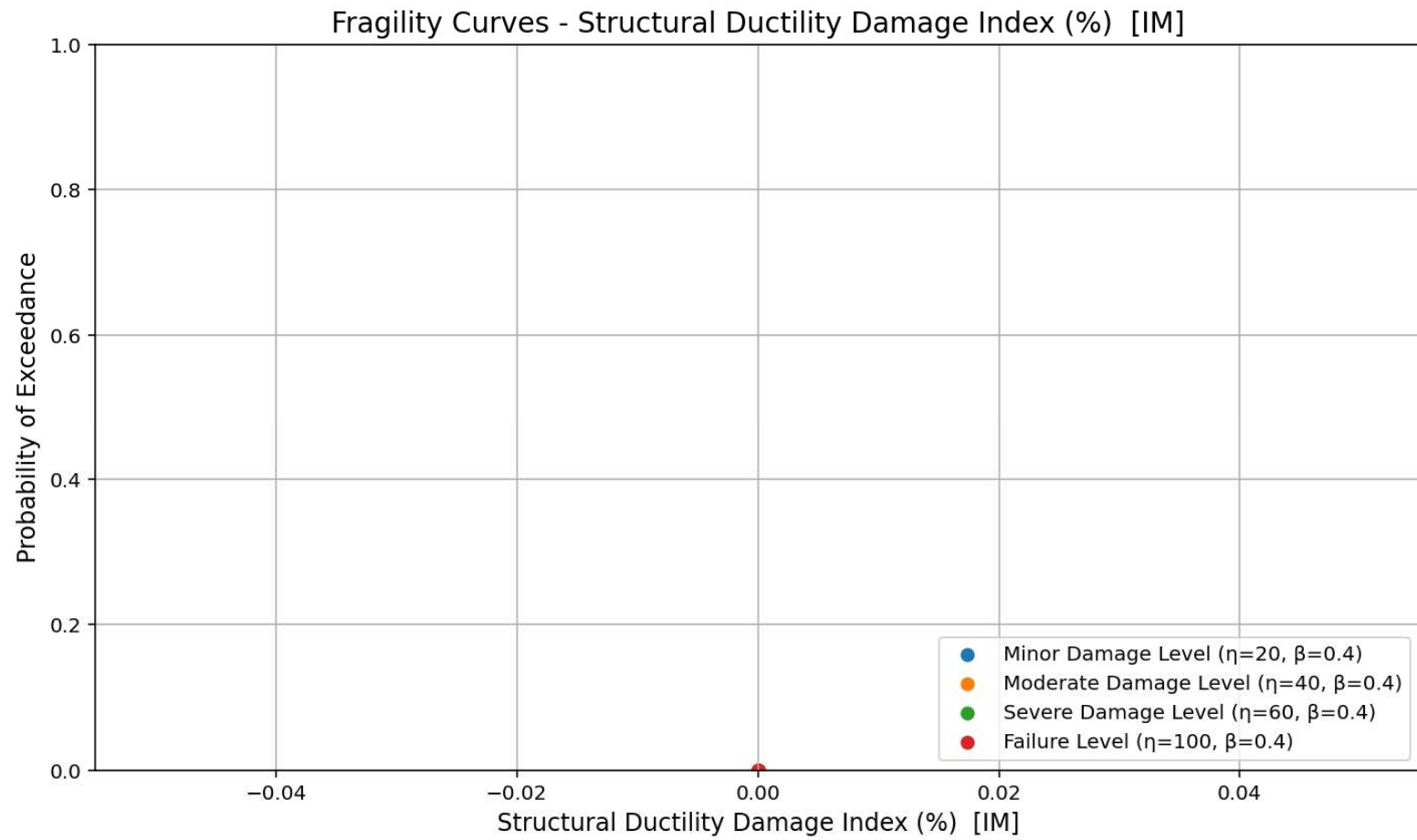
Over Strength Coefficient (Ω_0): 0.8982

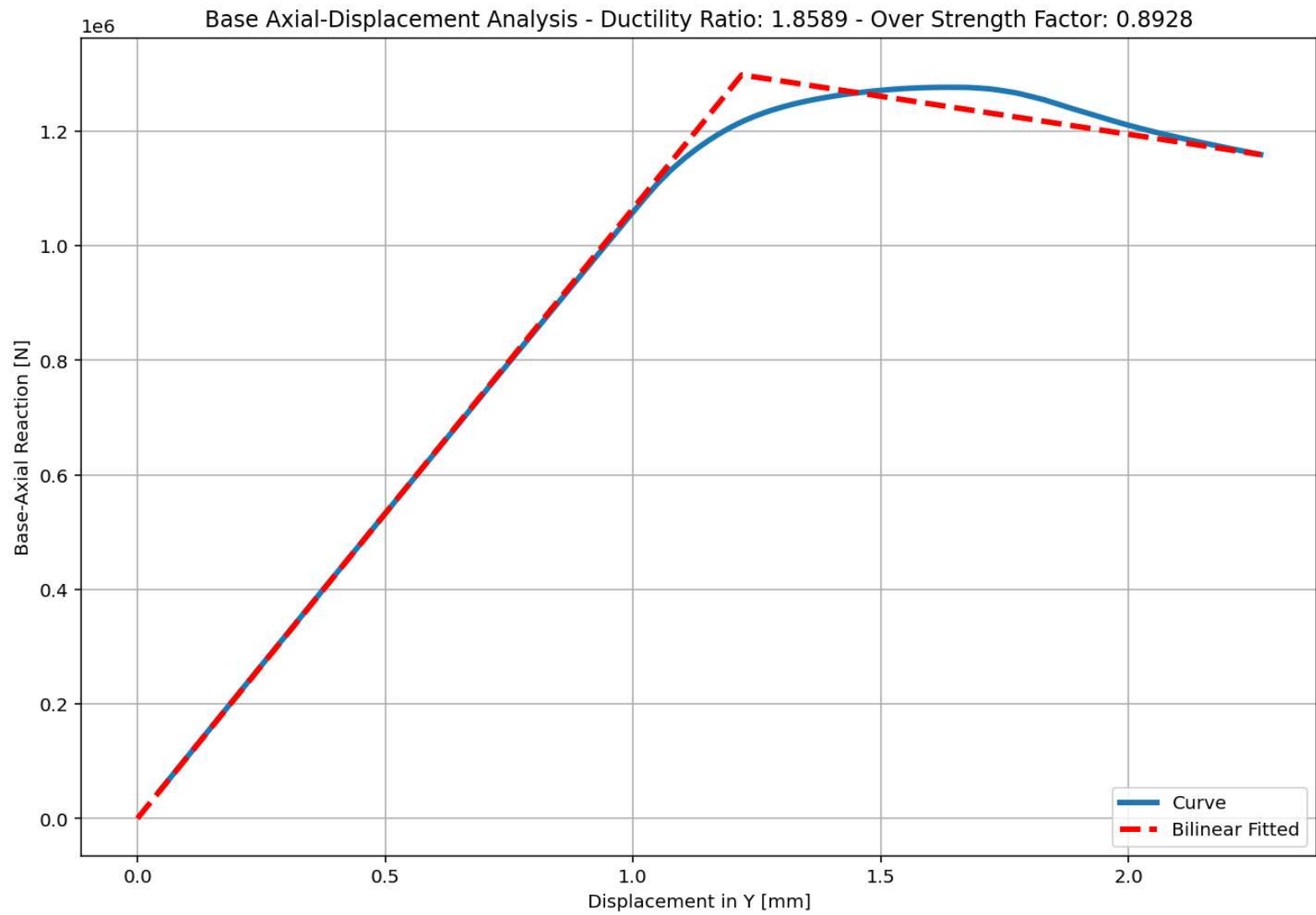
Displacement Ductility Ratio (μ): 9.5498

Ductility Coefficient (R_μ): 9.5498

Structural Behavior Coefficient (R): 8.5778

Structural Ductility Damage Index in X Direction: -5.9291 (%)





+=====+

= Analysis curve fitted =

Disp Base Shear

[[0.00000000e+00 0.00000000e+00]

[1.21954389e+00 1.29801277e+06]

[2.26702882e+00 1.15890539e+06]]

+=====+

+-----+

Structure Elastic Stiffness : 1064342.81

Structure Plastic Stiffness : 511200.11

Structure Tangent Stiffness : -132801.33

Structure Ductility Ratio : 1.86

Structure Over Strength Factor: 0.89

+-----+

Over Strength Coefficient (Ω_0): 0.8928

Displacement Ductility Ratio (μ): 1.8589

Ductility Coefficient (R_μ): 1.8589

Structural Behavior Coefficient (R): 1.6597

Structural Ductility Damage Index in Y Direction: -82.2420 (%)

